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Acupuncture and Dry Needling

Number: 0135

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Scope of Policy

This Clinical Policy Bulletin addresses acupuncture and dry needling.

I. **Medical Necessity**

Aetna considers acupuncture (manual or electroacupuncture) medically necessary for *any* of the following indications:

A. Chronic (minimum 12 weeks duration) neck pain; *or*
B. Chronic (minimum 12 weeks duration) headache; *or*
C. Low back pain; *or*
D. Nausea of pregnancy; *or*
E. Pain from osteoarthritis of the knee or hip (adjunctive therapy); *or*
F. Post-operative and chemotherapy-induced nausea and vomiting; *or*
G. Post-operative dental pain; *or*
H. Temporomandibular disorders (TMD).

Maintenance treatment, where the member's symptoms are neither regressing or improving, is considered not medically necessary. If no clinical benefit is appreciated after four weeks of acupuncture, then the treatment plan should be reevaluated. Further acupuncture treatment is not considered medically necessary if the member does not demonstrate meaningful improvement in symptoms.

II. **Experimental, Investigational, or Unproven**

The following procedures are considered experimental, investigational, or unproven because the effectiveness of these approaches has not been established:

A. Acupuncture

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There is inadequate scientific research assessing the efficacy of acupuncture compared with placebo, sham acupuncture or other modalities of treatment in the following conditions (not an all inclusive list):

- Abdominal obesity
- Acne
- Acute pancreatitis
- Addiction
- AIDS
- Alcohol withdrawal syndrome
- Allergies
- Alzheimer’s disease
- Amblyopia
- Anorexia
- Asthma
- Attention deficit hyperactivity disorder (ADHD)
- Autism spectrum disorders
- Bell's palsy
- Benign prostatic hypertrophy
- Breast cancer-related hot flashes
- Breast cancer-related lymphedema
- Burning mouth syndrome
- Cancer-induced bone pain
- Cancer-related dyspnea
- Cancer-related fatigue
- Cardiovascular diseases (e.g., angina pectoris, heart failure, hypertension)
- Carpal tunnel syndrome
- Cerebral palsy
- Chemotherapy-induced leukopenia
- Chemotherapy-induced neuropathic pain
- Chronic hepatitis B
- Chronic ankle instability
- Chronic pain syndrome (e.g., RSD, facial pain)
- Chronic obstructive pulmonary disease (COPD)
- Chronic constipation
- Chronic fatigue syndrome
- Cognitive impairment
- Coronary heart disease
- Diabetic gastroparesis
- Diabetic peripheral neuropathy
- Diminished ovarian reserve
- Dry eyes
- Dysmenorrhea
- Endometriosis pain
- Epilepsy
- Erectile dysfunction
- Facial spasm
- Fetal breech presentation
- Fibromyalgia
- Fibrotic contractures
- Gastric ulcer
- Glaucoma
- Gout
- Hypoxic ischemic encephalopathy
- Induction of labor
- Infantile colic
- Infantile diarrhea
- Infertility (e.g., to assist oocyte retrieval and embryo transfer during IVF treatment cycle)
- Inflammatory bowel diseases (Crohn's disease and ulcerative colitis)
- Insomnia (including cancer-related insomnia)
- Intra-cerebral hemorrhage
- Irritable bowel syndrome
- Menopause-associated vasomotor symptoms
- Menopausal hot flashes
- Menstrual cramps/dysmenorrhea
- Mild cognitive impairment
- Multiple sclerosis
- Mumps
- Myofascial pain
- Myopia
- Neuropathic pain
- Nocturnal enuresis
- Non-alcoholic fatty liver disease
- Obesity / weight reduction
- Obstructive sleep apnea
- Occipital neuralgia
- Oligoasthenozoospermia
- Oral ulcer
- Osteoporosis
- Painful neuropathies
- Parkinson's disease
- Parkinson's disease-related fatigue, depression
- Peptic ulcer
- Peripheral arterial disease (e.g., intermittent claudication)
- Phantom leg pain
- Plantar fasciitis
- Polycystic ovary syndrome
- Post-herpetic neuralgia
- Post-operative ileus
- Post-prandial distress syndrome
- Post-stroke hiccup
- Post-stroke shoulder pain
- Post-surgical neuropathic pain after breast cancer surgery
- Post-traumatic stress disorder
- Premature ejaculation

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- Premenstrual syndrome/premenstrual dysphoric disorder
- Pruritus
- Psoriasis
- Psychiatric disorders (e.g., anxiety, depression, and schizophrenia)
- Raynaud's disease pain
- Renal colic
- Respiratory disorders
- Restless leg syndrome
- Rheumatoid arthritis
- Rhinitis
- Sensorineural deafness
- Sexual dysfunction
- Shoulder bursitis
- Sleep disturbance in individuals with cancer
- Smoking cessation
- Spasticity after stroke
- Stroke rehabilitation (e.g., dysphagia)
- Systemic lupus erythematosus
- Tennis elbow / epicondylitis
- Thoracic back pain
- Tic disorders (e.g., Tourette syndrome)
- Tinnitus
- Urinary incontinence
- Uterine fibroids
- Vascular dementia
- Xerostomia
- Whiplash;

B. Acupuncture point injection (also known as acupoint injection therapy, biopuncture) for the following conditions (not an all-inclusive list):

- Amyotrophic lateral sclerosis
- Cancer-related pain
- Cervical spondylosis
- Chronic daily headache
- Dysmenorrhea (menstrual pain)
- Lateral elbow pain (tennis elbow);

C. Acupotomy for the treatment of herpes zoster, low back/shoulder pain, paresthesia of skin, and trigger finger;
D. Dry needling;
E. Transcutaneous electronic acupoint stimulation for improvement of post-operative recovery after gynecologic surgery;
F. Use of the Teding Dianci Pu (TDP) lamp as an adjunct to acupuncture.

III. Policy Limitations and Exclusions

Note: Standard Aetna plans extend coverage of acupuncture for medically necessary indications when administered by a health care provider practicing within the scope of his/her license. Some Aetna plans limit coverage of acupuncture to when it is used in a lieu of other anesthesia for a surgical or dental procedure covered under the health benefits plan, and the health care provider administering it is a legally qualified physician practicing within the scope of his/her license. Some other plans may extend coverage of acupuncture for medically necessary indications, but only when administered by a health care provider who is a legally qualified physician practicing within the scope of his/her license. Please check benefit plan descriptions for details.

Table:

CPT Codes / HCPCS Codes / ICD-10 Codes

Code	Code Description

CPT codes covered if selection criteria are met:

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Code	Code Description
97810	Acupuncture, 1 or more needles; without electrical stimulation, initial 15 minutes of personal one-on-one contact with patient
+ 97811	without electrical stimulation, each additional 15 minutes of personal one-on-one contact with the patient, with re-insertion of needle(s) (List separately in addition to code for primary procedure)
97813	with electrical stimulation, initial 15 minutes of personal one-on-one contact with patient
+ 97814	with electrical stimulation, each additional 15 minutes of personal one-on-one contact with the patient, with re-insertion of needle(s) (List separately in addition to code for primary procedure)

CPT codes not covered for indications listed in the CPB:

***Teding Dianci Pu (TDP) lamp, transcutaneous electronic acupoint stimulation* -no specific code**

20560	Needle insertion(s) without injection(s); 1 or 2 muscle(s)
20561	3 or more muscles [dry needling]

Other CPT codes related to CPB:

19120	Excision of cyst, fibroadenoma, or other benign or malignant tumor, aberrant breast tissue, duct lesion, nipple or areolar lesion (except 19300), open, male or female, 1 or more lesions
19125	Excision of breast lesion identified by preoperative placement of radiological marker, open; single lesion
+19126	each additional lesion separately identified by a preoperative radiological marker (List separately in addition to code for primary procedure)
19301	Mastectomy, partial (eg, lumpectomy, tylectomy, quadrantectomy, segmentectomy)
19302	with axillary lymphadenectomy
19303	Mastectomy, simple, complete
19305	Mastectomy, radical, including pectoral muscles, axillary lymph nodes
19306	Mastectomy, radical, including pectoral muscles, axillary and internal mammary lymph nodes (Urban type operation)
19307	Mastectomy, modified radical, including axillary lymph nodes, with or without pectoralis minor muscle, but excluding pectoralis major muscle
19316	Mastopexy
19340	Insertion of breast implant on same day of mastectomy (ie, immediate)
19342	Insertion or replacement of breast implant on separate day from mastectomy
19357	Tissue expander placement in breast reconstruction, including subsequent expansion(s)
19361	Breast reconstruction; with latissimus dorsi flap
19364	with free flap (eg, fTRAM, DIEP, SIEA, GAP flap)
19367	with single-pedicled transverse rectus abdominis myocutaneous (TRAM) flap
19368	with single-pedicled transverse rectus abdominis myocutaneous (TRAM) flap, requiring separate microvascular anastomosis (supercharging)
19369	with bipedicled transverse rectus abdominis myocutaneous (TRAM) flap

HCPCS codes covere if selection criteria are met:

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Code	Code Description
S8930	Electrical stimulation of auricular acupuncture points; each 15 minutes of personal one-on-one contact with the patient

ICD-10 codes covered if selection criteria are met (not all-inclusive):

G43.001 - G43.E19	Migraine
K08.9	Disorder of teeth and supporting structures, unspecified [postoperative dental pain]
M16.0 - M16.12	Primary osteoarthritis of hip
M16.2 - M16.7	Secondary osteoarthritis, hip
M16.9	Osteoarthritis of hip, unspecified
M17.0 - M17.12	Osteoarthritis of knee
M17.2 - M17.5	Secondary osteoarthritis, knee
M17.9	Osteoarthritis of knee, unspecified
M26.601 - M26.69	Temporomandibular joint disorders
M54.2	Cervicalgia [chronic neck pain]
M54.50 - M54.59	Low back pain
O21.0 - O21.9	Excessive vomiting in pregnancy
R11.2	Nausea with vomiting [postoperative] [chemotherapy-induced]
R51.0- R51.9	Headache
T45.1x5A - T45.1x5S	Adverse effect of antineoplastic and immunosuppressive drugs [chemotherapy-induced nausea and vomiting]
Z98.890 - Z98.891	Other specified postprocedural status [dental, with pain]

ICD-10 codes not covered for indications listed in the CPB:

B02.21 - B02.29	Zoster with other nervous system involvement
B18.0 - B18.1	Chronic (viral) hepatitis B
B20	Human immunodeficiency virus [HIV] disease
B26.0 - B26.9	Mumps
C50.011 - C50.929	Malignant neoplasm of breast
C79.81	Secondary malignant neoplasm of breast
D05.00 - D05.92	Carcinoma in situ of breast
D25.0 - D25.9	Leiomyoma of uterus (fibroids)
D72.819	Decreased white blood cell count, unspecified [leukocytopenia]
E28.2	Polycystic ovaries syndrome
E28.310	Symptomatic premature menopause
E28.8	Other ovarian dysfunction [diminished ovarian reserve]
E66.01 - E66.1 E66.3 - E66.9	Overweight and obesity
E89.41	Symptomatic postprocedural ovarian failure
F01.50 - F99	Mental disorders [including addiction, insomnia, tension headache, smoking cessation, autism spectrum disorders]
G12.21	Amyotrophic lateral sclerosis
G20.A1 - G20.C	Parkinson's disease
G25.81	Restless legs syndrome
G30.0 - G30.9	Alzheimer's disease
G31.84	Mild cognitive impairment, so stated

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Code	Code Description
G35	Multiple sclerosis
G40.001 - G40.919	Epilepsy and recurrent seizures
G44.1	Vascular headache, not elsewhere classified
G47.00 - G47.09	Insomnia
G47.30	Sleep apnea, unspecified
G47.33	Obstructive sleep apnea (adult) (pediatric)
G47.8	Other sleep disorders [sleep disturbance in individuals with cancer]
G50.0 - G64	Trigeminal nerve disorders, facial nerve disorders, disorders of other cranial nerves, nerve root and plexus disorders, mononeuritis of upper limb and mononeuritis multiplex, mononeuritis of lower limb, hereditary and idiopathic peripheral neuropathy or inflammatory and toxic neuropathy
G80.0 - G80.9	Cerebral palsy
G89.18	Other acute postprocedural pain
G89.28	Other chronic postprocedural pain
G90.50 - G90.59	Complex regional pain syndrome I (CRPS I)
H04.121 - H04.129	Dry eye syndrome
H40.001 - H40.9	Glaucoma
H52.10 - H52.13	Myopia
H53.001 - H53.039	Amblyopia ex anopsia
H90.3 - H90.5	Sensorineural hearing loss
H93.11 - H93.19	Tinnitus
H93.A1 - H93.A9	Pulsatile tinnitus
I00 - I99.9	Diseases of the circulatory system
I10 - I1A.0	Hypertensive disease
I20.0 - I20.9	Angina pectoris
I21.01 - I25.9	Coronary heart disease
I50.1 - I50.9	Heart Failure
I60.00 - I69.993	Cerebrovascular disease
I69.998	Other sequelae following unspecified cerebrovascular disease [spasticity after stroke]
I70.0 - I75.89 I77.0 - I79.8	Diseases of arteries, arterioles and capillaries
I89.0	Lymphedema, not elsewhere classified
I97.2	Postmastectomy lymphedema syndrome [breast cancer-related]
J00 - J99	Diseases of the respiratory system
K11.7	Disturbance of salivary secretion [xerostomia]
K12.30 - K12.39	Oral mucositis (ulcerative)
K14.6	Glossodynia [burning mouth syndrome]
K25.0 - K25.9	Gastric ulcer
K27.0 - K27.9	Peptic ulcer
K30	Functional dyspepsia [post-prandial distress syndrome]
K31.84	Gastroparesis [diabetic]
K50.00 - K51.919	Inflammatory bowel diseases [Crohn's disease and ulcerative colitis]
K58.0 - K58.9	Irritable bowel syndrome
K59.00 - K59.09	Constipation
K75.81	Nonalcoholic steatohepatitis (NASH)
K76.0	Fatty (change of) liver, not elsewhere classified [non-alcoholic fatty liver disease]
K85.00 - K85.92	Acute pancreatitis

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Code	Code Description
K91.89	Other postprocedural complications and disorders of digestive system [post-operative ileus]
L29.0 - L29.9	Pruritus
L40.0 - L40.9	Psoriasis
L70.0 - L70.9	Acne
M05.00 - M05.09 M05.20 - M06.9 M08.00 - M08.99	Rheumatoid arthritis
M10.00 – M10.9	Gout
M1A.00X0 – M1A.09X1	Chronic gout
M25.371 – M25.373	Other instability, ankle
M25.721 - M25.729	Osteophyte, elbow
M30.0 - M31.9	Polyarteritis nodosa and related conditions
M32.0 – M32.9	Systemic lupus erythematosus (SLE)
M43.20 - M43.28, M43.27 - M43.28 M48.00 - M48.08 M51.14 - M51.17, M53.2x7 - M53.2x8 M53.3, M54.03 - M54.09, M54.14 - M54.17, M62.830	Other and unspecified disorders of back
M43.6, M48.01 - M48.03, M48.8x2 M50.10 - M50.13, M53.0 - M53.1 M53.81 - M53.83, M54.00 - M54.02 M54.11 - M54.13, M54.81 M99.20 - M99.21, M99.30 - M99.31 M99.40 - M99.41, M99.50 - M99.51 M99.60 - M99.61, M99.70 - M99.71	Other disorders of cervical region
M47.11 - M47.13, M47.21 - M47.23 M47.811 - M47.813, M47.891 - M47.893	Cervical spondylosis without myelopathy or with myelopathy
M50.00 - M50.03	Cervical disc disorder with myelopathy
M50.20 - M50.23	Other cervical disc displacement
M54.6	Pain in thoracic spine
M60.80 - M60.9	Other myositis
M62.40 - M62.49	Contracture of muscle [fibrotic contracture]
M72.2	Plantar fascial fibromatosis
M75.00 - M75.02	Adhesive capsulitis of shoulder
M75.100 - M75.122 M75.50 - M75.52	Rotator cuff tear or rupture, not specified as traumatic, and bursitis of shoulder
M77.10 - M77.12	Lateral epicondylitis

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Code	Code Description
M79.10 - M79.18	Myalgia
M79.2	Neuralgia and neuritis, unspecified [neuropathic pain]
M80.00xA - M81.8	Osteoporosis with/without current pathological fracture
M96.1	Postlaminectomy syndrome, not elsewhere classified
N23	Unspecified renal colic
N39.3 - N39.8	Urinary incontinence
N46.01 - N46.9	Male infertility
N52.01 - N52.9	Male erectile dysfunction
N53.11 – N53.9	Other male sexual dysfunction
N80.00 - N80.9, N80.A0 - N80.D9	Endometriosis
N94.4 - N94.6	Dysmenorrhea
N95.1	Menopausal and female climacteric states
N97.0 - N97.9	Female infertility
Numerous options	Injuries to musculoskeletal and connective tissues, skin and cutaneous tissues, nervous system and other and unspecified injuries, sequela
O32.1xx+	Maternal care for breech presentation
O99.210 - O99.215	Obesity complicating pregnancy, childbirth, and the puerperium
P11.3	Birth injury to facial nerve
P19.0 - P28.9	Fetal distress, birth asphyxia, respiratory distress syndrome, and other respiratory conditions of fetus and newborn
P91.60 - P91.63	Hypoxic ischemic encephalopathy [HIE]
R06.00 - R06.09	Abnormalities of breathing[cancer-related]
R07.82 - R07.89	Other chest pain
R10.83	Colic
R19.7	Diarrhea [infantile]
R32	Unspecified urinary incontinence
R37	Sexual dysfunction, unspecified
R53.81 - R53.83	Other malaise and fatigue [cancer related]
R63.0	Anorexia [pediatric]
R63.2	Polyphagia
R63.5	Abnormal weight gain
S13.4xxA-S13.4xxS, S13.8xxA - S13.8xxS, S13.9xxA - S13.9xxS, S16-1xxA - S16.1xxS	Whiplash injury
T45.1x5A - T45.1x5S	Adverse effects of antineoplastic and immunosuppressive drugs
T78.40XA - T78.40XS	Other and unspecified allergies
Z68.30 - Z68.45	Body Mass Index 30.0-70 and over, adult

Acupotomy:

CPT codes not covered for indications listed in the CPB:

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Code	Code Description

Acupotomy **–no specific code**

ICD-10 codes not covered for indications listed in the CPB:

B02.0 - B02.9	Zoster [herpes zoster]
M25.511 - M25.519	Pain in shoulder
M54.50 - M54.59	Low back pain
M65.30 - M65.359	Trigger finger
R20.2	Paresthesia of skin

Transcutaneous electronic acupoint stimulation:

CPT codes not covered for indications listed in the CPB:

Transcutaneous electronic acupoint stimulation **–no specific code**

Other CPT codes related to CPB:

56405 - 58999	Female Genital System

ICD-10 codes not covered for indications listed in the CPB:

Z48.816	Encounter for surgical aftercare following surgery on the genitourinary system [Female]

Background

Acupuncture

Acupuncture as a therapeutic intervention is widely practiced in the United States. The general theory of acupuncture is based on the premise that there are patterns of energy flow (Qi) through the body that are essential for health. Disruptions of this flow are believed to be responsible for disease. Acupuncture may correct imbalances of flow at identifiable points close to the skin. Findings from basic research have begun to elucidate the mechanisms of action of acupuncture, including the release of opioids and other peptides in the central nervous system and the periphery and changes in neuroendocrine function.

While there have been many studies of its potential usefulness, the vast majority of papers studying acupuncture in the biomedical literature consist of case reports, case series, or intervention studies. One of the difficulties with drawing conclusions from the existing literature is that the term acupuncture is used to describe a variety of treatments that differ in many important aspects according to level of effect (e.g., local, segmental, generalized) and type of acupuncture treatment (e.g., manual versus electrical acupuncture). Many of these studies provide equivocal results because of design, sample size, and other factors. The issue is further complicated by inherent difficulties in the use of appropriate controls, such as placebos and sham acupuncture groups, and by absence of studies comparing acupuncture with conventional biomedical treatments. Some factors needing investigation include frequency, number, and duration of treatments, depth

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of puncture, number of acupuncture points used, combination with other therapies, sample size, setting, blinding factors, and needle size. Be that as it may, promising results have emerged on the efficacy of acupuncture in adult post-operative and chemotherapy nausea and vomiting and in postoperative dental pain.

There is insufficient evidence of the efficacy of acupuncture as a treatment for fibromyalgia. The U.S. Department of Health and Human Services, Public Health Service, Agency for Healthcare Research and Quality (AHRQ) performed a technology assessment (2003) on *Acupuncture for the Treatment of Fibromyalgia*; it stated that "[a]t this time, therefore, there is insufficient evidence to conclude that acupuncture has efficacy for the treatment of fibromyalgia."

There is evidence to support the use of acupuncture in migraine. In a large randomized controlled study (n = 401), Vickers et al (2004) examined the effects of a policy of "use acupuncture" on headache (predominantly migraine), health status, days off sick, and use of resources in patients with chronic headache compared with a policy of "avoid acupuncture". Patients were randomly allocated to receive up to 12 acupuncture treatments over 3 months or to a control intervention offering usual care. Headache score, SF-36 health status, and use of medication were assessed at baseline, 3, and 12 months. Use of resources was assessed every 3 months. Headache score at 12 months, the primary end point, was lower in the acupuncture group (16.2, SD 13.7, n = 161, 34 % reduction from baseline) than in controls (22.3, SD 17.0, n = 140, 16 % reduction from baseline). The adjusted difference between means is 4.6 (95 % confidence interval [CI]: 2.2 to 7.0; p = 0.0002). This result is robust to sensitivity analysis incorporating imputation for missing data. Patients in the acupuncture group experienced the equivalent of 22 fewer days of headache per year (8 to 38). SF-36 data favored acupuncture, although differences reached significance only for physical role functioning, energy, and change in health. Compared with controls, patients randomized to acupuncture used 15 % less medication (p = 0.02), made 25 % fewer visits to general practitioners (p = 0.10), and took 15 % fewer days off sick (p = 0.2). The authors concluded that acupuncture leads to persisting, clinically relevant benefits for primary care patients with chronic headache, particularly migraine.

The results of the study by Vickers et al (2004) are in agreement with recent findings of Allais et al (2003) who reported that acupuncture is effective in reducing the frequency of migraine attacks as well as those by Linde et al (2009) who reported that acupuncture was more effective than a placebo injection in the early treatment of an acute migraine attack.

Facco and colleagues (2008) examined the effectiveness of a true acupuncture treatment according to traditional Chinese medicine (TCM) in migraine without aura, comparing it to a standard mock acupuncture protocol, an accurate mock acupuncture healing ritual, and untreated controls. A prospective, randomized, controlled study was performed in 160 patients suffering from migraine without aura, assessed according to the ICD-10 classification. Patients were divided into the following 4 groups:

I. group TA, treated with true acupuncture (according to TCM) plus rizatriptan;
II. group RMA, treated with ritualized mock acupuncture plus rizatriptan;
III. group SMA, treated with standard mock acupuncture plus rizatriptan; and
IV. group R, without prophylactic treatment with relief therapy only (rizatriptan).

The MIDAS Questionnaire was administered before treatment (T0), at 3 (T1) and 6 months (T2) from the beginning of treatment, and the MIDAS Index (MI) was calculated. Rizatriptan intake was also checked in all groups of patients at T0, T1, and T2. Group TA and RMA were evaluated according to TCM as well; then, the former was submitted to true acupuncture and the latter to mock acupuncture treatment resembling the same as TA. The statistical analysis was conducted with factorial ANOVA and multiple tests with a Bonferroni adjustment. A total of 127 patients completed the study (33 dropouts): 32 belonged to group TA, 30 to group RMA, 31 to group SMA, and 34 to group R. Before treatment the MI (T(0)) was moderate-to-severe with no significant inter-group differences. All groups underwent a decrease of MI at T(1) and T(2), with a significant group difference at both T(1) and T(2) compared to T(0) ($p < 0.0001$). Only TA provided a significant improvement at both T(1) and T(2) compared to R ($p < 0.0001$). Patients in the RMA group underwent a transient improvement of MI at T(1). The intake of rizatriptan paralleled the MI in all groups. The authors concluded that TA was the only treatment able to provide a steady outcome improvement in comparison to the use of only rizatriptan, while RMA showed a transient placebo effect at T1.

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There is insufficient evidence for acupuncture as a treatment for insomnia. Sok and colleagues (2003) stated that further investigation, using a randomized clinical trial design, is necessary to determine the effectiveness of acupuncture for the treatment of insomnia. Furthermore, additional work is also needed to promote the long-term therapeutic effects of acupuncture and to compare it with other therapies for insomnia.

There is limited and insufficient evidence for acupuncture for treatment of dysmenorrhea, infertility and other women's reproductive indications. White (2003) performed a review of controlled studies of acupuncture for women's reproductive health care. The author concluded that in view of the small number of studies and their variable quality, doubt remains about the effectiveness of acupuncture for gynecological conditions. Acupuncture appears promising for dysmenorrhea and infertility, and further studies are justified.

There is insufficient evidence for acupuncture to improve outcomes of in vitro fertilization. In a Cochrane review, Cheong et al (2008) determined the effectiveness of acupuncture in the outcomes of assisted reproductive treatment (ART). Randomized controlled trials (RCTs) of acupuncture for couples who were undergoing ART comparing acupuncture treatment alone or acupuncture with concurrent ART versus no treatment, placebo or sham acupuncture plus ART for the treatment of primary and secondary infertility were selected. Women with medical illness deemed contra-indications for ART or acupuncture were excluded. A total of 16 RCTs that involved acupuncture and assisted conception were identified; 13 trials were included in the review and 3 were excluded. Quality assessment and data extraction were performed independently by 2 review authors. Meta-analysis was performed using odds ratio (OR) for dichotomous outcomes. The outcome measures were live birth rate, clinical ongoing pregnancy rate, miscarriage rate, and any reported side effects of treatment. There is evidence of benefit when acupuncture is performed on the day of embryo transfer (ET) on the live birth rate (OR 1.89, 95 % CI: 1.29 to 2.77) but not when it is performed 2 to 3 days after ET (OR 1.79, 95 % CI: 0.93 to 3.44). There is no evidence of benefit on pregnancy outcomes when acupuncture is performed around the time of oocyte retrieval. The authors concluded that acupuncture performed on the day of ET shows a beneficial effect on the live birth rate; however, with the present evidence this could be attributed to placebo effect and the small number of women included in the trials. They stated that acupuncture should not be offered during the luteal phase in routine clinical practice until further evidence is available from sufficiently powered RCTs. This is in agreement with the observation of El-Toukhy et al (2008) who stated that currently available literature does not provide sufficient evidence that adjuvant acupuncture improves in vitro fertilization clinical pregnancy rate. In addition, Ng et al (2008) noted that although acupuncture has gained increasing popularity in the management of sub-fertility, its effectiveness has remained controversial.

There is some evidence to support the use of acupuncture for treatment of hip and knee osteoarthritis. An earlier AHRQ technology assessment (2003) on *Acupuncture for Osteoarthritis* concluded that "The currently available evidence is insufficient to determine whether acupuncture has a specific beneficial effect in osteoarthritis." However, a Cochrane review of acupuncture for peripheral joint osteoarthritis (Manheimer et al, 2010) concluded that sham-controlled trials show statistically significant benefits; the authors stated, however, that these benefits are small, do not meet the authors' pre-defined thresholds for clinical relevance, and are probably due at least partially to placebo effects from incomplete blinding. The authors found that waiting list-controlled trials of acupuncture for peripheral joint osteoarthritis suggest statistically significant and clinically relevant benefits, much of which may be due to expectation or placebo effects.

Acupuncture has also been employed to relieve pain and improve movement in people with osteoarthritis (OA) of the knee. In the largest clinical study of acupuncture reported to date, Berman et al (2004) studied 570 patients with an average age of 65 who had OA of the knee. Subjects were randomly assigned to receive one of three treatments for 26 weeks, in addition to standard care such as anti-inflammatory medications and pain relievers:

I. 190 received acupuncture,
II. 191 underwent sham acupuncture and
III. 189 participants attended 6, 2-hour group sessions over 12 weeks based on the Arthritis Foundation's Arthritis Self-Help Course.

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Patients' progress was assessed at 4, 8, 14, and 26 weeks. At week 8, patients receiving acupuncture began showing a significant increase in function and by week 14 a significant decrease in pain, compared with the sham and control groups. Overall those who received acupuncture had a 40 % decrease in pain and a nearly 40 % improvement in function compared to baseline assessments. The authors concluded that acupuncture seems to provide improvement in function and pain relief as an adjunctive therapy for OA of the knee when compared with credible sham acupuncture and education control groups. This finding is in agreement with the recent observations of Vas et al (2004), Tukmachi et al (2004), as well as that of Stener-Victorin et al (2004).

In a randomized, controlled, single-blind trial on the use of acupuncture as a complementary therapy to the pharmacological treatment of OA of the knee (n = 97), Vas and colleagues (2004) concluded that acupuncture plus diclofenac is more effective than placebo acupuncture plus diclofenac for the symptomatic treatment of OA of the knee. Tukmachi and associates (2004), in a randomized controlled trial (n = 30), reported that manual and electroacupuncture causes a significant improvement in the symptoms of OA of the knee, either on its own or as an adjunctive therapy, with no loss of benefit after one month.

In a randomized controlled study, Stener-Victorin et al (2004) evaluated the therapeutic effect of electro-acupuncture (EA) and hydrotherapy, both in combination with patient education or with patient education alone, in the treatment of OA in the hip (n = 45). These investigators found that EA and hydrotherapy, both in combination with patient education, induce long-lasting effects, shown by reduced pain and ache and by increased functional activity and quality of life, as demonstrated by differences in the pre- and post-treatment assessments. This finding is in agreement with that of Haslam (2001) who reported that acupuncture is more effective than advice and exercises in the symptomatic treatment of OA of the hip (n = 32) as well as that of Fink and co-workers (2001) who found that placement of acupuncture needle in the area of the affected hip is associated with improvement in the symptoms of OA (n = 67).

Guidelines from the American College of Physicians (Qaseem, et al., 2017) recommend the use of acupuncture in the management of acute and subacute low back pain. The guidelines state that, "[g]iven that most patients with acute or subacute low back pain improve over time regardless of treatment, clinicians and patients should select nonpharmacologic treatment with superficial heat (moderate-quality evidence), massage, acupuncture, or spinal manipulation (low-quality evidence). If pharmacologic treatment is desired, clinicians and patients should select nonsteroidal anti-inflammatory drugs or skeletal muscle relaxants (moderate-quality evidence). (Grade: strong recommendation). The guidelines also recommend acupuncture for chronic low back pain. "For patients with chronic low back pain, clinicians and patients should initially select nonpharmacologic treatment with exercise, multidisciplinary rehabilitation, acupuncture, mindfulness-based stress reduction (moderate-quality evidence), tai chi, yoga, motor control exercise, progressive relaxation, electromyography biofeedback, low-level laser therapy, operant therapy, cognitive behavioral therapy, or spinal manipulation (low-quality evidence). (Grade: strong recommendation).

There is evidence to support the use of acupuncture in treating chronic low back pain (LBP). In a prospective cohort study, Kukuk et al (2005) ascertained the long-term effects 3 and 6 months after the end of a course of acupuncture treatment for chronic LBP or chronic pain caused by gonarthrosis. A total of 1,096 eligible patients with chronic LBP or gonarthrosis pain were identified (68.1 % female) and invited by letter to participate in the study. Ultimately 249 patients remained, with no loss of representativeness. Two telephone interviews were conducted 3 and 6 months after the last acupuncture session using standardized questionnaires, available as electronic case report forms. The primary target criteria were self-assessment of pain tolerability before the start of acupuncture and after the end of treatment, and pain intensity (GCPS) over time. Secondary target criteria were changes to functional impairment (HFAQ for chronic LBP, WOMAC for gonarthrosis), quality of life (SF12), depression (CES-D) and patient global assessment of treatment effectiveness (PGA). For the indication chronic LBP, pain-related fear avoidance beliefs (FABQ) were also queried. These investigators found that pain tolerability was significantly improved after acupuncture and remained so up to 6 months after treatment. The mean scores of almost all questionnaires did not change significantly between 3 and 6 months. They concluded that acupuncture had a long-term effect on important aspects of cognitive and emotional pain coping.

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In a multi-center, randomized controlled trial, Thomas et al (2005) examined whether patients with persistent non-specific LBP, when offered access to traditional acupuncture care alongside conventional primary care, gained more long-term relief from pain than those offered conventional care only, for equal or less cost. Safety and acceptability of acupuncture care to patients, and the heterogeneity of outcomes were also tested. Patients in the experimental arm were offered the option of referral to the acupuncture service comprising 6 acupuncturists. The control group received usual care from their general practitioner (GP). Eligible patients were randomised in a ratio of 2:1 to the offer of acupuncture to allow between-acupuncturist effects to be tested. Patients were 18 to 65 years of age with non-specific LBP of 4 to 52 weeks' duration, and were assessed as suitable for primary care management by their general practitioner. The trial protocol allowed up to 10 individualized acupuncture treatments per patient. The acupuncturist determined the content and the number of treatments according to patient need. Main outcome measures included the Short Form 36 (SF-36) Bodily Pain dimension (range of 0 to 100 points), assessed at baseline, and 3, 12 and 24 months. Cost-utility analysis was conducted at 24 months using the EuroQOL 5 Dimensions (EQ-5D) and a preference-based single index measure derived from the SF-36 (SF-6D). Secondary outcomes included the McGill Present Pain Index (PPI), Oswestry Pain Disability Index (ODI), all other SF-36 dimensions, medication use, pain-free months in the past year, worry about back pain, satisfaction with care received, as well as safety and acceptability of acupuncture care. A total of 159 patients were in the acupuncture offer arm and 80 in the usual care arm. All 159 patients randomized to the offer of acupuncture care chose to receive acupuncture treatment, and received an average of 8 acupuncture treatments within the trial. These investigators found that traditional acupuncture care delivered in a primary care setting was safe and acceptable to patients with non-specific LBP. Acupuncture care and usual care were both associated with clinically significant improvement at 12- and 24-month follow-up. Acupuncture care was significantly more effective in reducing bodily pain than usual care at 24-month follow-up. No benefits relating to function or disability were identified. They concluded that GP referral to a service providing traditional acupuncture care offers a cost-effective intervention for reducing LBP over a 2-year period.

In a meta-analysis, Manheimer et al (2005) evaluated the effectiveness of acupuncture for treating LBP. These researchers concluded that acupuncture effectively relieves chronic LBP. However, no evidence suggests that acupuncture is more effective than other active therapies. This is in agreement with the findings of a Cochrane review on acupuncture for LBP by Furlan et al (2005) who stated that the data do not allow firm conclusions about the effectiveness of acupuncture for acute LBP. For chronic LBP, acupuncture is more effective for pain relief and functional improvement than no treatment or sham treatment immediately after treatment and in the short-term only. Acupuncture is not more effective than other conventional and alternative treatments. They concluded that the data suggest that acupuncture may be useful adjuncts to other therapies for chronic LBP.

Standaert et al (2011) sought to answer the following clinical questions:

I. Is structured exercise more effective in the treatment of chronic LBP than spinal manipulative therapy (SMT)?
II. Is structured exercise more effective in the treatment of chronic LBP than acupuncture?
III. Is SMT more effective in the treatment of chronic LBP than acupuncture?
IV. Do certain subgroups respond more favorably to specific treatments? and
V. Are any of these treatments more cost-effective than the others?

A systematic review of the literature was performed to identify RCTs comparing a structured exercise program, SMT, or acupuncture with one another in patients with chronic LBP. Two studies were identified comparing the use of structured exercise with SMT that met the inclusion criteria. Although these studies utilized different approaches for the exercise and SMT treatment groups, patients in both groups improved in terms of pain and function in both studies. Using random-effects modeling, there was no difference between the exercise and SMT groups when the data from these studies were pooled. These researchers identified no studies meeting the inclusion criteria that compared acupuncture with either structured exercise or SMT or that addressed the relative cost-effectiveness of these approaches in the treatment of patients with chronic LBP. The authors concluded that studies identified indicate that structured exercise and SMT appear to offer equivalent benefits in terms of pain and functional improvement for those with chronic LBP with clinical benefits evident within 8 weeks of care. However, the level of evidence is low. There is insufficient

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evidence to comment on the relative benefit of acupuncture compared with either structured exercise or SMT or to address the differential effects of structured exercise, SMT, or acupuncture for specific subgroups of individuals with chronic LBP. There is also insufficient evidence regarding the relative cost-effectiveness of structured exercise, SMT, or acupuncture in the treatment of chronic LBP. Structured exercise and SMT appear to offer equivalent benefits in the management of pain and function for patients with non-specific chronic LBP. If no clinical benefit is appreciated after using one of these approaches for 8 weeks, then the treatment plan should be re-evaluated and consideration should be given to modifying the treatment approach or using alternate forms of care. There is insufficient evidence regarding the relative benefits of the acupuncture compared with either structured exercise or SMT in the treatment of chronic LBP. There is insufficient evidence to address differential effects of structured exercise, SMT, or acupuncture for specific subgroups of individuals with chronic LBP. There is insufficient evidence regarding the relative cost-effectiveness of structured exercise, SMT, or acupuncture in the treatment of chronic LBP.

There is insufficient evidence that acupuncture, alone or in combination with moxibustion, may be effective in the treatment of fetal breech presentation. Moxibustion refers to a type of Chinese medicinal practice that involves burning a herb close to the skin of the acupuncture point – urinary bladder 67 (BL67, Chinese name Zhiyin), located at the tip of the 5th toe. Available guidelines have yielded conflicting recommendations on the use of moxibustion in fetal breech presentation.

Evidence based clinical guidelines from the New Zealand Guidelines Group (2004) state that "[m]oxibustion is an acupuncture technique that involves burning herbal preparations to stimulate the acupoint by the 5th toe. It may be offered to women with breech presentation". Their conclusions were based primarily on a study by Cardini and Weixin (1998), which assessed the safety and effectiveness of moxibustion on acupoint BL67 to increase fetal activity and correct breech presentation in a randomized, controlled, open clinical trial ($n = 260$). The 130 primigravidas in the 33rd week of gestation with normal pregnancy and an ultrasound diagnosis of breech presentation randomized to the intervention group received stimulation of acupoint BL 67 by moxa (Japanese term for Artemisia vulgaris) rolls for 7 days, with treatment for an additional 7 days if the fetus persisted in the breech presentation. The 130 subjects randomized to the control group received routine care but no interventions for breech presentation. Subjects with persistent breech presentation after 2 weeks of treatment could undergo external cephalic version (ECV) anytime between 35 weeks' gestation and delivery. The intervention group experienced a mean of 48.45 fetal movements versus 35.35 in the control group ($p < 0.001$). During the 35th week of gestation, 98 (75.4 %) of 130 fetuses in the intervention group were cephalic versus 62 (47.7 %) of 130 fetuses in the control group ($p < 0.001$). Despite the fact that 24 subjects in the control group and 1 subject in the intervention group underwent ECV, 98 (75.4 %) of the 130 fetuses in the intervention group were cephalic at birth versus 81 (62.3 %) of the 130 fetuses in the control group ($p = 0.02$). The authors concluded that among primigravidas with breech presentation during the 33rd week of gestation, moxibustion for 1 to 2 weeks increased fetal activity during the treatment period and cephalic presentation after the treatment period and at delivery.

Kanakura et al (2001) discussed their findings on the use of moxibustion or electrical stimulation for the treatment of breech. Only patients with breech pregnancies at the 28th week or later were entered into the study. With moxibustion treatment, the control group had a spontaneous correction rate of 165/224 (73.7 %), and the treatment group had a correction rate of 123/133 (92.5 %) ($p < 0.0001$). With low-frequency percutaneous electrical stimulation, the correction rate was 20/941 (83.9 %) in the control group and 171/191 (89.5 %) in the treatment group ($p = 0.094$). The controls in the moxibustion study did no exercises and received no external manipulation to correct breech presentation whereas those in the electrical stimulation study experienced both. Acupuncture stimulation, especially with moxibustion, is expected to serve as a safe and effective modality in the management of breech presentation in a clinical setting.

Habek et al (2003) evaluated the value of acupuncture in the conversion of fetal breech presentation into vertex presentation in a randomized prospective controlled clinical study that included 67 pregnant women with fetal breech presentation: 34 women with singleton pregnancies treated with manual acupuncture (Zhiyin) and a control group which included 33 women with singleton pregnancies without acupuncture treatment. The acupuncture treatment lasted 30 minutes a day, and was conducted during and after 34 weeks of pregnancy with simultaneous cardiotocography. The success rate of the acupuncture correction of fetal breech presentation is 76.4 % (26 women), and spontaneous conversion without acupuncture in vertex

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presentation is observed in 15 women (45.4 %; $p < 0.001$). The authors concluded that acupuncture correction of fetal malpresentation is a relatively simple, efficacious and inexpensive method associated with a lower percentage of operatively completed deliveries, which definitely reflects in improved parameters of vital and perinatal statistics.

In a controlled study by Neri et al (2004), a total of 240 women at 33 to 35 weeks of gestation carrying a fetus in breech presentation were randomized to receive active treatment (acupuncture plus moxibustion) or to be assigned to the observation group. Bilateral acupuncture plus moxibustion was applied at the BL67 acupoint. The primary outcome of the study was fetal presentation at delivery. Fourteen cases dropped out. The final analysis was thus made on 226 cases, 114 randomized to observation and 112 to acupuncture plus moxibustion. At delivery, the proportion of cephalic version was lower in the observation group (36.7 %) than in the active-treatment group (53.6 %) ($p = 0.01$). Hence, the proportion of Cesarean sections indicated for breech presentation was significantly lower in the treatment group than in the observation group (52.3 % versus 66.7 %, $p = 0.03$). The authors concluded that acupuncture plus moxibustion is more effective than observation in revolving fetuses in breech presentation. Such a method appears to be a valid option for women willing to experience a natural birth.

While the majority of evidence supports the use of acupuncture/moxibustion in correcting fetal breech presentation, recent publications are less clear in its role for the management of this condition. In a single-blind randomized controlled study, Cardini et al (2005) assessed the effectiveness of moxibustion for the correction of fetal breech presentation in a non-Chinese population. Healthy non-Chinese nulliparous pregnant women at 32 to 33 weeks + 3 days of gestational age with the fetus in breech presentation were randomly assigned to treatment or observation. Treatment consisted of moxibustion (stimulation with heat from a stick of Artemisia vulgaris) at the Zhiyin for 1 or 2 weeks. Subjects in the control group received no moxibustion but were observed. Two weeks after recruitment, each participant was subjected to an ultrasonic examination of the fetal presentation. The main outcome measure was number of participants with cephalic presentation in the 35th week. The study was interrupted when 123 participants had been recruited (46 % of the planned sample). Intermediate data monitoring revealed a high number of treatment interruptions. At this point no difference was found in cephalic presentation in the 35th week (treatment group: 22/65, 34 %; control group: 21/58, 36 %). The authors stated that the results underline the methodological problems evaluating of a traditional treatment transferred from a different cultural context. They do not support either the effectiveness or the ineffectiveness of moxibustion in correcting fetal breech presentation.

In a Cochrane review, Coyle and colleagues (2005) examined the safety and effectiveness of moxibustion on changing the presentation of an unborn baby in the breech position, the need for ECV, mode of birth, and perinatal morbidity and mortality for breech presentation. These investigators concluded that there is insufficient evidence from randomized controlled clinical trials to support the use of moxibustion to correct a breech presentation. The authors stated that moxibustion may be beneficial in reducing the need for ECV, and decreasing the use of oxytocin; however there is a need for well-designed randomised controlled trials to evaluate moxibustion for breech presentation which report on clinically relevant outcomes as well as the safety of the intervention.

Women with a 3rd trimester breech presentation often receive Cesarean section as the mode of delivery of 1st choice, especially when ECV has failed to turn the fetus to cephalic (Tiran, 2004). According to the American College of Obstetricians and Gynecologists (ACOG, 2002), ECV may not be for some women and it can pose risks including pre-term labor, placental abruption, umbilical cord entanglement, premature rupture of the membranes, as well as severe maternal discomfort. Currently, ACOG does not have a policy statement/recommendation on the use of acupuncture/moxibustion for managing fetal breech presentation.

The Royal College of Obstetricians and Gynaecologists has concluded that "moxibustion should not be recommended as a method of promoting spontaneous version over ECV." The guidelines explain that moxibustion, burnt at the tip of the 5th toe (acupuncture point BL67) has been used to promote spontaneous version of the breech, with some success, and appears to be safe. However, citing the Cochrane systematic evidence review (Coyle et al, 2005) and the study by Cardini et al (2005), RCOG concluded that pooled and recent data conclude that "there is insufficient evidence to support its use, highlighting the need for good quality studies."

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A randomized controlled study by Smith et al (2008) found acupuncture to be ineffective at inducing labor. Women who were scheduled for a post-term induction with a singleton pregnancy and cephalic presentation were eligible for the study. Subjects received 2 acupuncture or sham acupuncture sessions over a 2-day period before the planned medical/pharmacological induction. The principal primary outcomes related to the need for induction methods and time from the administration of the intervention to delivery. A total of 364 women were randomly assigned to the trial (treatment n = 181 and control n = 183). Subjects did not differ in their need for induction methods between groups: prostaglandin induction: relative risk (RR) 1.20, 95 % CI: 0.96 to 1.51, p = 0.11; artificial rupture of membranes only: RR 0.93, 95 % CI: 0.72 to 1.20, p = 0.57; oxytocin only: RR 0.89, 95 % CI: 0.60 to 1.32, p = 0.55; artificial rupture of membranes plus oxytocin: RR 0.87, 95 % CI: 0.57 to 1.33, p = 0.52; prostaglandins, artificial rupture of membranes, and oxytocin: RR 0.84, 95 % CI: 0.37 to 1.91, p = 0.68. The median time from acupuncture to delivery was 68.6 hours (interquartile range of 53.9 to 79.5) compared with 65 hours (interquartile range of 49.3 to 76.3) for women in the control group. The authors concluded that 2 sessions of manual acupuncture, using local and distal acupuncture points, administered 2 days before a scheduled induction of labor did not reduce the need for induction methods or the duration of labor for women with a post-term pregnancy.

A systematic review found no reliable evidence for the effectiveness of acupuncture in the management of xerostomia. Jedel (2005) evaluated the effectiveness of acupuncture in the management of xerostomia. Articles of controlled clinical studies assessing the effectiveness of acupuncture in the management of xerostomia were obtained by searching through the databases MEDLINE and Cochrane Central Register of Controlled Trials. Three articles met the criteria for inclusion and a criteria list was used to assess the quality of these studies. The studies were considered to be of high quality or low quality in accordance with the criteria list utilized. The results of the trials were considered positive, negative or indifferent based on statistically significant between group differences. The criteria list utilized indicate that one of the three studies was of high quality and it presents indifferent results. One of the two studies of low quality presents positive results and one presents indifferent results. An analysis of the results degree of evidence resulted in no evidence for the effectiveness of acupuncture in the management of xerostomia. The authors concluded that this systematic review showed that there is no evidence for the effectiveness of acupuncture in the management of xerostomia, and there is a need for future high quality randomized controlled trials.

A Cochrane review found insufficient evidence for acupuncture in irritable bowel syndrome. Lim et al (2006) examined if acupuncture is more effective than no treatment, more effective than "sham" (placebo) acupuncture, and as effective as other interventions used to treat irritable bowel syndrome. The authors concluded that most of the trials included in this review were of poor quality and were heterogeneous in terms of interventions, controls, and outcomes measured. Thus, it is still inconclusive if acupuncture is more effective than sham acupuncture or other interventions for treating irritable bowel syndrome.

A systematic evidence review found no clear evidence of the effectiveness of acupuncture in allergic rhinitis and asthma. Passalacqua et al (2006) noted that complementary-alternative medicines (CAM) are extensively used in the treatment of allergic rhinitis and asthma, but evidence-based recommendations are lacking. These researchers carried out a systematic review on CAM for these two indications. Meta-analyses provided no clear evidence for the effectiveness of acupuncture in rhinitis and asthma. Some positive results were described with homeopathy in good-quality trials in rhinitis, but a number of negative studies were also found. Therefore, it is not possible to provide evidence-based recommendations for homeopathy in the treatment of allergic rhinitis, and further trials are needed. A limited number of studies of herbal remedies showed some effectiveness in rhinitis and asthma, but the studies were too few to make recommendations. There are also unresolved safety concerns. The authors concluded that the effectiveness of CAM (e.g., acupuncture) for rhinitis and asthma is not supported by currently available evidence.

There is insufficient evidence of the effectiveness of acupuncture for chemotherapy-induced leukopenia and neutropenia. Lu et al (2007) stated that chemotherapy-induced leukopenia and neutropenia are common side effects during cancer treatment. Acupuncture has been reported as an adjunct therapy for this complication. These researchers reviewed randomized controlled trials of acupuncture's effect and explored the acupuncture parameters used in these trials. The study populations were cancer patients who were undergoing or had just completed chemotherapy or chemo-radiotherapy, randomized to either acupuncture therapy or usual care. The methodologic quality of trials was assessed. From 33 reviewed articles, 682

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patients from 11 eligible trials were included in analyses. All trials were published in non-PubMed journals from China. The methodologic quality of these trials was considerably poor. The median sample size of each comparison group was 45, and the median trial duration was 21 days. The frequency of acupuncture treatment was once-daily, with a median of 16 sessions in each trial. In the seven trials in which white blood cell (WBC) counts were available, acupuncture use was associated with an increase in leukocytes in patients during chemotherapy or chemo-radiotherapy, with a weighted mean difference of 1,221 WBC/muL on average (95 % CI: 636 to 1,807; $p < 0.0001$). Acupuncture for chemotherapy-induced leukopenia is an intriguing clinical question. However, the inferior quality and publication bias present in these studies may lead to a false-positive estimation. Meta-analysis based on these published trials should be treated in an exploratory nature only.

In a review on the safety and effectiveness of various interventions for the treatment of neck pain, Binder (2008) stated that compared with sham treatment, inactive treatment, or waiting list control, acupuncture may be more effective than some types of sham treatment (not further defined) or inactive treatment (not further defined) at improving pain relief at the end of treatment or in the short-term (less than 3 months), but not in the intermediate-term (not defined) or in the long-term (not defined) in people with chronic mechanical disorders. The author also noted that acupuncture may be more effective than sham TENS at improving pain at 1 week after treatment, and at 6 months, in people with chronic neck pain. Needle acupuncture may be more effective than no acupuncture at improving a composite outcome of neck pain and disability (not further defined) at 3 months in people with chronic neck pain (very low-quality evidence). Furthermore, compared with sham treatment, inactive treatment, or waiting list control, needle acupuncture may be more effective than no acupuncture at improving quality of life (measured by SF-36) at 3 months in people with chronic neck pain (very low-quality evidence).

There is no evidence of benefit of acupuncture for dyspnea palliation in cancer patients. Ben-Aharon and associates (2008) conducted a systematic review of RCTs assessing all pharmacological and non-pharmacological interventions for dyspnea palliation in cancer patients. Two reviewers independently appraised the quality of trials and extracted data. The search yielded 18 trials; 14 evaluated pharmacological interventions: 7 assessing opioids (n = 256 patients), 5 assessing oxygen (n = 137 patients), 1 assessing helium-enriched air, and 1 assessing furosemide. Four trials evaluated non-pharmacological interventions (n = 403 patients). The administration of subcutaneous morphine resulted in a significant reduction in dyspnea visual analog scale (VAS) compared with placebo. No difference was observed in dyspnea VAS score when nebulized morphine was compared with subcutaneous morphine, although patients preferred the nebulized route. The addition of benzodiazepines to morphine was significantly more effective than morphine alone, without additional adverse effects. Oxygen was not superior to air for alleviating dyspnea, except for patients with hypoxemia. Nursing-led interventions improved breathlessness. Acupuncture was not beneficial. The authors concluded that their review supports the use of opioids for dyspnea relief in cancer patients. The use of supplemental oxygen to alleviate dyspnea can be recommended only in patients with hypoxemia. Nursing-led non-pharmacological interventions seem valuable. Only a few studies addressing this question were performed. Thus, the investigators concluded, further studies evaluating interventions for alleviating dyspnea are warranted.

A systematic evidence review by Bausewein et al (2008) reached similar conclusions about the lack of adequate evidence to support the use of acupuncture for cancer-associated dyspnea.

There is a lack of reliable evidence for acupuncture treatment of Parkinson's disease. Lam and co-workers (2008) evaluated the safety and effectiveness of acupuncture therapy (monotherapy or adjuvant therapy), compared with placebo, conventional interventions, or no treatment in treating patients with idiopathic Parkinson's disease (IPD). All RCTs of any duration comparing monotherapy and adjuvant acupuncture therapy with placebo or no intervention were included. Data were abstracted independently by 2 investigators onto standardized forms, and disagreements were resolved by discussion. A total of 10 trials were included, each using a different set of acupoints and manipulation of needles. None of them reported the concealment of allocation. Only 2 studies mentioned the number of dropouts; 2 used a non-blind method while others did not mention their blinding methods. Nine studies claimed a statistically significant positive effect from acupuncture as compared with their control; only 1 indicated that there were no statistically significant differences for all variables measured. Only 2 studies described details about adverse events. The

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authors concluded that there is evidence indicating the potential effectiveness of acupuncture for treating IPD. However, results were limited by the methodological flaws, unknowns in concealment of allocation, number of dropouts, and blinding methods in the studies. They stated that large, well-designed, placebo-controlled RCTs with rigorous methods of randomization and adequately concealed allocation, as well as intention-to-treat data analysis are needed to ascertain the clinical value of acupuncture in the treatment of IPD.

There is insufficient evidence for the use of acupuncture in polycystic ovary syndrome. Stener-Victorin and colleagues (2008) described the etiology and pathogenesis of polycystic ovary syndrome (PCOS) and evaluated the use of acupuncture to prevent and reduce symptoms related with PCOS. This syndrome is the most common female endocrine disorder and it is strongly associated with hyper-androgenism, ovulatory dysfunction and obesity. It increases the risk for metabolic disturbances such as hyper-insulinemia and insulin resistance, which can lead to type 2 diabetes, hypertension and an increased likelihood of developing cardiovascular risk factors and impaired mental health later in life. Despite extensive research, little is known about the etiology of PCOS. The syndrome is associated with peripheral and central factors that influence sympathetic nerve activity. Therefore, the sympathetic nervous system may be an important factor in the development and maintenance of PCOS. Many women with PCOS require prolonged treatment. Current pharmacological approaches are effective but have adverse effects. Thus, non-pharmacological treatment strategies need to be evaluated. Acupuncture may affect PCOS via modulation of endogenous regulatory systems, including the sympathetic nervous system, the endocrine and the neuroendocrine system. Experimental observations in rat models of steroid-induced polycystic ovaries and clinical data from studies in women with PCOS suggested that acupuncture exert long-lasting beneficial effects on metabolic and endocrine systems and ovulation.

In a randomized controlled trial, Stener-Victorin and associates (2009) examined the effect of low-frequency electroacupuncture (EA) and physical exercise on sympathetic nerve activity in women with PCOS. A total of 20 women with PCOS were randomly allocated to one of 3 groups:

I. low-frequency EA (n = 9),
II. physical exercise (n = 5), or
III. untreated control (n = 6) during a 16-wk study period.

Direct recordings of multi-unit efferent post-ganglionic muscle sympathetic nerve activity (MSNA) in a muscle fascicle of the peroneal nerve before and following 16 wks of treatment were carried out. Biometric, hemodynamic, endocrine, and metabolic parameters were measured. Low-frequency EA ($p = 0.036$) and physical exercise ($p = 0.030$) decreased MSNA burst frequency compared with the untreated control group. The low-frequency EA group reduced sagittal diameter ($p = 0.001$), while the physical exercise group reduced body weight ($p = 0.004$) and body mass index ($p = 0.004$) compared with the untreated control group. Sagittal diameter was related to MSNA burst frequency ($Rs = 0.58$, $p < 0.005$) in the EA group. No correlation was found for body mass index and MSNA in the exercise group. There were no differences between the groups in hemodynamic, endocrine, and metabolic variables. For the first time, these researchers showed that low-frequency EA and physical exercise lowers high sympathetic nerve activity in women with PCOS. Thus, treatment with low-frequency EA or physical exercise with the aim to reduce MSNA may be of importance for women with PCOS.

There is insufficient evidence of the effectiveness of acupuncture for toxic neuropathy. Zhou et al (2009) noted that thalidomide and bortezomib are effective in the treatment of multiple myeloma. Unfortunately, their use can cause sensory neuropathy that frequently limits dose and duration of treatment. Although the relationship between peripheral neuropathy and therapeutic dose is controversial, many researchers have demonstrated a positive correlation between neuropathy and cumulative dose, dose intensity, and length of therapy. Peripheral neuropathic pain is the most troublesome symptom of neuropathy. Spontaneous pain, allodynia, hyperalgesia, and hyperpathia are often associated with decreased physical activity, increased fatigue, mood, and sleep problems. Symptoms are often difficult to manage, and available treatment options rarely provide total relief. Moreover, the adverse effects of these treatments often limit their use. Several studies have reported the efficacy of acupuncture, with fewer adverse effects than analgesic drugs, in the treatment of painful diabetic and human immunodeficiency virus-related neuropathy. However, the

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effectiveness of acupuncture in treating toxic neuropathy has not been assessed. Although its putative mechanisms remain elusive, acupuncture has strong potential as an adjunctive therapy in thalidomide- or bortezomib-induced painful neuropathy, and a better understanding might guide its use in the management of chemotherapy-induced neuropathic pain. The authors concluded that well-designed clinical trials with adequate sample size and power are warranted.

There is no reliable evidence for the use of acupuncture as a treatment for erectile dysfunction. Lee and colleagues (2009) evaluated the current evidence for the use of acupuncture to treat erectile dysfunction (ED). Systematic searches were conducted in 15 electronic databases, with no language restrictions. Hand-searches included conference proceedings and our files. All clinical studies of acupuncture as a treatment for ED were considered for inclusion, and their methodological quality was assessed using the Jadad score. Of the 4 studies included, 1 randomized controlled trial (RCT) showed beneficial effects of acupuncture compared with sham acupuncture in terms of response rate, while another RCT found no effects of acupuncture. The remaining 2 studies were uncontrolled clinical trials. Collectively these data showed that RCTs of acupuncture for ED are feasible but scarce. Most investigations had methodological flaws (e.g., inadequate study design, poor reporting of results, small sample size, and publication without appropriate peer review process). The authors concluded that the evidence is insufficient to suggest that acupuncture is an effective intervention for treating ED. They stated that further research is needed to investigate if there are specific benefits of acupuncture for men with ED.

A Cochrane review found insufficient evidence of the effectiveness of acupuncture in Bell's palsy. Chen et al (2010) examined the effectiveness of acupuncture in hastening recovery and reducing long-term morbidity from Bell's palsy. These investigators updated the searches of the Cochrane Neuromuscular Disease Group Trials Specialized Register (May 24, 2010), the Cochrane Central Register of Controlled Trials (CENTRAL) (issue 2, 2010), MEDLINE (January 1966 to May 2010), EMBASE (January 1980 to May 2010), AMED (January 1985 to May 2010), LILACS (from January 1982 to May 2010) and the Chinese Biomedical Retrieval System (January 1978 to May 2010) for RCTs using "Bell's palsy" and its synonyms, "idiopathic facial paralysis" or "facial palsy" as well as search terms including "acupuncture". Chinese journals in which the researchers thought they might find RCTs relevant to their study were hand-searched. These investigators reviewed the bibliographies of the randomized trials and contacted the authors and known experts in the field to identify additional published or unpublished data. They included all RCTs involving acupuncture by needle insertion in the treatment of Bell's palsy irrespective of any language restrictions. Two review authors identified potential articles from the literature search, extracted data and assessed quality of each trial independently. All disagreements were resolved by discussion between the review authors. The literature search and hand-searching identified 49 potentially relevant articles. Of these, 6 RCTs were included involving 537 participants with Bell's palsy. Two more possible trials were identified in the update than the previous version of this systematic review, but both were excluded because they were not real RCTs. Of the 6 included trials, 5 used acupuncture while the other 1 used acupuncture combined with drugs. No trial reported on the outcomes specified for this review. Harmful side effects were not reported in any of the trials. Poor quality caused by flaws in study design or reporting (including uncertain method of randomization, allocation concealment and blinding) and clinical differences between trials prevented reliable conclusions about the effectiveness of acupuncture. The authors concluded that the quality of the included trials was inadequate to allow any conclusion about the effectiveness of acupuncture. They stated that more research with high quality trials is needed.

There is insufficient evidence of the effectiveness of acupuncture for respiratory symptoms. Gibson and colleagues (2010) noted that anecdotal evidence from both clinicians and patients suggests there may be some beneficial effect of acupuncture in the treatment of respiratory symptoms, such as bronchospasm, breathlessness and hyper-ventilation syndromes. Some respiratory clinicians are introducing acupuncture as a treatment modality for the management of respiratory symptoms, despite the lack of available objective evidence to support this practice. The authors reviewed the available evidence on the use of acupuncture in respiratory disorders and discussed the methodological issues that are evident within this literature. In addition, they highlighted reasons for the lack of objective evidence to support acupuncture for respiratory conditions and the difficulties faced by acupuncture researchers when designing randomized, placebo-controlled trials. The authors concluded that presently, there is insufficient evidence to support a recommendation on the use of acupuncture in respiratory disorders.

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There is insufficient evidence of the effectiveness of acupuncture for treatment of uterine fibroids. Zhang et al (2010) evaluated the benefits and harms of acupuncture in women with uterine fibroids. All RCTs comparing acupuncture management with placebo acupuncture, no management, Chinese medication, Western medication or other managements of uterine fibroids were considered for inclusion. Acupuncture management included either traditional acupuncture or contemporary acupuncture, regardless of the source of stimulation (e.g., body, electro, scalp, elongated, fire, hand, fine needle, moxibustion). Acupuncture management without needling was excluded. Two review authors assessed trial risk of bias according to their a priori criteria. No trials were included in this version of the review, therefore no data was collected. No randomized double-blind controlled trials met the inclusion criteria. The authors concluded that the effectiveness of acupuncture for the management of uterine fibroids remains uncertain. They stated that more evidence is needed to establish the safety and effectiveness of acupuncture for uterine fibroids. There is a continued need for well-designed RCTs with long-term follow-up.

In a randomized, patient-assessor blinded, sham-acupuncture, controlled trial, Shin et al (2010) assessed the safety and effectiveness of acupuncture for ocular symptoms, tear film stability and tear secretion in dry eye patients. A total of 42 subjects with defined moderate to severe dry eye underwent acupuncture treatment 3 times a week for 3 weeks. Seventeen standard points (GV23; bilateral BL2, GB14, TE23, Ex1, ST1 and GB20; and unilateral SP3, LU9, LU10 and HT8 on the left for men and right for women) with "de qi" manipulation for the verum acupuncture group and seventeen sham points of shallow penetration without other manipulation for the sham group were applied during the acupuncture treatment. Differences were measured using the ocular surface disease index (OSDI), the VAS of ocular discomfort, the tear film break-up time (TFBUT) and the Schimer I test with anesthesia. In addition, adverse events were recorded. There were no statistically significant differences between results on the OSDI, VAS, TFBUT or Schimer I tests from baseline between the verum and sham acupuncture groups. However, results from the within-group analysis showed that the OSDI and VAS in both groups and the TFBUT in the verum acupuncture group were significantly improved after 3 weeks of treatment. No adverse events were reported during this trial. The authors concluded that both types of acupuncture improved signs and symptoms in dry-eye patients after a 4-week treatment. However, verum acupuncture did not result in better outcomes than sham acupuncture.

Lee and colleagues (2011) evaluated the effectiveness of acupuncture as a treatment option for treating the condition of dry eye. These investigators searched the literature using 14 databases from their inceptions to December 3, 2009, without language restrictions. They included RCTs comparing acupuncture with conventional treatment. Their risk of bias was assessed using Cochrane criteria. A total of 6 RCTs met all the inclusion criteria. Three RCTs compared the effects of acupuncture with artificial tears in patients with xerophthalmia or Sjögren syndrome. A meta-analysis of these data showed that acupuncture improved tear break-up times ($p < 0.0001$), Schirmer test scores ($p < 0.00001$), response rates ($p = 0.002$) and the region of cornea fluorescent staining ($p = 0.0001$) significantly more than artificial tears did. The other 3 RCTs compared the effects of acupuncture plus artificial tears with artificial tears alone – 2 of these studies failed to show significant effects of acupuncture, while 1 reported significant effects. For Schirmer test scores and frequency of artificial tear usage, 2 RCTs reported superior effects of acupuncture plus artificial tears, while 1 RCT failed to do so. The authors concluded that these findings provide limited evidence for the effectiveness of acupuncture for treating dry eye. However, the total number of RCTs, the total sample size and the methodological quality were too low to draw firm conclusions.

In a prospective, randomized, controlled, cross-over trial, Lam et al (2011) evaluated the safety and adjunctive effect of acupuncture added to refractive correction for anisometropic amblyopia in younger children. A total of 83 children aged 3 to less than 7 years with untreated anisometropic amblyopia and baseline best-corrected visual acuity (BCVA) of 20/40 to 20/200 in the amblyopic eye were included in this study. Participants were randomized to receive spectacles alone (group 1; $n = 42$) or spectacles + acupuncture (group 2; $n = 41$) for 15 weeks, and were then crossed-over to receive the other regimen for another 15 weeks. The BCVA in both eyes was measured at baseline and every 5 (+/- 1) weeks for the initial 45 weeks and at 60 (+/- 1) weeks. Main outcome measures were BCVA in the amblyopic eye at 15, 30, and 60 weeks. The mean baseline BCVA in the amblyopic eye was 0.50 and 0.49 logarithm of the minimum angle of resolution (logMAR) in groups 1 and 2, respectively. After 15 weeks of treatment, the BCVA had improved by a mean of 2.2 lines in group 1 and 2.9 lines in group 2. The mean difference in BCVA between groups was 0.77 lines (95 % CI: 0.29 to 1.3; $p = 0.0020$) with baseline adjustment. BCVA of less than or

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equal to 0.1 logMAR was achieved in 14.6 % of the patients in group 1 and 57.5 % in group 2 ($p < 0.00010$). After the regimens were crossed-over at 30 weeks, group 1 had a mean of 1.2 (95 % CI: 0.98 to 1.48) lines additional improvement from the 15-week BCVA, whereas in group 2 the mean improvement was 0.4 (95 % CI: 0.19 to 0.63) lines. The proportions of responders, resolution, and participants achieving a BCVA of less than or equal to 0.1 logMAR at 30 weeks were similar between groups. After completion of acupuncture, only 1 participant had greater than 1 line of VA decrease to 60 weeks. Acupuncture was well-tolerated by all children, and no severe adverse effect was encountered. The authors concluded that acupuncture is a potentially useful complementary treatment modality that may provide sustainable adjunctive effect to refractive correction for anisometropic amblyopia in young children. They stated that acupuncture has good potential to become a complimentary therapeutic modality for amblyopia, and further large-scale studies seem warranted.

In a Cochrane review, Cheuk et al (2011) examined the effectiveness of acupuncture for people with autism spectrum disorders (ASD) in improving core autistic features, as well as communication, cognition, overall functioning and quality of life, and established if it has any adverse effects. These investigators searched the following databases on September 30, 2010: CENTRAL (The Cochrane Library, 2010, Issue 3), MEDLINE (1950 to September 2010 Week 2), EMBASE (1980 to 2010 Week 38), PsycINFO, CINAHL, China Journal Full-text Database, China Master Theses Full-text Database, China Doctor Dissertation Full-text Database, China Proceedings of Conference Database, Index to Taiwan Periodical Literature System, metaRegister of Controlled Trials and the Chinese Clinical Trials Registry. They also searched AMED (February 26, 2009) and Dissertation Abstracts International (March 3, 2009), but these were no longer available to the authors or editorial base at the date of the most recent search. TCMLARS (Traditional Chinese Medical Literature Analysis and Retrieval System) was last searched on March 3, 2009. These researchers included RCTs and quasi-RCTs. They included studies comparing an acupuncture group with at least one control group that used no treatment, placebo or sham acupuncture treatment in people with ASD. They excluded trials that compared different forms of acupuncture or compared acupuncture with another treatment. Two review authors independently extracted trial data and assessed the risk of bias in the trials. They used relative risk (RR) for dichotomous data and mean difference (MD) for continuous data. The authors included 10 trials that involved 390 children with ASD. The age range was 3 to 18 years and the treatment duration ranged from 4 weeks to 9 months. The studies were carried out in Hong Kong, mainland China and Egypt. Two trials compared needle acupuncture with sham acupuncture and found no difference in the primary outcome of core autistic features (RFRLRS total score: MD 0.09; 95 % CI: -0.03 to 0.21, $p = 0.16$), although results suggested needle acupuncture might be associated with improvement in some aspects of the secondary outcomes of communication and linguistic ability, cognitive function and global functioning. Six trials compared needle acupuncture plus conventional treatment with conventional treatment alone. The trials used different primary outcome measures and most could not demonstrate effectiveness of acupuncture in improving core autistic features in general, though 1 trial reported patients in the acupuncture group were more likely to have improvement on the Autism Behavior Checklist (RR 1.53; 95 % CI: 1.09 to 2.16, $p = 0.02$) and had slightly better post-treatment total scores (MD -5.53; 95 % CI: -10.76 to -0.31, $p = 0.04$). There was no evidence that acupuncture was effective for the secondary outcome of communication and linguistic ability, though there seemed to be some benefit for the secondary outcomes of cognitive function and global functioning. Two trials compared acupressure plus conventional treatment with conventional treatment alone and did not report on the primary outcome. Individual study results suggested there may be some benefit from acupressure for certain aspects of the secondary outcomes of communication and linguistic ability, cognitive function and global functioning. Four trials reported some adverse effects, though there was little quantitative information, and at times both intervention and control groups experienced them. Adverse effects included bleeding, crying due to fear or pain, irritability, sleep disturbance and increased hyperactivity. None of the trials reported on quality of life. There are a number of problems with the evidence base: the trials were few in number and included only children; 6 of the trials were at high-risk of bias; they were heterogeneous in terms of participants and intervention; they were of short duration and follow-up; they reported inconsistent and imprecise results, and, due to carrying out large numbers of analyses, they were at risk of false positivity. The authors concluded that current evidence does not support the use of acupuncture for treatment of ASD. There is no conclusive evidence that acupuncture is effective for treatment of ASD in children and no RCTs have been carried out with adults. They stated that further high quality trials of larger size and longer follow-up are needed.

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In a Cochrane review, Wei et al (2011) evaluated the safety and effectiveness of acupuncture in slowing the progression of myopia in children and adolescents. These investigators searched CENTRAL (which contains the Cochrane Eyes and Vision Group Trials Register) (The Cochrane Library 2011, Issue 7), MEDLINE (January 1950 to July 2011), EMBASE (January 1980 to July 2011), the Allied and Complementary Medicine Database (AMED) (January 1985 to July 2011), Latin American and Caribbean Health Sciences Literature Database (LILACS) (January 1982 to July 2011), the metaRegister of Controlled Trials (mRCT), ClinicalTrials.gov, the National Center for Complementary and Alternative Medicine (NCCAM) (The first issue to August 2010), the Chinese Biological Medicine Database (CBM) (1978 to April 2011), China National Knowledge Infrastructure (CNKI) (1994 to April 2011) and VIP (1989 to April 2011). There were no date or language restrictions in the electronic searches for trials. CENTRAL, MEDLINE, EMBASE, AMED, LILACS, mRCT and ClinicalTrials.gov were last searched on 9 July 2011. NCCAM was searched up to August 2010 and CBM, CNKI, and VIP were last searched on April 6, 2011. These researchers included RCTs that included any type of acupuncture treatment for myopia in children and adolescents. Two authors independently evaluated the search results according to the inclusion and exclusion criteria. Two authors extracted and assessed data independently. They contacted the study investigator for missing data. The authors included 2 RCTs conducted in Taiwan with a total of 131 participants. They did not perform a meta-analysis as the trials were assessing different outcomes. Neither trial met the pre-defined primary outcome criteria of myopia progression defined as 1 diopter mean change. Only 1 trial reported the changes of axial length without non-significant difference among groups and both trials reported that several children experienced mild pain during acupuncture stimulation. Two trials were included in this review but no conclusions can be drawn for the benefit of co-acupressure for slowing progress of myopia in children. The authors concluded that further evidence in the form of RCTs are needed before any recommendations can be made for the use of acupuncture treatment in clinical use. These trials should compare acupuncture to placebo and have large sample sizes. Other types of acupuncture (such as auricular acupuncture) should be explored further as well as compliance with treatment for at least 6 months or longer. Axial length elongation of the eye should be investigated for at least 1 year. The potential to reduce/eliminate pain from acupuncture experienced by children should also be reviewed.

The Canadian Thoracic Society's clinical practice guideline on "Managing dyspnea in patients with advanced chronic obstructive pulmonary disease" (Marciniuk et al, 2011) noted that dyspnea is a cardinal symptom of chronic obstructive pulmonary disease (COPD), and its severity and magnitude increases as the disease progresses, leading to significant disability and a negative effect on quality of life. Refractory dyspnea is a common and difficult symptom to treat in patients with advanced COPD. There are many questions concerning optimal management and, specifically, whether various therapies are effective in this setting. These investigators addressed these important clinical issues using an evidence-based systematic review process led by a representative inter-professional panel of experts. The evidence supported the benefits of oral opioids, neuromuscular electrical stimulation, chest wall vibration, walking aids and pursed-lip breathing in the management of dyspnea in the individual patient with advanced COPD. Oxygen is recommended for COPD patients with resting hypoxemia, but its use for the targeted management of dyspnea in this setting should be reserved for patients who receive symptomatic benefit. There is insufficient evidence to support the routine use of anxiolytic medications, nebulized opioids, acupuncture, acupressure, distractive auditory stimuli (music), relaxation, hand-held fans, counseling programs or psychotherapy. There is also no evidence to support the use of supplemental oxygen to reduce dyspnea in non-hypoxemic patients with advanced COPD.

Williams et al (2012) stated that acne is a chronic inflammatory disease of the pilo-sebaceous unit resulting from androgen-induced increased sebum production, altered keratinization, inflammation, and bacterial colonization of hair follicles on the face, neck, chest, and back by Propionibacterium acnes. Although early colonization with P acnes and family history might have important roles in the disease, exactly what triggers acne and how treatment affects the course of the disease remain unclear. Other factors such as diet have been implicated, but not proven. Facial scarring due to acne affects up to 20 % of teenagers. Acne can persist into adulthood, with detrimental effects on self-esteem. There is no ideal treatment for acne, although a suitable regimen for reducing lesions can be found for most patients. Good quality evidence on comparative effectiveness of common topical and systemic acne therapies is scarce. Topical therapies including benzoyl peroxide, retinoids, and antibiotics when used in combination usually improve control of mild to moderate acne. Treatment with combined oral contraceptives can help women with acne. Patients with more severe

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inflammatory acne usually need oral antibiotics combined with topical benzoyl peroxide to decrease antibiotic-resistant organisms. Oral isotretinoin is the most effective therapy and is used early in severe disease, although its use is limited by teratogenicity and other side-effects. Availability, adverse effects, and cost, limit the use of photodynamic therapy. New research is needed into the therapeutic comparative effectiveness and safety of the many products available, and to better understand the natural history, subtypes, and triggers of acne. Moreover, the authors stated that complementary and alternative medicine (including acupuncture) can not be recommended for the treatment of acne because it is not supported by good evidence.

Yan et al (2012) noted that burning mouth syndrome (BMS) is a common chronic pain condition that lacks a satisfactory treatment approach. These researchers examined the effects of acupuncture or acupoint injection on the management of BMS and evaluated the evidence supporting the use of acupuncture therapy for BMS in clinical practice. The following databases were searched for relevant articles: Cochrane Oral Health Group Trials Register (July 2011), Cochrane Central Register of Controlled Trials (issue 7, 2011), MEDLINE (1966 to June 2011), and electronic medical database from the China-National Knowledge Infrastructure (1979 to June 2011). Articles were screened, and the quality of the included trials was assessed independently by 2 reviewers. After screening, 9 studies with 547 randomized patients were included in this review. All 9 articles were published in Chinese and were clinical trial studies with a Jadad score of less than 3. Their results showed that acupuncture/acupoint injection may benefit patients with BMS. The evidence supported the efficacy of acupuncture/acupoint injection therapy in reducing BMS pain and related symptoms. The authors concluded that in light of the positive outcomes reported, the use of acupuncture therapy for BMS patients warrants further research.

Bo and colleagues (2012) evaluated the reports' qualities which are about RCTs of acupuncture treatment on diabetic peripheral neuropathy (DPN). A total of 8 databases including The Cochrane Library(1993 to Sept.,2011), PubMed (1980 to Sept., 2011), EMbase (1980 to Sept.,2011), SCI Expanded (1998 to Sept.,2011), China Biomedicine Database Disc (CBMdisc, 1978 to Sept., 2011), China National Knowledge Infrastructure (CNKI, 1979 to Sept., 2011), VIP (a full text issues database of China, 1989 to Sept., 2011), Wan Fang (another full text issues database of China 1998 to Sept., 2011) were searched systematically. Hand-search for further references was conducted. Language was limited to Chinese and English. These investigators identified 75 RCTs that used acupuncture as an intervention and assessed the quality of these reports with the Consolidated Standards for Reporting of Trials statement 2010 (CONSORT2010) and Standards for Reporting Interventions Controlled Trials of Acupuncture 2010(STRICTA2010). A total of 24 articles (32 %) applied the method of random allocation of sequences. No article gave the description of the mechanism of allocation concealment, no experiment applied the method of blinding. Only 1 article (1.47 %) could be identified directly from its title as about the RCTs, and only 4 articles gave description of the experimental design. No article mentioned the number of cases lost or eliminated. During 1 experiment, acupuncture syncope led to temporal interruption of the therapy. Two articles (2.94 %) recorded the number of needles, and 8 articles (11.76 %) mentioned the depth of needle insertion. None of articles reported the base of calculation of sample size, or has any analysis about the metaphase of an experiment or an explanation of its interruption. One (1.47 %) mentioned intentional analysis (ITT). The authors concluded that the quality of the reports on RCTs of acupuncture for diabetic peripheral neuropathy is moderate to low. They stated that the CONSORT2010 and STRICTA2010 should be used to standardize the reporting of RCTs of acupuncture in future.

In a meta-analysis, Wang et al (2012) evaluated the effectiveness of acupuncture in facial spasm. The research team categorized results from each of the reviewed studies in 2 ways:

I. the number of participants who showed a positive response to therapy (total effectiveness rate), and
II. the number of participants who made a full recovery (clinical cure rate).

The research team reviewed a total of 13 studies involving 1,262 participants with facial spasm. Researchers in China had conducted all studies, and most studies were poor in methodological quality. All studies reported that acupuncture was superior to other treatments, including carbamazepine, mecobalamin, and massage, and the meta-analysis on these low-quality studies yielded similar results. The authors concluded that present trials evaluating the effectiveness of acupuncture in treatment of facial spasm are mostly poor in

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methodological quality. These studies showed that acupuncture was superior to other treatments for facial spasm; however, in its meta-analysis, the research team could not draw an affirmative conclusion as to the benefits of acupuncture due to the poor methodological quality and localized population of the included trials. The authors concluded that the field needs large international, well-conducted RCTs.

In a Cochrane review, He and colleagues (2012) evaluated the safety and effectiveness of acupuncture for children with mumps. These investigators searched CENTRAL (2012, Issue 4), MEDLINE (1950 to April week 4, 2012), EMBASE (1974 to May 2012), CINAHL (1981 to May 2012), AMED (1985 to May 2012), the Chinese BioMedicine Database (CBM) (1979 to May 2012), China National Knowledge Infrastructure (CNKI) (1979 to May 2012), Chinese Technology Periodical Database (CTPD) (1989 to May 2012) and WANFANG database (1982 to May 2012). They also hand-searched a number of journals (from first issue to current issue). These researchers included RCTs comparing acupuncture with placebo acupuncture, no management, Chinese medication, Western medication or other treatments for mumps. Acupuncture included either traditional acupuncture or contemporary acupuncture, regardless of the source of stimulation (body, electro, scalp, fire, hand, fine needle, moxibustion). Two review authors independently extracted data and assessed the quality of included studies. They calculated risk ratios (RR) with their 95 % CI for the effective percentage and standardized mean differences (SMD) with 95 % CIs for the time to cure. Only 1 study with 239 participants met the inclusion criteria. There were a total of 120 participants in the acupuncture group, of which 106 recovered, with their temperature returning back to normal and no swelling or pain of the parotid gland; the condition of 14 participants improved, with a drop in temperature and alleviation of swelling or pain of the parotid gland. There were 119 participants in the Western medicine group, of which 56 recovered and the condition of 63 improved. The acupuncture group had a higher recovery rate than the control group. The relative RR of recovery was 1.88 (95 % CI: 1.53 to 2.30). However, the acupuncture group had a longer time to cure than the control group. The mean was 4.20 days and the standard deviation (SD) was 0.46 in the acupuncture group, while in the control group the mean was 3.78 days and the SD was 0.46. There was a potential risk of bias in the study because of low methodological quality. The authors concluded that they could not reach any confident conclusions about the safety and effectiveness of acupuncture based on 1 study. They stated that more high-quality research is needed.

The American Academy of Otolaryngology-Head and Neck Surgery Foundation (AAO-HNSF)'s clinical practice guideline on "Bell's palsy" (Baugh et al, 2013) stated that "no recommendation can be made regarding the effect of acupuncture in patients with Bell's palsy".

Zhang et al (2014) systematically examined published reviews and meta-analyses in order to determine if and when acupuncture is an effective treatment for stroke and stroke-related disorders. These investigators also hoped to identify the best directions for future research in this area. Systematic reviews and meta-analyses of RCTs and quasi-RCTs evaluating the effectiveness of acupuncture to treat stroke or stroke-related conditions were included. Electronic searches were conducted in the Cochrane Database of Systematic Reviews, Ovid MEDLINE, CINAHL, Ovid EMBASE, EBSCO Allied and Complementary Medicine (AMED) database, Chinese Biological Medicine Database, and Chinese National Knowledge Infrastructure Database. Two authors independently assessed the compliance of studies with eligibility criteria, and extracted data from included studies. The quality of systematic reviews was assessed according to the Overview Quality Assessment Questionnaire. A total of 24 systematic reviews were included, of which 4 (16.7 %) were Cochrane systematic reviews and 20 (83.3 %) were non-Cochrane reviews. Acupuncture was analyzed as an acute stroke intervention in 3 reviews (12.5 %), as an approach to stroke rehabilitation in 6 (25 %), and as an intervention to treat various stroke-related disorders in the remaining 15 (62.5 %). Reviews analyzing death or dependency/disability as the primary outcome reported no statistically significant difference between acupuncture and non-acupuncture control treatments. In contrast, reviews in which the outcome was improvement in global neurological deficit scores or performance on the video-fluoroscopic swallowing study test or water-swallowing test often reported that acupuncture was superior to control treatment. The quality of 10 reviews was "poor", 6 reviews were "moderate" and 8 were "good". The authors concluded that the available evidence suggests that acupuncture may be effective for treating post-stroke neurological impairment and dysfunction such as dysphagia, although these reported benefits should be verified in large, well-controlled studies. On the other hand, the available evidence does not clearly indicate that acupuncture

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can help prevent post-stroke death or disability, or ameliorate other aspects of stroke recovery, such as post-stroke motor dysfunction. These findings suggest that researchers should focus on the potential application of acupuncture to treat post-stroke neurological impairment and dysfunction and on the development of more precise tools to assess these improvements after stroke.

Zeng and associates (2014) evaluated the effectiveness of acupuncture for cancer-related fatigue (CRF). Five databases (Medline, CINAHL, Scopus, the Cochrane Library, and CAJ Full-text Database) were searched up to May 2013. Randomized controlled trials of acupuncture for the treatment of CRF were considered for inclusion. A total of 7 RCTs were included for meta-analysis, involving a total of 689 subjects. Three studies compared acupuncture with sham acupuncture for CRF with follow-up at 10 weeks; the standardized mean difference (SMD) for general CRF change values was -0.82 (95 % CI: -1.90 to 0.26). When acupuncture plus education intervention was compared with usual care, there was a statistically significant difference for the change score of general CRF (SMD = -2.12; 95 % CI: -3.21 to -1.03). The SMD for general CRF change scores between acupuncture with no treatment or wait-list control was -1.46 (95 % CI: -3.56 to 0.63). Finally, the SMD for general CRF change scores between acupuncture with acupressure or self-acupuncture was -1.12 (95 % CI: -3.03 to 0.78). Three trials reported data for general quality of life and functioning status, reporting enough data for statistical pooling but showing no statistically significant difference (: score = 1.15, : = 0.25, SMD = 0.99, 95 % CI: -0.70 to 2.68 and : score = 1.13, : = 0.26, SMD = 1.38, 95 % CI: -1.02 to 3.79, respectively). The : 2 statistics of all statistically pooled data were higher than 50 %, indicating heterogeneity between the trials. The authors concluded that there were 4 sets of comparison for the effectiveness of acupuncture for CRF; statistical pooling of the reduction in CRF from baseline to follow-up showed in favor of acupuncture. However, 3 sets of comparison for the pooled estimates of effect sizes had no statistical significance. Although one set of comparison (acupuncture plus education interventions versus usual care) had statistically significant differences, it is unclear whether this pooled positive outcome is attributable to the effects of acupuncture or to the education intervention. In addition, the duration of follow-up in these included trials was up to 10 weeks, and some RCTs had methodological flaws. They stated that further rigorously designed RCTs adhering to acceptable standards of trial methodology are needed to determine the effectiveness of acupuncture and its long-term effects on CRF.

Cao and colleagues (2013) evaluated the safety and effectiveness of acupuncture for patients with vascular mild cognitive impairment (VMCI). A total of 7 electronic databases were searched for RCTs that investigated the effects of acupuncture compared with no treatment, placebo or conventional therapies on cognitive function or other clinical outcomes in patients with VMCI. The quality of the trials selected was evaluated according to the "risk of bias" assessment provided by the Cochrane Handbook for Systematic Reviews of Interventions. RevMan V.5.1 software was employed for data analysis. A total of 12 trials with 691 participants were included. The methodological quality of all included trials was unclear and/or they had a high-risk of bias. Meta-analysis showed acupuncture in conjunction with other therapies could significantly improve Mini-Mental State Examination scores (mean difference 1.99, 95 % CI: 1.09 to 2.88, random model, $p < 0.0001$, 6 trials). No included trials mentioned any adverse events of the treatment. The authors concluded that the current clinical evidence is not of sufficient quality for wider application of acupuncture to be recommended for the treatment of VMCI; they stated that further large, rigorously designed trials are warranted.

Yang et al (2013) conducted a systematic review of RCTs to evaluate the effectiveness of acupuncture for diabetic gastroparesis (DGP). These investigators searched PubMed, EMbase, Cochrane Central Register of Controlled Trials (CENTRAL) and 4 Chinese databases including China National Knowledge Infrastructure (CNKI), VIP Database for Chinese Technical Periodicals, Chinese Biomedical Literature Database (CBM) and WanFang Data up to January 2013 without language restriction. Eligible RCTs designed to examine the effectiveness of acupuncture in improving dyspeptic symptoms and gastric emptying in DGP were selected for analysis. Risk of bias, study design and outcomes were extracted from trials. Relative risk (RR) was calculated for dichotomous data. Mean difference (MD) and standardized mean difference (SMD) were selected for continuous data to pool the overall effect. These investigators searched 744 studies, among which 14 RCTs were considered eligible. Overall, acupuncture treatment had a high response rate than controls (RR, 1.20 [95 % CI: 1.12 to 1.29], $p < 0.00001$), and significantly improved dyspeptic symptoms compared with the control group. There was no difference in solid gastric emptying between acupuncture and control. Acupuncture improved single dyspeptic symptom such as nausea and vomiting, loss of appetite

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and stomach fullness. Most studies were in unclear and high-risk of bias and with small sample size (median = 62). The majority of the RCTs reported positive effect of acupuncture in improving dyspeptic symptoms. The authors concluded that the results suggested that acupuncture might be effective to improve dyspeptic symptoms in DGP, while a definite conclusion about whether acupuncture was effective for DGP could not be drawn due to the low quality of trials and possibility of publication bias. They stated that further large-scale, high-quality RCTs are needed to validate this claim and translate this result to clinical practice.

In a prospective, blinding-validated, randomized controlled multi-center trial, Skjeie et al (2013) tested the hypothesis that acupuncture treatment has a clinically relevant effect for infantile colic. Research assistants and parents were blinded. Intervention was 3 days of bilateral needling of the acupuncture point ST36, with no treatment as control. A total of 113 patients were recruited; 23 patients were excluded, and 90 randomized; 79 diaries and 84 interviews were analyzed. Main outcome measures were difference in changes in crying time during the trial period between the intervention and control group. The blinding validation questions showed a random distribution with $p = 0.41$ and 0.60, indicating true blinding. These researchers found no statistically significant difference in crying time reduction between acupuncture and control group at any of the measured intervals, nor in the main analysis of differences in changes over time ($p = 0.26$). There was a tendency in favor of the acupuncture group, with a non-significant total baseline-corrected mean of 13 minutes (95 % CI: -24 to + 51) difference in crying time between the groups. This was not considered clinically relevant, according to protocol. The authors concluded that this trial of acupuncture treatment for infantile colic showed no statistically significant or clinically relevant effect; they suggested that acupuncture for infantile colic should be restricted to clinical trials.

Zhang et al (2013) analyzed the effectiveness of acupoint application therapy for infantile diarrhea. The authors of the present paper did a literature retrieval using the China National Knowledge Infrastructure (CNKI) database, Chinese bio-medical database and Wanfang database covering the period of January 1, 1990 to June 30, 2012, and made a systemic evaluation on the retrieved RCTs of acupoint application therapy for infantile diarrhea using Cochrane system evaluation method. Following excluding the repetitive, irrelevant and non-RCTs, those meeting the standards of RCTs were collected. Trial quality was assessed using the Jadad score that evaluates the randomization process, blinding, and the description of withdrawals or drop-outs. The RevMan 5. 1 software was used to make statistical analysis. A total of 16 papers (2,151 patients) were included in the meta-analysis. The homogeneity test was better ($chi2 = 8.09$, $p = 0.92$, $I2 = 0$ %), displaying a homogenicity of most studies. Meta- analysis showed the merger effect quantity $OR = 4.68$, and 95 % CIs: 3.41 to 6.42, and the merger effect value test $Z = 9.58$, $p < 0.00001$. Statistical difference indicates a better therapeutic effect of acupoint application group than the control group, providing evidence in favor of acupoint application therapy for infantile diarrhea. Funnel chart displays that the researched object distribution is symmetric, being smaller in the bias. But the potential publication bias still possibly exists. The authors concluded that acupoint application therapy for infantile diarrhea has some advantages, which needs further confirmation due to lower quality of the collected literatures. They stated that larger sample, high quality RCTs are highly recommended.

Zhu and colleagues (2013) examined the safety and effectiveness of acupuncture for pain in endometriosis. These investigators searched the Cochrane Menstrual Disorders and Subfertility Group (MSDG) Specialized Register of controlled trials, Cochrane Central Register of Controlled Trials (CENTRAL) (The Cochrane Library), MEDLINE, EMBASE, CINAHL, AMED, PsycINFO, CNKI and TCMDS (from inception to 2010) and reference lists of retrieved articles. Randomized single or double-blind controlled trials enrolling women of reproductive age with a laparoscopically confirmed diagnosis of endometriosis and comparing acupuncture (body, scalp or auricular) to either placebo or sham, no treatment, conventional therapies or Chinese herbal medicine were selected for analysis. Three authors independently assessed risk of bias and extracted data; they contacted study authors for additional information. Meta-analyses were not performed as only 1 study was included. The primary outcome measure was decrease in pain from endometriosis. Secondary outcome measures included improvement in quality of life scores, pregnancy rate, adverse effects and rate of endometriosis recurrence. A total of 24 studies were identified that involved acupuncture for endometriosis; however only 1 trial, enrolling 67 participants, met all the inclusion criteria. The single included trial defined pain scores and cure rates according to the Guideline for Clinical Research on New Chinese Medicine. Dysmenorrhea scores were lower in the acupuncture group (mean difference -4.81 points, 95 % CI: -6.25 to -3.37, $p < 0.00001$) using the 15-point Guideline for Clinical Research on New

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Chinese Medicine for Treatment of Pelvic Endometriosis scale. The total effective rate ("cured", "significantly effective" or "effective") for auricular acupuncture and Chinese herbal medicine was 91.9 % and 60 %, respectively (risk ratio 3.04, 95 % CI: 1.65 to 5.62, p = 0.0004). The improvement rate did not differ significantly between auricular acupuncture and Chinese herbal medicine for cases of mild-to-moderate dysmenorrhea, whereas auricular acupuncture did significantly reduce pain in cases of severe dysmenorrhea. Data were not available for secondary outcomes measures. The authors concluded that the evidence to support the effectiveness of acupuncture for pain in endometriosis is limited, based on the results of only a single study that was included in this review. This review highlighted the necessity for developing future studies that are well-designed, double-blinded, RCTs that assess various types of acupuncture in comparison to conventional therapies.

Acupoint Point Injection

Acupuncture point injection (also known as acupoint injection therapy, biopuncture) entails the injection of small amounts of saline or other substances (e.g., Chinese herbal extracts, drugs, homeopathic substances, vitamin B12, and vitamin K) into acupuncture points/trigger points by qualified acupuncturists. Similar to acupuncture, fine needles are inserted into acupuncture points/trigger points just under the skin or into muscles. However, there is insufficient evidence to support the effectiveness of this approach.

In a Cochrane review, Green et al (2002) evaluated the effectiveness of acupuncture in the treatment of adults with lateral elbow pain with respect to pain reduction, improvement in function, grip strength and adverse effects. These investigators searched MEDLINE, CINAHL, EMBASE and SCISEARCH and the Cochrane Clinical Trials Register and the Musculoskeletal Review Group's specialist trial database from 1966 to June 2001. Identified keywords and authors were searched in an effort to retrieve as many trials as possible. Two independent reviewers assessed all identified trials against pre-determined inclusion criteria. Randomized and pseudo-randomized trials in all languages were included in the review provided they were testing acupuncture compared to placebo or another intervention in adults with lateral elbow pain (tennis elbow). Outcomes of interest were pain, function, disability, quality of life, strength, participant satisfaction with treatment and adverse effect. For continuous variables means and standard deviations were extracted or imputed to allow the analysis of weighted mean difference, while for binary data numbers of events and total population were analyzed and interpreted as relative risks. Trial results were combined only in the absence of clinical and statistical heterogeneity. A total of 4 small RCTs were included but due to flaws in study designs (particularly small populations, uncertain allocation concealment and substantial loss to follow-up) and clinical differences between trials, data from trials could not be combined in a meta-analysis. One RCT found that needle acupuncture results in relief of pain for significantly longer than placebo (weighted mean difference [WMD] = 18.8 hours, 95 % CI: 10.1 to 27.5) and is more likely to result in a 50 % or greater reduction in pain after 1 treatment (RR 0.33, 95 % CI: 0.16 to 0.69) (Molsberger 1994). A second RCT demonstrated needle acupuncture to be more likely to result in overall participant reported improvement than placebo in the short term (RR = 0.09 95 % CI: 0.01 to 0.64) (Haker 1990a). No significant differences were found in the longer term (after 3 or 12 months). A RCT of laser acupuncture versus placebo demonstrated no differences between laser acupuncture and placebo with respect to overall benefit (Haker 1990b). A 4th RCT included trial published in Chinese demonstrated no difference between vitamin B12 injection plus acupuncture, and vitamin B12 injection alone (Wang 1997). The authors concluded that there is insufficient evidence to either support or refute the use of acupuncture (either needle or laser) in the treatment of lateral elbow pain. This review has demonstrated needle acupuncture to be of short-term benefit with respect to pain, but this finding is based on the results of 2 small trials, the results of which were not able to be combined in meta-analysis. No benefit lasting more than 24 hours following treatment has been demonstrated. No trial assessed or commented on potential adverse effect. They stated that further trials, utilizing appropriate methods and adequate sample sizes, are needed before conclusions can be drawn regarding the effect of acupuncture on tennis elbow.

In a prospective, observational, pilot study, Wang et al (2004) examined the effects of vitamin K acupuncture point injection on menstrual pain in young women aged 14 to 25 from different countries and cultural backgrounds who have had unmitigated severe primary dysmenorrhea for 6 months or more. All subjects were treated with bilateral acupuncture point injection of vitamin K on the first or second day of menstrual

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pain. Pain intensity, total duration, and average intensity of menstrual distress, hours in bed, normal daily activity restrictions, and numbers of analgesic tablets taken to relieve pain were recorded before the treatment and for 4 subsequent menstrual cycles. Noticeable pain relief was observed 2 mins after treatment, and subsequent pain reduction occurred at 30 mins ($p < 0.001$). Subjects reported significantly fewer daily life restrictions, fewer hours in bed, less consumption of analgesic tablets, and lower scores of menstrual pain duration and intensity ($p < 0.001$). There were no adverse events. Some women experienced mild, self-limited pain at the injection site. The authors concluded that acupuncture point injection with vitamin K alleviated acute menstrual pain, and relief extended through the non-treatment follow-up cycles in this uncontrolled pilot study conducted in 2 countries. They stated that further investigation employing controlled experimental designs is warranted.

Liang et al (2011) reported the findings of 2 patients with amyotrophic lateral sclerosis (ALS) who were treated with 4 weeks of acupuncture injection point therapy using Enercel. These patients were administered 0.25 to 0.5 cc Enercel Plus IM to specific acupuncture points for 5 days per week for 4 weeks. Patient 1 exhibited flaccid paralysis of all 4 extremities and impaired speech and swallowing. By Week 4, she demonstrated significant improvement in her motor strength in all 4 extremities (R>L) and improved speech and swallowing. However, she did not continue to receive the Enercel acupoint injections, and she subsequently demonstrated a slow, progressive loss of neurological function during the ensuing 3 months, as shown on follow-up examinations. Patient 2 had significantly impaired speech and mild motor loss in the upper extremities and the left leg. After 4 weeks of treatment, his voice had significantly improved to the point where his speech was understandable and his motor functions had returned to normal. He continued receiving Enercel acupoint injections during the 3-month follow-up period and his clinical improvements were maintained. Thus, these 2 patients with ALS showed clinical improvements after 4 weeks of Enercel acupoint injection therapy. Follow-up data suggested that ongoing therapy may be necessary in order to maintain these positive effects. The authors concluded that the findings of this preliminary study merits further study and confirmation.

In a Cochrane review, Paley et al (2011) evaluated effectiveness of acupuncture for relief of cancer-related pain in adults. CENTRAL, MEDLINE, EMBASE, PsycINFO, AMED, and SPORTDiscus were searched up to November 2010 including non-English language papers. Randomized controlled trials evaluating any type of invasive acupuncture for pain directly related to cancer in adults of 18 years or over were selected for analysis. It was planned to pool data to provide an overall measure of effect and to calculate the number needed to treat to benefit, but this was not possible due to heterogeneity. Two review authors independently extracted data adding it to data extraction sheets. Quality scores were given to studies. Data sheets were compared and discussed with a third review author who acted as arbiter. A total of 3 RCTs (204 participants) were included. One high-quality study investigated the effect of auricular acupuncture compared with auricular acupuncture at "placebo" points and with non-invasive vaccaria ear seeds attached at "placebo" points. Participants in 2 acupuncture groups were blinded but blinding wasn't possible in the ear seeds group because seeds were attached using tape. This may have biased results in favor of acupuncture groups. Participants in the real acupuncture group had lower pain scores at 2-month follow-up than either the placebo or ear seeds group. There was high-risk of bias in 2 studies because of low methodological quality. One study comparing acupuncture with medication concluded that both methods were effective in controlling pain, although acupuncture was the most effective. The second study compared acupuncture, point-injection and medication in participants with stomach cancer. Long-term pain relief was reported for both acupuncture and point-injection compared with medication during the last 10 days of treatment. Although both studies have positive results in favor of acupuncture, they should be viewed with caution due to methodological limitations, small sample sizes, poor reporting and inadequate analysis. The authors concluded that there is insufficient evidence to judge whether acupuncture is effective in treating cancer-related pain in adults.

In a pilot study, park et al (2011) examined the possibility of Carthami-Semen (CS, Safflower seed) acupuncture point injection as a new promising treatment for chronic daily headache (CDH). A total of 40 subjects with CDH were recruited and randomized to a CS acupuncture point injection group or a normal saline (NS) acupuncture point injection group. Acupuncture point injections were applied twice-weekly during a 4-week period to the bilateral Fengchi (GB20), Jianjing (GB21) and Taiyang (EX-HN5) acupoints with CS extract or NS. The primary outcome measure was headache-related QOL (HR-QOL), assessed

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using the Headache Impact Test (HIT). Secondary outcome measures were the changes in the number of headache-free days and health status as assessed with the Short Form (36) Health Survey (SF-36). HIT scores decreased by 14.9 points in the CS acupuncture point injection group compared with 7.9 points in the NS acupuncture point injection group ($p = 0.013$). Headache-free days increased by 32.6 % in the CS acupuncture point injection group compared with 17.4 % in the NS acupuncture point injection group ($p = 0.045$). There were significant increases in SF-36 scores compared with baseline in both groups, but the mean improvement was greater in the CS acupuncture point injection group. No serious adverse events were reported. The authors stated that these findings suggested that the CS acupuncture point injection could be a new safe and promising treatment for CDH. They stated that a larger and long-term follow-up trial is needed to determine more definitely the efficacy of CS acupuncture point injection and to elucidate how long the effect lasts.

Zhang et al (2012) examined the effects of acupoint injection on cervical spondylosis. Electronic retrieval was carried out on literatures from the period of May 1, 2006 to June 1, 2011 in databases of PubMed, ISI web of knowledge and CNKI. The selected literatures were summarized and classified from 3 aspects of acupoints selection, medication selection and manipulations. The authors noted that cervical Jiaji (EX-B 2) points, Fengchi (GB 20) and Ashi points are common acupoints. The medications contain simple Chinese herbs (e.g., Danggui injection, etc.) and compound Chinese herbs (e.g., compound Danggui injection, etc.), simple western medicine (e.g., vitamin B family) and Chinese herbs combined with western medicine (compound Danggui combined with vitamin B12). Disposable syringes were used for injection equipment. The authors stated that while acupoint injection in treating cervical spondylosis is effective, however, mechanism studies are still deficient since most of the researches focused on clinical observation. They concluded that manipulation of acupoint injection is not standardized; laws of clinical effect are unclear. Thus, they stated that "the above mentioned defects are still remained for further improvement".

Bae et al (2014) conducted a meta-analysis of an array of appropriate studies to evaluate the pre-operative anxiolytic efficacy of acupuncture therapy. Four electronic databases (MEDLINE, EMBASE, CENTRAL, and CINAHL) were searched up to February 2014. In the meta-analysis, data were included from RCT studies in which groups receiving pre-operative acupuncture treatment were compared with control groups receiving a placebo for anxiety. A total of 14 publications ($n = 1,034$) were included. Six publications, using the State-Trait Anxiety Inventory-State (STAI-S), reported that acupuncture interventions led to greater reductions in pre-operative anxiety relative to sham acupuncture (mean difference = 5.63, $p < 0.00001$, 95 % CI: 4.14 to 7.11). Further 8 publications, employing VAS, also indicated significant differences in pre-operative anxiety amelioration between acupuncture and sham acupuncture (MD = 19.23, $p < 0.00001$, 95 % CI: 16.34 to 22.12). The authors concluded that acupuncture therapy aiming at reducing pre-operative anxiety has a statistically significant effect relative to placebo or non-treatment conditions. They stated that well-designed and rigorous studies that employ large sample sizes are needed to corroborate this finding.

Ryu et al (2014) stated that to explore the pain mechanism, numerous animal models have been developed to simulate specific human pain conditions, including cancer-induced bone pain (CIBP). In this study, these researchers analyzed the current research methodology of acupuncture for the treatment of CIBP. They electronically searched the PubMed database for animal studies published from 2000 onward using these search terms: (bone cancer OR cancer) AND (pain OR analgesia) AND (acupuncture OR pharmacopuncture OR bee venom). They selected articles that described cancer pain in animal models. These investigators analyzed the methods used to induce cancer pain and the outcome measures used to assess the effects of acupuncture on CIBP in animal models. They reviewed articles that met their inclusion criteria. Injection of mammary cancer cells into the cavity of the tibia was the most frequently used method for inducing CIBP in the animal models. Among the 8 selected studies, 5 demonstrated the effects of electroacupuncture on CIBP. The effects of acupuncture were assessed by measuring pain-related behavior. The authors concluded that future researches will be needed to ascertain the effectiveness of acupuncture for treating CIBP and to explore the specific mechanism of CIBP in animal models.

In a Cochrane review, Shen et al (2014) examined the effects of acupuncture, alone or in combination treatments compared with placebo (or no treatment) or any other treatments for people with schizophrenia or related psychoses. These investigators searched Cochrane Schizophrenia Group's Trials Register (February 2012), which was based on regular searches of CINAHL, BIOSIS, AMED, EMBASE, PubMed, MEDLINE,

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PsycINFO and clinical trials registries. They also inspected references of identified studies and contacted relevant authors for additional information. They included all relevant RCTs involving people with schizophrenia-like illnesses, comparing acupuncture added to standard dose anti-psychotics with standard dose anti-psychotics alone, acupuncture added to low dose anti-psychotics with standard dose anti-psychotics, acupuncture with anti-psychotics, acupuncture added to Traditional Chinese Medicine (TCM) drug with TCM drug, acupuncture with TCM drug, electric acupuncture convulsive therapy with electroconvulsive therapy. These researchers reliably extracted data from all included studies, discussed any disagreement, documented decisions and contacted authors of studies when necessary. They analyzed binary outcomes using a standard estimation of RR and its 95 % CI. For continuous data, they calculated MDs with 95 % CI. For homogeneous data they used fixed-effect model. They assessed risk of bias for included studies and created "Summary of findings" tables using Grading of Recommendations, Assessment, Development and Evaluation (GRADE). After an update search in 2012 the review included 30 studies testing different forms of acupuncture across 6 different comparisons. All studies were at moderate risk of bias. When acupuncture plus standard anti-psychotic treatment was compared with standard anti-psychotic treatment alone, people were at less risk of being "not improved" (n = 244, 3 RCTs, medium-term RR 0.40 CI: 0.28 to 0.57, very low quality evidence). Mental state findings were mostly consistent with this finding as was time in hospital (n = 120, 1 RCT, days MD -16.00 CI: -19.54 to -12.46, moderate quality evidence). If anything, adverse effects were less for the acupuncture group (e.g., central nervous system, insomnia, short-term, n = 202, 3 RCTs, RR 0.30 CI: 0.11 to 0.83, low quality evidence). When acupuncture was added to low dose anti-psychotics and this was compared with standard dose anti-psychotic drugs, relapse was less in the experimental group (n = 170, 1 RCT, long-term RR 0.57 CI: 0.37 to 0.89, very low quality evidence) but there was no difference for the outcome of "not improved". Again, mental state findings were mostly consistent with the latter. Incidences of extra-pyramidal symptoms – akathisia, were less for those in the acupuncture added to low dose anti-psychotics group (n = 180, 1 RCT, short-term RR 0.03 CI: 0.00 to 0.49, low quality evidence) – as dry mouth, blurred vision and tachycardia. When acupuncture was compared with anti-psychotic drugs of known efficacy in standard doses, there were equivocal data for outcomes such as "not improved" using different global state criteria. Traditional acupuncture added to TCM drug had benefit over use of TCM drug alone (n = 360, 2 RCTs, RR no clinically important change 0.11 CI: 0.02 to 0.59, low quality evidence), but when traditional acupuncture was compared with TCM drug directly there was no significant difference in the short-term. However, these researchers found that participants given electroacupuncture were significantly less likely to experience a worsening in global state (n = 88, 1 RCT, short-term RR 0.52 CI: 0.34 to 0.80, low quality evidence). In the 1 study that compared electric acupuncture convulsive therapy with electroconvulsive therapy there were significantly different rates of spinal fracture between the groups (n = 68, 1 RCT, short-term RR 0.33 CI: 0.14 to 0.81, low quality evidence). Attrition in all studies was minimal. No studies reported death, engagement with services, satisfaction with treatment, quality of life, or economic outcomes. The authors concluded that limited evidence suggested that acupuncture may have some anti-psychotic effects as measured on global and mental state with few adverse effects. They stated that better designed large studies are needed to fully and fairly test the effects of acupuncture for people with schizophrenia.

Park et al (2014) reviewed the available literature on the use of acupuncture as a treatment for spasticity in patients with stroke. Randomized trials assessing the effects of acupuncture for the treatment of spasticity after stroke were identified by searching the Cochrane Library, PubMed, ProQuest, EBSCOhost, SCOPUS, CINAHL, EMBASE, Alternative Medicine Database, and Chinese and Korean medical literature databases. Two reviewers independently extracted data on study characteristics, patient characteristics, and spasticity outcomes. A total of 8 trials with 399 patients met all the inclusion criteria. Compared with controls without acupuncture, acupuncture had no effect on improving clinical outcomes (as measured by validated instruments such as the Modified Ashworth Scale) or physiologic outcomes (assessed by measures such as the H-reflex/M-response [H/M] ratio at the end of the treatment period); H/M ratios did decrease significantly immediately after the first acupuncture treatment. Methodological quality of all evaluated trials was considered inadequate. The authors concluded that the effect of acupuncture for spasticity in patients with stroke remains uncertain, primarily because of the poor quality of the available studies. They stated that larger and more methodologically sound trials are needed to confirm or refute any effect of acupuncture as a treatment for spasticity after stroke.

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Li et al (2014) noted that spontaneous intra-cerebral hemorrhage (ICH) is the most devastating subtype of stroke, but there is currently no evidence-based treatment strategy. Acupuncture is a well-known traditional Chinese therapy for stroke-induced disability, and GV20 is the commonly used acupuncture point. These researchers evaluated the effectiveness of GV20-based acupuncture in animal models of acute ICH. Studies of GV20-based acupuncture in animal models of acute ICH were identified from 6 databases up to July 2013. Study quality for each included article was evaluated according to the CAMARADES 10-item checklist. Outcome measures were neurological deficit scores and brain water content. All the data were analyzed using RevMan V.5.1 software. A total of 19 studies were identified describing procedures involving 1,628 animals. The quality score of the studies ranged from 3 to 6, with a mean of 4.6. The global estimate of the effect of GV20-based acupuncture was 0.19 (95 % CI: 0.13 to 0.25, $p < 0.001$) SDs improvement in outcome compared with controls. In subgroup analyses, size of effect was higher where the outcome was measured as the neurological deficit score than the brain water content or both ($p < 0.001$). The authors concluded that these findings showed the possible efficacy of GV20-based acupuncture in animal models of acute ICH, suggesting it as a candidate therapy for acute ICH.

Other Experimental, Investigational, or Unproven Indications for Acupuncture

Ji and colleagues (2013) noted that inflammatory bowel diseases (IBD) are recurrent and refractory; it includes Crohn's disease (CD) and ulcerative colitis (UC). Clinical researches about acupuncture and moxibustion treatments for IBD are increasing, while systematic reviews about their effectiveness remains in a shortage. These investigators evaluated the effectiveness of acupuncture and moxibustion for IBD. A total of 7 significant databases both in and abroad were searched for RCTs that compared acupuncture and moxibustion as the main intervention to pharmacotherapy in treating IBD. A meta-analysis was performed. A total of 43 RCTs were included. Among the 43 included trials, 10 trials compared oral sulphasalazine (SASP) with acupuncture and/or moxibustion treatments. A meta-analysis of the 10 trials indicated that acupuncture and moxibustion therapy was superior to oral SASP. The authors concluded that acupuncture and moxibustion therapy demonstrated better effectiveness than oral SASP in treating IBD. However, given the limitations of this systematic review and the included literature, definitive conclusions regarding the exact effectiveness of acupuncture and moxibustion treatment for IBD cannot be drawn. They stated that extant RCTs still cannot provide sufficient evidence and multi-center, double-blind RCTs with large sample sizes are needed to provide higher-quality evidence.

Kim et al (2013) evaluated the current evidence on the effectiveness of acupuncture for post-traumatic stress disorder (PTSD) in the form of a systematic review. These researchers performed a systematic literature search in 23 electronic databases. Grey literature was also searched. The key search terms were "acupuncture" and "PTSD". No language restrictions were imposed. They included all RCTs or prospective clinical trials that evaluated acupuncture and its variants against a wait-list, sham acupuncture, conventional therapy control for PTSD, or without control. A total of 4 RCTs and 2 uncontrolled clinical trials (UCTs) out of 136 articles in total were systematically reviewed. One high-quality RCT reported that acupuncture was superior to wait-list control and therapeutic effects of acupuncture and cognitive-behavioral therapy (CBT) were similar based on the effect sizes. One RCT showed no statistical difference between acupuncture and selective serotonin reuptake inhibitors (SSRIs); 1 RCT reported a favorable effect of acupoint stimulation plus CBT against CBT alone. A meta-analysis of acupuncture plus moxibustion versus SSRI favored acupuncture plus moxibustion in 3 outcomes. The authors concluded that this systematic review and meta-analysis suggested that the evidence of effectiveness of acupuncture for PTSD is encouraging but not cogent. They stated that further qualified trials are needed to confirm whether acupuncture is effective for PTSD.

Yang et al (2015) stated that In recent years, acupuncture has increasingly being integrated into pediatric health care. It was used on approximately 150,000 children (0.2 %). The authors updated the evidence for the safety and effectiveness of acupuncture for children and evaluated the methodological qualities of these studies to improve future research in this area. They included 24 systematic reviews, comprising 142 RCTs with 12,787 participants. Only 25 % (6/24) reviews were considered to be high quality (10.00 ± 0.63). High-quality systematic reviews and Cochrane systematic reviews tended to yield neutral or negative results ($p = 0.052$, 0.009, respectively). The effectiveness of acupuncture for 5 diseases (cerebral palsy (CP), nocturnal

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enuresis (NE), tic disorders, amblyopia, and pain reduction) is promising. It was unclear for hypoxic ischemic encephalopathy, attention deficit hyperactivity disorder, mumps, autism spectrum disorder (ASD), asthma, nausea/vomiting, and myopia. Acupuncture is not effective for epilepsy. Only 6 reviews reported adverse events (AEs) and no fatal side effects were reported. The authors concluded that the effectiveness of acupuncture for some diseases is promising and there have been no fatal side effects reported. They stated that further high-quality studies are justified, with 5 diseases in particular as research priorities.

Lv and colleagues (2015) stated that NE is recognized as a widespread health problem in young children and adolescents. Clinical researches about acupuncture therapy for NE are increasing, while systematic reviews assessing the effectiveness of acupuncture therapy are still lacking. These researchers evaluated the effectiveness of acupuncture therapy for NE. A comprehensive literature search of 8 databases was performed up to June 2014; RCTs which compared acupuncture therapy and placebo treatment or pharmacological therapy were identified. A meta-analysis was conducted. This review included 21 RCTs and a total of 1,590 subjects. The overall methodological qualities were low. The results of meta-analysis showed that acupuncture therapy was more effective for clinical effectiveness when compared with placebo or pharmacological treatment; AEs associated with acupuncture therapy were not documented. The authors concluded that based on the findings of this study, they cautiously suggested that acupuncture therapy could improve the clinical effectiveness. However, they stated that the beneficial effect of acupuncture might be over-stated due to low methodological qualities; rigorous high quality RCTs are needed.

Yu and colleagues (2015) noted that pruritus is a sensitive state that provokes the desire to scratch. It is not only a common symptom of skin diseases but it also occurs in some systemic diseases. Clinical studies on the effectiveness of the acupuncture therapy in alleviating itch are increasing, while systematic reviews assessing the effectiveness of acupuncture therapy are still lacking. These investigators evaluated the effectiveness of acupuncture therapy for pruritis. A comprehensive literature search of 8 databases was performed up to June 2014, and RCTs that compared acupuncture therapy and placebo acupuncture or no treatment group were identified. Accordingly, a meta-analysis was conducted. This review included 3 articles of RCTs from a total of 2,530 articles. The results of the meta-analysis showed that acupuncture therapy was effective to alleviate pruritis compared with placebo acupuncture and no treatment group. The authors concluded that based on the findings of this systematic review, they cautiously suggested that acupuncture therapy could improve the clinical effectiveness of pruritis. However, they stated that this conclusion needs more studies on various ethnic samples to confirm the final conclusion.

Acute Pancreatitis

In a systematic review and meta-analysis, Zhang and colleagues (2019) examined the safety and efficacy of acupuncture plus routine treatment (RT) for the treatment of acute pancreatitis (AP). These researchers carried out a literature search in 8 databases up to October 31, 2018; RCTs comparing acupuncture plus RT with RT alone for AP were included. A total of 12 eligible studies were included finally. The meta-analysis showed that acupuncture plus RT compared with RT alone could significantly improve the total effective rate and gastro-intestinal (GI) function and reduce the Acute Physiology, Age, Chronic Health Evaluation II score, tumor necrosis factor-alpha (TNF-α) count, the time of resuming to diets, and the length of hospital stay. Only 3 of the studies reported AEs or reactions. The authors concluded that the findings of this study suggested that acupuncture combined with RT may be effective for AP. However, these researchers stated that more rigorously designed RCTs are needed to confirm these findings.

Alcohol Withdrawal Syndrome

Li and co-workers (2018) noted that acupuncture has been used as a potential therapy for alcohol withdrawal syndrome (AWS), but evidence for its effects on this condition is limited. These researchers examined the safety and effectiveness of acupuncture for AWS. Central Register of Controlled Trials (CENTRAL), PubMed, Embase, the Cochrane Library, PsycINFO, Chinese Biomedicine Literature (CBM), China National Knowledge Infrastructure (CNKI) and Wan-Fang Database were searched from their inception to August 2016; RCTs of drug plus acupuncture or acupuncture alone for the treatment of AWS were included. Continuous data were expressed as MD with 95 % CI. Dichotomous data were expressed as RR with 95 %

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CI. A total of 11 RCTs with 875 participants were included. In the acute phase, 2 trials reported no difference between drug plus acupuncture and drug plus sham acupuncture in the reduction of craving for alcohol; however, 2 positive trials reported that drug plus acupuncture was superior to drug alone in the alleviation of psychological symptoms. In the protracted phase, 1 trial reported acupuncture was superior to sham acupuncture in reducing the craving for alcohol, 1 trial reported no difference between acupuncture and drug (disulfiram), and 1 trial reported acupuncture was superior to sham acupuncture for the alleviation of psychological symptoms; adverse effects were tolerable and not severe. The authors concluded that there was no significant difference between acupuncture (plus drug) and sham acupuncture (plus drug) with respect to the primary outcome measure of craving for alcohol among participants with AWS, and no difference in completion rates (pooled results). There was limited evidence from individual trials that acupuncture may reduce alcohol craving in the protracted phase and help alleviate psychological symptoms; however, given concerns about the quantity and quality of included studies, further large-scale and well-conducted RCTs are needed.

Alzheimer's Disease

In a randomized, controlled, parallel-group, exploratory study, Jia and colleagues (2017) examined the safety and efficacy of acupuncture in patients with mild-to-moderate Alzheimer's disease (AD) with a 4-week baseline (T0), a 12-week treatment phase (T1), and a 12-week follow-up period (T2). Patients with mild-to-moderate AD meeting inclusion criteria were randomly allocated into either

I. acupuncture or
II. donepezil hydrochloride groups.

Subjects in the acupuncture group (AG; n = 43) were administered acupuncture 3 times/week and subjects in the donepezil group (DG; n = 44) was given donepezil once-daily (5 mg/day for the first 4 weeks and 10 mg/day thereafter). Primary efficacy was measured using Alzheimer's disease Assessment Scale-Cognitive (ADAS-cog) and Clinician's Interview-Based Impression of Change-Plus (CIBIC-Plus). The second outcomes were measured with 23-Item Alzheimer's disease Cooperative Study Activities of Daily Living Scales (ADAS-ADL23) and Neuropsychiatric Index (NPI). Of 87 participants enrolled in the study, 79 patients finished their treatment and follow-up processes. The ADAS-cog scores for AG group showed obvious decreases at T2 and Δ(T2-T0) when compared with DG group, and significant between-group differences were detected (all $p < 0.05$). The mean CIBIC-Plus values for the AG group at T1 and T2 were much lower than that for the DG group, and there were significant differences between the 2 groups ($p < 0.05$). There were no significant between-group differences in the scores of ADAS-ADL23 and NPI during the study period. Treatment discontinuations due to AEs were 0 (0 %) and 4 (9.09 %) for the AG and DG groups, respectively. The authors concluded that acupuncture treatment was efficacious and well-tolerated in improving cognitive function and global clinical status; it offered a promising therapeutic option for the treatment of AD with few adverse effects or contraindications.

This study had several drawbacks:

I. the small sample size (n = 43 for AG; n = 44 for DG) led to under-powered results,
II. subjects who were glad to participate in this study appeared to have a positive attitude toward treatment (including acupuncture and medicine). It was possible that patients with high treatment expectations would introduce positive bias into the results of the study, and
III. it was unclear if the findings in Tianjin city can be extrapolated to other districts.

Anorexia

In a systematic review and meta-analysis, Chen et al (2023) examined the safety and effectiveness of acupuncture at Sifeng (acupuncture point) for the treatment of pediatric anorexia. RCTs of acupuncture at Sifeng for pediatric anorexia were queried in 7 databases from their inception to October 2022. The Cochrane risk of bias evaluation tool was employed to perform the risk of bias analysis of the included studies. A meta-analysis of the total efficiency, score of food intake reduction, time to normalize food intake,

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body-weight, leptin levels, and blood zinc levels was carried out using Review Manager 5.3 software. The GRADE criteria were applied to evaluate the evidence's quality. A total of 24 RCTs were included, involving 2,202 children. The allocation of concealment, blinding, and selective reporting had a high or unclear risk of bias. All experiments compared acupuncture at Sifeng with traditional Chinese medicine or Western medicine. The results showed that, compared with medicine, acupuncture at Sifeng could significantly improve the total efficiency (OR = 6.44, 95 % CI: 4.78 to 8.66), lower the score of food intake reduction (MD = -0.69, 95 % CI: -1.00 to -0.39]), decrease leptin levels (MD = -5.19, 95 % CI: -8.09 to -2.29]) and time to normal food intake (MD = -2.22, 95 % CI: -2.42 to -2.01), increase blood zinc (MD = 0.79, 95 % CI: 0.21 to 1.37]) and body-weight (MD = 1.28, 95 % CI: 0.85 to 1.72]; 7 studies found that the treatment was safe both during and after. Based on the GRADE criteria, the quality of the evidence for the majority of indicators was extremely poor. The authors concluded that the low certainty of evidence suggested that acupuncture at Sifeng was safe and effective in the treatment of pediatric anorexia. Moreover, these researchers stated that future high-quality clinical studies are needed to provide more reliable evidence of the safety and effectiveness of the therapy. They stated that the mechanism of action of acupuncture at Sifeng and its optimal strategy for treating anorexia with different traditional Chinese medicine syndromes need to be further examined.

Autism Spectrum Disorder

Yu et al (2023) noted that the effects of acupuncture and behavioral therapy, 2 non-pharmacological interventions, on social function in autism spectrum disorder (ASD) patients are still controversial. In a systematic review and meta-analysis, Yu et al (2023) examined the impact of these 2 treatments and compared their effects for the management of ASD. A total of 7 electronic databases were searched to identify RCTs on the use of acupuncture or behavioral therapy for ASD. A meta-analysis was performed using Review Manager 5.4 software. Continuous data were reported as MDs or SMDs with 95 % CIs. An assessment of methodological quality using the Cochrane risk-of-bias (ROB) tool for trials was performed. The GRADE was employed to examine the quality (certainty) of evidence for results regarding social function indicators. A total of 30 RCTs on acupuncture and 36 on behavioral therapy were included. Compared with the control condition, body acupuncture (SMD: 0.76, 95 % CI: 0.52 to 1.01; low certainty), modern acupuncture technology (SMD: 0.84, 95 % CI: 0.32 to 1.35; low certainty), cognitive behavioral therapy (SMD: 0.42, 95 % CI: 0.26 to 0.58; high certainty), the Denver model (SMD: 0.61, 95 % CI: 0.23 to 0.99; moderate certainty) and social skills training (SMD: 0.56, 95 % CI: 0.41 to 0.71; moderate certainty) improved social functioning. The authors concluded that behavioral therapies (such as CBT, the Denver model, social skills training), improved the social functioning of patients with ASD in the short- and long-term, as supported by high- and moderate-quality evidence. Acupuncture (including scalp acupuncture, body acupuncture, and use of modern acupuncture technology) also improved social functioning, as supported by low- and very low-quality evidence. These researchers stated that more high-quality evidence is needed to confirm the effect of acupoint catgut embedding (ACE) and early intensive behavioral intervention (EIBI).

The authors stated that this study had several drawbacks. First, given limited time and human resources, these researchers focused on only the effect of the 8 intervention types on social functioning in ASD according to studies published in Chinese or English. Second, few studies applying ACE or EIBI met the inclusion criteria; therefore, it was difficult to examine the effect of the 2 treatments on social functioning among patients with ASD. Third, there were insufficient eligible studies to compare the effects of acupuncture and behavioral therapy on social functioning. These investigators stated that further advanced and multi-dimensional studies or more in-depth analyses, such as NMAs, are needed to examine the effectiveness of these 2 critical non-pharmacological interventions and to determine differences in the effectiveness of these 2 intervention types for improving social functioning.

Breast Cancer-Related Hot Flashes

Salehi and colleagues (2016) evaluated the effectiveness of acupuncture for treatment of hot flash in women with breast cancer. The aspects considered in this study included searching for 12 data bases until April 2015 and consulting reference lists of reviews and related articles. Additional features studied comprised all articles on human patients with breast cancer treated with needle acupuncture with or without electrical

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stimulation for the treatment of hot flashes. The methodological quality was assessed using the modified Jadad score. The searches identified 12 relevant articles for inclusion. The meta-analysis without any subgroup or moderator failed to show favorable effects of acupuncture on reducing the frequency of hot flashes after intervention (n = 680, SMD = - 0.478, 95 % CI: -0.397 to 0.241, p = 0.632); but exhibited marked heterogeneity of the results (Q value = 83.200, p = 0.000, I^2 = 83.17, τ^2 = 0.310). The authors concluded that this meta-analysis showed contradictory results and yielded no convincing evidence to suggest that acupuncture was an effective treatment of hot flash in patients with breast cancer. They stated that multi-center studies with large sample size are needed to ascertain the effectiveness of acupuncture for treating hot flash in patients with breast cancer.

Breast Cancer-Related Lymphedema

Chien and colleagues (2019) stated that breast cancer-related lymphedema (BCRL) is hard to control. Management may include lymphatic drainage, skin care, bandaging, or even surgery. Since acupuncture has been proven to affect the neurophysiology and neuroendocrine systems, it has the potential to control BCRL. In a systematic review and meta-analysis, these researchers examined the effect of acupuncture in BCRL in RCTs. They carried out a literature search, following the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) statement and without language restrictions. A total of 5 databases were searched from inception through September 2018. Only studies that fulfilled the eligibility criteria of evaluating the effect of acupuncture on lymphedema in breast cancer were included. The methodological quality of these trials was assessed using the Cochrane criteria, and meta-analysis software (RevMan 5.3) was used for analysis. These investigators examined 178 breast cancer patients from 6 trials. All included RCTs had medium-to-high quality, based on the modified Jadad scale. The systematic review showed that acupuncture was safe and had a trend to improve symptoms, but trials did not consistently measure outcomes. The meta-analysis showed that acupuncture produced no significant improvement in the extent of lymphedema as compared with the control intervention (-1.90; 95 % CI: -5.39 to 1.59, p = 0.29). None of the studies reported severe AEs. The authors concluded that acupuncture is safe and has a trend to improve the lymphedema related to breast cancer, yet it did not significantly change arm circumference in BCRL. These researchers stated that future studies should include both subjective and objective measurements and large-scale studies are needed.

Hu and associates (2019) performed a comprehensive analysis of clinical studies on acupuncture treatment for BCRL to examine the safety and efficacy of acupuncture treatment and provide evidence for the clinical decision-making. Public databases, mainly including China Academic Journals Full-text Database, Database of Chinese Sci-Tech Journal, Wanfang, PubMed, Embase and the Cochrane Library, from the establishment of databases to December 2018 were searched for RCTs of acupuncture for BCRL. Clinical RCT on the treatment of BCRL with acupuncture combined with drugs or functional exercise were enrolled for the analysis. Bias risk and quality were assessed by 2 investigators according to the Cochrane Handbook 5.1.0 standard, and the Revman 5.3 software was used for meta-analysis. A total of 13 studies were enrolled, comprising 747 patients (377 in the treatment group and 370 in the control group). The results of meta-analysis showed that acupuncture intervention could improve the total effective rate for the treatment of BCRL (OR = 4.62; 95 % CI: 2.61 to 8.17). Recent studies suggest that acupuncture therapy could alleviate the upper limb swelling and improve the subjective pain and discomfort in patients with BCRL, regardless of the control intervention used. However, the number of high-quality RCT was low. Moreover, most of the studies adopted inconsistent efficacy indicators. These researchers stated that additional blinded, large-sample, randomized, well-controlled studies with objective and uniform efficacy indicators are needed, especially in China, to confirm these findings.

In a systematic review and meta-analysis, Yu and co-workers (2020) examined if acupuncture was a good method for treating limb edema in women after breast cancer surgery. These investigators reviewed published RCTs to evaluate the effectiveness of acupuncture in the prevention of BCRL. Searching strategies were performed with the following keywords: "Breast cancer", "acupuncture", "neoplasm," and "lymphoedema" with derivations and different combinations of these keywords. The following databases were searched: PubMed, Cochrane Library, Embase, Web of Science, CNKI, WanFang, and CBM. Studies published in English and Chinese were considered for inclusion in this study. Study selection, risk of bias

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assessment and data extraction were independently conducted. Statistical analyses were conducted with RevMan software (version 5.3). A total of 8 studies were identified by the search strategy, and 519 patients were included in this study. The effective rate was higher (OR: 4.23; 95 % CI: 2.11 to 8.49; Z=4.07, p < 0.0001) in the experimental group than that in the control group. There were no significant improvements in the front flexion (MD: 0.19; 95 % CI: -3.68 to 4.06; Z=0.09, p=0.92) or the back extension (MD: 0.42; 95 % CI: -2.22 to 3.06; Z=0.31, p=0.75) movements of the shoulder between the experimental and control groups. The authors concluded that acupuncture may be an effective method for improving the condition of BCRL. However, due to the high risk of bias and the low quality of the available studies, further high-quality RCTs are needed to confirm the efficacy of acupuncture for patients with BCRL.

Cancer-Related Fatigue

Li and Liu (2021) stated that increasing attention has been paid to acupuncture and auricular acupressure as alternative strategies for cancer-related fatigue (CRF) management. These investigators described the protocol of a systematic review and meta-analysis that will examine the safety and effectiveness of acupuncture and auricular acupressure for relieving CRF in patients (great than or equal to 18 years of age) during lung cancer chemotherapy. They searched the Web of Science, Embase, PubMed, and Cochrane Library electronic databases from inception to August 2021, using the key phrases "acupuncture", "auricular acupressure", and "lung cancer" for all relevant trials. Studies that compared acupuncture (including electro-acupuncture) and auricular acupressure with acupuncture alone were included. The primary outcome was the measurement of the CRF symptoms; secondary outcome measures were physical activity, QOL, and AEs. A “p” value of < 0.05 was considered to be statistically significant. It will be the first of such study and will obtain evidence for using acupuncture and auricular acupressure for CRF management in lung cancer patients. The authors concluded that combined acupuncture and auricular acupressure may be effective for relieving CRF in patients during lung cancer chemotherapy.

Cardiovascular Diseases (e.g., angina pectoris, heart failure, hypertension)

Lee and colleagues (2016) reviewed RCTs on acupuncture for heart failure (HF) and evaluated the clinical evidence. Electronic databases such as Medline, Embase, Cochrane Central Register of Controlled Trials (CENTRAL) and certain Chinese & Korean databases were searched until October 2015. The main outcomes assessed were mortality, New York Heart Association (NYHA) function classifications, and acupuncture-related AEs. The details of acupuncture intervention were also investigated. Among 4,107 publications, 7 RCTs were included; most of them showed considerable methodological flaws. These researchers could not conduct a meta-analysis because of the heterogeneity of the included studies. In 1 acute HF study, acupuncture shortened intensive care unit (ICU) stay by 2.2days (95 % CI: 1.26 to 3.14) and reduced the RR of re-admission to 0.53 (95 % CI: 0.28 to 0.99). However, mortality was not affected. Hemodynamic parameters also showed improvement. Another study reported an improved left ventricular ejection fraction (LVEF) by 9.95 % (95 % CI: 3.24 to 16.66). In 5 chronic HF studies, acupuncture improved exercise capacity, QOL, hemodynamic parameters, and time domain heart rate variability parameters. Acupuncture decreased NT-pro BNP levels by 292.20 (95 % CI: -567.36 to -17.04); no AEs were reported. The author concluded that the effectiveness of acupuncture as a therapy for HF is currently inconclusive. They stated that further large and rigorous clinical trials are needed to establish its clinical utility.

de Lima Pimentel and colleagues (2019) stated that acupuncture, for the Westerns countries, is an innovative and low-cost therapy for treatment and prevention of cardiovascular diseases (CVDs). However, most of its effects and mechanisms are poorly understood. These investigators systematically reviewed the literature regarding the clinical effects of acupuncture for the treatment and prevention of CVDs. A search for papers published, in English or Portuguese in the past 20 years, was conducted at PubMed, SciELO, and PEDro databases. Clinical trials conducted on the effects of acupuncture were included in this review. Two reviewers extracted the data independently remaining 17 articles after screening. The most used acupoint was PC6 (10 studies, 64.7 %), followed by ST36 (6 studies, 35.3 %), and auricular acupoints (4 studies, 23.5 %). Among the clinical applications, hypertension was the most studied CVD, with acupuncture being the most reported method among the studies (70.6 %). Only 3 articles reported no benefit in the treatment of CVDs for the methodology used. The authors concluded that although several studies indicated an

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improvement in the response of the cardiovascular system in CVDs by acupuncture, electro-acupuncture, or electro-stimulation treatment, the heterogeneity of the studies did not allow a standardization of its application for each specific disease, making further studies necessary for its use to become a reality.

Shen and colleagues (2021) performed an overview of systematic reviews/meta-analyses (SRs/MAs) on acupuncture for the treatment of patients with stable angina pectoris (SAP). A total of 8 databases were searched for SRs/MAs of acupuncture on SAP. The methodological quality, reporting quality, and evidence quality were evaluated by Assessing the Methodological Quality of Systematic Reviews 2 (AMSTAR-2), the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) checklist, and the Grading of Recommendations Assessment, Development and Evaluation (GRADE) system, respectively. A total of 7 published SRs/MAs met the inclusion criteria. According to the evaluation results of AMSTAR-2, 2 studies were considered as of moderate quality; the remaining 5 were considered as of very low quality. According to the evaluation results of the PRISMA checklist, only 1 study reported the checklist in its entirety, while others had reporting deficiencies. According to GRADE, a total of 18 outcome indicators extracted from the included studies were evaluated. The evidence quality was very low in 3, low in 3, moderate in 8, and high in 4. The authors concluded that acupuncture may be beneficial for SAP from the currently published evidence; however, this conclusion must be interpreted cautiously due to the generally low methodological quality, reporting quality, and evidence quality of the included studies. These researchers stated that more rigorous, more standardized and comprehensive SRs/MAs are needed to provide strong evidence for convincing conclusions.

Chronic Ankle Instability

Luan et al (2023) stated the effectiveness of acupuncture in the treatment of chronic ankle instability (CAI) is unclear. In a systematic review and meta-analysis, these investigators examined the effectiveness of acupuncture or similar needling therapy on pain, proprioception, balance, and self-reported function in individuals with CAI. A total of 9 databases (PubMed, Embase, Cochrane Library, Web of Science, EBSCO, PEDro, CNKI, WanFang, and CQVIP) were systematically searched from inception to April 2023. This study included RCTs involving acupuncture or similar needling therapy as an intervention for individuals with CAI. Data were extracted independently by 2 assessors using a standardized form. Literature quality and risk bias were assessed by using the PEDro scale. A total of 12 studies (n = 571) were found, of which the final meta-analysis was conducted with 8. Different studies employ varying treatments, including specific needle types, techniques, and therapeutic frameworks. Compared to control without acupuncture or similar needling therapy, acupuncture or similar needling intervention resulted in improved pain (WMD 1.33, 95 % CI:0.14 to 2.52, $I^2 = 90$ %, $p = 0.03$), proprioception (active joint position sense, WMD 1.71, 95 % CI: 0.95 to 2.48, $I^2 = 0$ %, $p < 0.0001$), balance (SMD 0.54, 95 % CI: 0.03 to 1.04, $I^2 = 46$ %, $p = 0.04$), and self-reported function (Cumberland Ankle Instability Tool (WMD 2.92, 95 % CI: 0.94 to 4.90, $I^2 = 78$ %, $p = 0.004$); American Orthopedic Foot and Ankle Society (WMD 9.36, 95 % CI: 6.57 to 12.15, $I^2 = 0$ %, $p < 0.001$); Foot and Ankle Ability Measure: activities of daily living (ADL) (WMD 5.09, 95 % CI: 1.74 to 8.44, $I^2 = 0$ %, $p = 0.003$)) for individuals with CAI. The authors concluded that the available evidence suggests that acupuncture or similar needling therapy may improve pain, proprioception, balance, and self-reported function in individuals with CAI; however, further investigations are needed to verify these findings. Furthermore, various needles and techniques using in different studies have resulted in methodologic limitations that should be addressed in the future.

Chronic Constipation

In a randomized, parallel, sham-controlled, clinical trial, Liu and associates (2016) the effectiveness of electro-acupuncture (EA) for the treatment of chronic severe functional constipation (CSFC). Patients with CSFC and no serious underlying pathologic cause for constipation were included in this study; interventions included 28 sessions of EA at traditional acupoints or sham EA (SA) at non-acupoints over 8 weeks. The primary outcome was the change from baseline in mean weekly complete spontaneous bowel movements (CSBMs) during weeks 1 to 8; subjects were followed until week 20. A total of 1,075 patients (536 and 539 in the EA and SA groups, respectively) were enrolled. The increase from baseline in mean weekly CSBMs during weeks 1 to 8 was 1.76 (95 % CI: 1.61 to 1.89) in the EA group and 0.87 (95 % CI: 0.73 to 0.97) in the

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SA group (between-group difference, 0.90 [95 % CI: 0.74 to 1.10]; p < 0.001). The change from baseline in mean weekly CSBMs during weeks 9 to 20 was 1.96 (95 % CI: 1.78 to 2.11) in the EA group and 0.89 (95 % CI: 0.69 to 0.95) in the SA group (between-group difference, 1.09 [95 % CI: 0.94 to 1.31]; p < 0.001). The proportion of patients having 3 or more mean weekly CSBMs in the EA group was 31.3 % and 37.7 % over the treatment and follow-up periods, respectively, compared with 12.1 % and 14.1 % in the SA group (p < 0.001). Acupuncture-related AEs during treatment were infrequent in both groups, and all were mild or transient. The authors concluded that 8 weeks of EA increased CSBMs and was safe for the treatment of CSFC. Moreover, they stated that additional study is needed to evaluate a longer-term treatment and follow-up. The main drawbacks of this study were:

I. longer-term follow-up was not assessed, and
II. acupuncturists could not be blinded.

Chronic Fatigue Syndrome

Wang and colleagues (2014) stated that there is no curative treatment for chronic fatigue syndrome (CFS); and traditional Chinese medicine (TCM) is widely used in the treatment of CFS in China. These investigators evaluated the effectiveness and safety of TCM for CFS. The protocol of this review is registered at PROSPERO. These investigators searched 6 main databases for randomized clinical trials (RCTs) on TCM for CFS from their inception to September 2013. The Cochrane risk of bias tool was used to assess the methodological quality. They used RevMan 5.1 to synthesize the results. A total of 23 RCTs involving 1,776 participants were identified. The risk of bias of the included studies was high. The types of TCM interventions varied, including Chinese herbal medicine, acupuncture, qigong, moxibustion, and acupoint application. The results of meta-analyses and several individual studies showed that TCM alone or in combination with other interventions significantly alleviated fatigue symptoms as measured by Chalder's fatigue scale, fatigue severity scale, fatigue assessment instrument by Joseph E. Schwartz, Bell's fatigue scale, and guiding principle of clinical research on new drugs of TCM for fatigue symptom. There was no enough evidence that TCM could improve the quality of life for CFS patients. The included studies did not report serious adverse events. The authors concluded that TCM appeared to be effective to alleviate the fatigue symptom for people with CFS. However, they stated that due to the high risk of bias of the included studies, larger, well-designed studies are needed to confirm the potential benefit in the future.

In a multi-center, non-blinded, randomized controlled trial, Kim et al (2015) examined the effectiveness of 2 forms of acupuncture added to usual care for CFS and idiopathic chronic fatigue (ICF) compared to usual care alone. A 3-arm parallel, non-blinded, randomized controlled trial was performed in 4 hospitals. These researchers divided 150 participants into treatment and control groups at the same ratio. The treatment groups (Group A, body acupuncture; Group B, Sa-am acupuncture) received 10 sessions for 4 weeks. The control group (Group C) continued usual care alone. The primary outcome was the Fatigue Severity Scale (FSS) at 5 weeks after randomization. Secondary outcomes were the FSS at 13 weeks and a short form of the Stress Response Inventory (SRI), the Beck Depression Inventory (BDI), the Numeric Rating Scale (NRS), and the EuroQOL-5 Dimension (EQ-5D) at 5 and 13 weeks. Group A showed significantly lower FSS scores than Group C at 5 weeks (p = 0.023); SRI scores were significantly lower in the treatment groups than in the control group at 5 (Group A, p = 0.032; B, p < 0.001) and 13 weeks (Group A, p = 0.037; B, p < 0.001). Group B showed significantly lower BDI scores than Group C at 13 weeks (p = 0.007). NRS scores from the treatment groups were significantly reduced compared to control at 5 (Group A and B, p <0.001) and 13 weeks (Group A, p = 0.011; B, p = 0.002). The authors concluded that body acupuncture for 4 weeks in addition to usual care may help improve fatigue in CFS and ICF patients. This was a relatively small study (150 subjects divided into 3 groups) with short-term follow-up (13 weeks) and its findings were confounded by the combinational use of acupuncture and usual care. These preliminary findings need to be validated by well-designed studies

Furthermore, an UpToDate review on "Treatment of systemic exertion intolerance disease (chronic fatigue syndrome)" (Gluckman, 2016) does not mention acupuncture as a therapeutic option.

Chronic Hepatitis B

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Kong and colleagues (2019) stated that chronic hepatitis B is a liver disease associated with high morbidity and mortality. Chronic hepatitis B requires long-term management aiming to reduce the risks of hepato-cellular inflammatory necrosis, liver fibrosis, decompensated liver cirrhosis, liver failure, and liver cancer, as well as to improve health-related QOL. Acupuncture is being used to decrease discomfort and improve immune function in people with chronic hepatitis B. However, the benefits and harms of acupuncture still need to be established in a rigorous way. In a Cochrane review, these researchers examined the benefits and harms of acupuncture versus no intervention or sham acupuncture in people with chronic hepatitis B. They undertook electronic searches of the Cochrane Hepato-Biliary Group Controlled Trials Register, CENTRAL, Medline, Embase, LILACS, Science Citation Index Expanded, Conference Proceedings Citation Index - Science, China National Knowledge Infrastructure (CNKI), Chongqing VIP (CQVIP), Wanfang Data, and SinoMed to March 1, 2019. They also searched the World Health Organization International Clinical Trials Registry Platform (www.who.int/ictrp), ClinicalTrials.gov (www.clinicaltrials.gov/), and the Chinese Clinical Trial Registry (ChiCTR) for ongoing or unpublished trials until March 1, 2019. These investigators included randomized clinical trials, irrespective of publication status, language, and blinding, comparing acupuncture versus no intervention or sham acupuncture in people with chronic hepatitis B. They included subjects of any sex and age, diagnosed with chronic hepatitis B as defined by the trialists or according to guidelines. They allowed co-interventions when the co-interventions were administered equally to all intervention groups. Review authors in pairs individually retrieved data from reports and through correspondence with investigators. Primary outcomes were all-cause mortality, proportion of subjects with 1 or more serious AEs, and health-related QOL. Secondary outcomes were hepatitis B-related mortality, hepatitis B-related morbidity, and AEs considered not to be serious. These researchers presented the pooled results as RRs with 95 % CIs. They examined the risks of bias using risk of bias domains with pre-defined definitions. They put more weight on the estimate closest to zero effect when results with fixed-effect and random-effects models differed. These investigators evaluated the certainty of evidence using GRADE. A total of 8 randomized clinical trials with 555 randomized subjects were included in this analysis. All included trials compared acupuncture versus no intervention. These trials assessed heterogeneous acupuncture interventions. All trials used heterogeneous co-interventions applied equally in the compared groups; 7 trials included subjects with chronic hepatitis B, and 1 trial included subjects with chronic hepatitis B with co-morbid tuberculosis. All trials were evaluated at overall high risk of bias, and the certainty of evidence for all outcomes was very low due to high risk of bias for each outcome, imprecision of results (the CIs were wide), and publication bias (small sample size of the trials, and all trials were conducted in China). Additionally, 79 trials lacked the necessary methodological information to ensure their inclusion in this review. None of the included trials aimed to assess all-cause mortality, serious AEs, health-related QOL, hepatitis B-related mortality, and hepatitis B-related morbidity. These investigators were uncertain whether acupuncture, compared with no intervention, had an effect regarding AEs considered not to be serious (RR 0.67, 95 % CI: 0.43 to 1.06; $I^2 = 0$ %; 3 trials; 203 subjects; very low-certainty evidence) or detectable hepatitis B e-antigen (HBeAg) (RR 0.64, 95 % CI: 0.11 to 3.68; $I^2 = 98$ %; 2 trials; 158 subjects; very low-certainty evidence). Acupuncture showed a reduction in detectable hepatitis B virus (HBV) DNA (a non-validated surrogate outcome; RR 0.45, 95 % CI: 0.27 to 0.74; 1 trial, 58 subjects; very low-certainty evidence). These investigators were uncertain whether acupuncture had an effect regarding the remaining separately reported AEs considered not to be serious; 3 of the 8 included trials received academic funding from government or hospital. None of the remaining 5 trials reported information on funding. The authors concluded that the clinical effects of acupuncture for chronic hepatitis B remain unknown. The included trials lacked data on all-cause mortality, health-related QOL, serious AEs, hepatitis-B related mortality, and hepatitis-B related morbidity. The vast number of excluded trials lacked clear descriptions of their design and conduct. Whether acupuncture influenced AEs considered not to be serious was uncertain. It remained unclear if acupuncture affected HBeAg, and if it was associated with reduction in detectable HBV DNA. Based on available data from only 1 or 2 small trials on AEs considered not to be serious and on the surrogate outcomes HBeAg and HBV DNA, the certainty of evidence was very low. In view of the wide usage of acupuncture, any conclusion that one might try to draw in the future should be based on data on patient and clinically relevant outcomes, assessed in large, high-quality randomized sham-controlled trials with homogeneous groups of subjects and transparent funding.

Diabetic Gastroparesis

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In a systematic review and meta-analysis, Li et al (2022b) examined the safety and effectiveness of traditional Chinese acupoint therapies versus standard therapies, nursing or recovery treatments in the treatment of diabetic gastroparesis; RCTs were included in this meta-analysis. The RR and WMD with 95 % CI for the total effective rate, gastrin level, gastric-emptying time, fasting blood glucose level, 2-hour blood glucose level, and glycosylated hemoglobin level were evaluated using RevMan 5.3 software. Bias assessment was carried out using the Cochrane risk-of-bias tool. A total of 59 articles were included in the analysis. In comparison with the control groups, the acupoint therapy groups showed higher total effective rates ($p < 0.00001$), enhanced gastric-emptying rates ($p < 0.00001$), and reduced glycosylated hemoglobin levels. The authors concluded that in comparison with Western medicine or conventional care, traditional Chinese acupoint therapies showed a significant advantage in the treatment of diabetic gastroparesis. Moreover, these researchers stated that considering the low-quality and high-risk of the included studies, more high-quality RCTs are needed to confirm these findings.

These investigators stated that considering the low-quality of the included studies and the non-negligible publication bias, for example, the possibility of selective data reporting by the authors, they were cautious regarding the adverse reaction rate. Furthermore, blinding was difficult to implement due to the nature of acupoint therapy interventions. Moreover, the inclusion of different acupoint therapies may result in significant heterogeneity, and the studies included were only in English and Chinese, which were potential limitations in the meta-analysis.

Diminished Ovarian Reserve

Diminished ovarian reserve (DOR) denotes reduced number and quality of oocytes in the ovary. In a systematic review and meta-analysis, Lin et al (2023) examined the effectiveness of acupuncture in the treatment of DOR based on available RCTs. A total of 9 databases from their inception to December 6, 2022, were comprehensively searched to retrieve RCTs related to the effectiveness of acupuncture for the treatment of DOR. The outcomes of interest were sex hormones level and antral follicle count (AFC). Risk of Bias (RoB) was adopted to examine the quality of the included trials. A total of 13 RCTs involving 787 patients were included in this meta-analysis. The review of available evidence revealed acupuncture produced a significant effectiveness in decreasing follicle-stimulating hormone (FSH) levels (SMD = -1.07, 95 % CI: -1.79 to -0.36, $p = 0.003$), FSH/LH ratio (MD = -0.31, 95 % CI: -0.54 to -0.09, $p = 0.006$) and increasing anti-Mullerian hormone (AMH) levels (SMD = 0.25, 95 % CI: -0.00 to 0.49, $p = 0.05$), along with AFC (MD = 1.87, 95 % CI: 0.96 to 2.79, $p < 0.0001$) compared to controls. Compared with electro-acupuncture treatment, manual acupuncture was superior in reducing FSH levels, FSH/LH ratio, and increasing AMH levels and AFC ($p < 0.05$). A notable association was also observed when acupuncture was combined with traditional Chinese medicine therapy for improving FSH levels, FSH/LH ratio, and AFC ($p < 0.05$). In addition, a high-dose of acupuncture (10 acupoints or more) was more conducive in ameliorating FSH levels, FSH/LH ratio, and AFC ($p < 0.05$) than a low-dose of acupuncture (less than 10 acupoints). Substantial heterogeneity existed among studies. The authors concluded that acupuncture may have significant clinical potential for patients with DOR in terms of improving sex hormones level and increasing AFC, although the evidence was drawn with high heterogeneity. Moreover, these researchers stated that more rigorous trials carried out in diverse regions worldwide are needed to ascertain the effectiveness of acupuncture for patients diagnosed with DOR.

In a scoping review, Xie et al (2023) examined the clinical research status of acupuncture and moxibustion in the treatment of patients with DOR. PubMed, Cochrane Library, Excerpta Medica database, Allied and Complementary Medicine Database, Chinese Biological Medicine, China National Knowledge Infrastructure, VIP Database for Chinese Technical Periodicals, and Wanfang database were searched from January 2010 to May 2022 using keywords and medical subject heading terms. After applying the inclusion and exclusion criteria, relevant studies were selected. Structured tables and descriptive charts were made to visually express research features by using Excel, Original, IBM SPSS Model 18.0, Adobe Illustrator and other software packages. Report quality was evaluated for Cochrane bias using Review Manager 5.3. A total of 851 studies were identified; of these, 90 met the inclusion criteria. The results extracted from these studies were classified into 4 categories: research characteristics, study type, acupuncture and moxibustion prescriptions, and efficacy observation. The authors concluded that the quality assessment of acupuncture

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and moxibustion for DOR is not ideal; thus, standardization and normalization should be strengthened, and high-quality evidence is needed to further examine the effectiveness of this approach. Moreover, these investigators stated that due to heterogeneity in DOR diagnosis, the observation index should be updated with reference to the latest research to improve efficacy evaluation.

Dysmenorrhea

Liu and co-workers (2017) stated that primary dysmenorrhea (PD) is one of the most common complaints among young women. Acupuncture has been widely applied as a therapeutic modality in China and abroad for PD; however, the evidence for its benefits is still not convincing. These researchers conducted a systematic review of RCTs to evaluate the evidence regarding the use of acupuncture in treating PD. The research team retrieved reports for RCTs published in 7 databases from their inception to March 2016, with no language restrictions: PubMed, Medline, Embase, the Cochrane Central Register of Controlled Trials, the Chinese National Knowledge Infrastructure database, the Chinese Biomedical database, and the Wanfang database. The study was conducted at the Beijing University of Traditional Chinese Medicine (Beijing, China). Participants in the reviewed studies were women aged 14 to 49 years who had received a diagnosis of PD in the absence of any visible pelvic pathology. The types of acupuncture included traditional acupuncture, electro-acupuncture, ear acupuncture, scalp acupuncture, superficial acupuncture, electro-superficial acupuncture, wrist-ankle acupuncture, and abdominal acupuncture. The primary outcome was pain relief measured using a VAS, a verbal rating scale (VRS), or a NRS. The secondary outcomes included

I. overall improvement as measured by the short-form McGill pain questionnaire or symptom scale based on the Clinical Study Guideline for New Developed Chinese Medicine,
II. menstrual distress as measured by the Menstrual Distress Questionnaire,
III. QOL as measured by a validated scale (e.g., the short-form 36), and (iv) adverse effects.

A total of 23 trials enrolling 2,770 patients were included in the review. Overall, most trials were of poor quality. Among the trials, only 6 were evaluated as having a low risk of bias, 3 of which indicated that acupuncture was statistically more effective than sham acupuncture – mean difference (MD), -3.51; 95 % CI: -5.27 to -1.75; $p < 0.0001$; I^2, 0 % – or no treatment – MD, -21.95; 95 % CI: -25.45 to -18.45; $p < 0.00001$; I^2, 0 % – on the VAS (0 to 100 mm). Acupuncture also showed superiority to the control arms on the VRS, the NRS, and the McGill pain questionnaire, but those findings had been influenced by methodological flaws. The authors concluded that the available evidence suggested that acupuncture may be effective for PD; and justified future high-quality studies.

Gastric Ulcer

Wang and colleagues (2021a) noted that gastric ulcer (GU) is a clinically common disease of the digestive system that adversely affects patients' QOL and work ability. Although some studies have reported that acupuncture could improve the clinical symptoms of GU, the effectiveness of acupuncture has not been scientifically or methodically examined. These investigators described the protocol of a systematic review and meta-analysis that will evaluate the safety and effectiveness of acupuncture for the treatment of patients with GUs. The following electronic databases will be searched from the respective dates of database inception to March 23, 2021: The Cochrane Library, Web of Science, Embase, Medline, China National Knowledge Infrastructure, Chinese Biomedical Literature Database, Wanfang database, the Chinese Scientific Journal Database, and other sources; RCTs comparing acupuncture with other interventions or sham acupuncture were included. Two independent researchers will perform article retrieval, duplication removal, screening, quality evaluation, and data analyses by Review Manager (V.5.3.5). Meta-analyzes, subgroup analysis, and/or descriptive analyses will be carried out based on the included data conditions. The protocol of this study will systematically evaluate the safety and effectiveness of acupuncture for patients with GUs. The primary outcome is the effective rate, and the secondary outcomes include negative conversing rate of Helicobacter pylori infection, untoward effect, recurrence rate, QOL, and symptom scores. The authors concluded that the findings of this study will provide evidence of whether acupuncture is a safe and effective intervention for the treatment of patients with gastric ulcers.

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Gout

In a systematic review and meta-analysis, Hwang et al (2022) examined the safety and effectiveness of heat stimuli (e.g., fire needling, and warm needling) in acupuncture for the treatment of acute gout. A total of 4 international online databases (PubMed, Cochrane, Embase, and Chinese National Knowledge Infrastructure) were searched to identify RCTs that used fire needling and warm needling for the treatment of acute gout. The methodological quality of the RCTs was assessed using the Cochrane risk-of-bias (RoB) tool. A total of 13 RCTs (840 patients) were included and analyzed; 3 evaluation tools (total effective rate, UA level, and pain score) were employed. Comparisons were made between Western medicine (WM) and fire needling or warm needling treatment alone; fire needling and blood-letting combination treatment; combination of fire needling, blood-letting, and herbal medicine; as well as warm needling (concurrently). Heat stimuli in acupuncture alone or in combination treatment were more effective in terms of the total effectiveness rates, UA levels, and pain scores than WM alone. In all the evaluation tools, the treatment effects in the fire needling alone or warm needling alone treatment group and the fire needling and blood-letting combination intervention group were significantly better than those in the WM control group. The warm needling and WM combination intervention groups also experienced significantly better treatment effects in terms of total effectiveness rates and UA levels. Only the pain scores in the fire needling, blood-letting, and herbal medicine combination groups showed significant improvement. Only 4 studies mentioned adverse reactions – 1 reported loss of appetite; 3 reported none. According to the Cochrane RoB tool, most studies showed either high or uncertain RoB. The authors concluded that heat stimuli during acupuncture could be effective for acute gout; however, due to limitations such as the small number of included studies and regional bias, higher-quality, large-scale RCT studies are needed to confirm the level of evidence.

The authors stated that a drawback of this systematic review and meta-analysis was that several types of WMs were set as one control group without any distinction between single and parallel drugs. Further investigations are needed to select the optimal WM drug as a control or combination treatment drug. In addition, the number of studies included in this review/analysis was small; and there were many studies in which all evaluation scales have not been used. Furthermore, this review exhibited regional bias because the studies retrieved from the Chinese database were all RCTs performed in China.

In-Vitro Fertilization

Zhang et al (2023a) noted that more and more infertility couples are opting for combined acupuncture to improve success rate of in-vitro fertilization (IVF); however, evidence from acupuncture for improving IVF pregnancy outcomes remains a matter of debate. In a systematic review and meta-analysis, these investigators examined the effectiveness of acupuncture among women undergoing IVF. A total of 4 English (PubMed, Web of Science, Embase, and Cochrane Register of Controlled Clinical Trials) and 4 Chinese databases (Wanfang Databases, Chinese National Knowledge Infrastructure, Chinese Science and Technology Periodical Database, and SinoMed) were searched from database inception until July 2, 2023; RCTs that evaluated the acupuncture's effects for women undergoing IVF were included. The subgroup analysis was carried out with respect to the age of participants, different acupuncture types, type of control, acupuncture timing, geographical origin of the study, whether or not repeated IVF failure, and acupuncture sessions. Sensitivity analyses were pre-defined to examine the robustness of results. The primary outcomes were clinical pregnancy rate (CPR) and live-birth rate (LBR), and the secondary outcomes were ongoing pregnancy rate and miscarriage rate. Random effects model with I2 statistics were used to quantify heterogeneity. Publication bias was estimated by funnel plots and Egger's tests. A total of 58 eligible RCTs representing 10,968 women undergoing IVF for pregnant success were identified. Pooled CPR and LBR showed a significant difference between acupuncture and control groups [69 comparisons, RR 1.19, 95 % CI: 1.12 to 1.25, I2 = 0], extremely low evidence; 23 comparisons, RR 1.11, 95 % CI: 1.02 to 1.21, I2 = 14.6, low evidence, respectively). Only transcutaneous electrical acupoint stimulation showed a positive effect on both CPR (16 comparisons, RR 1.17, 95 % CI: 1.06 to 1.29; I2 = 0, moderate evidence) and LBR (9 comparisons, RR 1.20, 95 % CI: 1.04 to 1.37; I2 = 8.5, extremely low evidence). Heterogeneity across studies was found and no studies were graded as high-quality evidence. The authors concluded that the findings of this study showed that the convincing evidence levels on the associations between acupuncture

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and IVF pregnant outcomes were relatively low, and the varied methodological design and heterogeneity might influence these findings.

Mild Cognitive Impairment

In a network meta-analysis (NMA), Li et al (2022a) compared the clinical effects of various acupuncture and acupuncture-related therapies on patients with mild cognitive impairment (MCI). These investigators examined the direct and indirect evidence from relevant research. A total of 7 databases [PubMed, Web of Science, Cochrane Library, Embase, China National Knowledge Infrastructure (CNKI), VIP Database, and Wanfang database] were examined to find RCTs of acupuncture-related therapies for individuals with MCI. Two researchers independently reviewed the literature, retrieved the data, and evaluated the risk of bias in the included studies. The data were analyzed using Stata15.0 and R3.6.1 software. A total of 27 RCTs entailing 2,210 patients were included. Bayesian NMA showed that manual acupuncture combined with conventional therapy, moxibustion combined with conventional therapy, manual acupuncture, and electro-acupuncture were most effective in improving the Mini-mental State Examination (MMSE) score. The most effective interventions related to the Montreal Cognitive Assessment (MoCA) score were moxibustion combined with conventional therapy, followed by manual acupuncture combined with conventional therapy, acupressure combined with conventional therapy, and manual acupuncture combined with moxibustion. Manual acupuncture combined with moxibustion was dominant in the cluster ranking. The results of the node-splitting method revealed that direct and indirect evidence were consistent ($p > 0.05$). In addition, publication bias was detected. The authors concluded that this study would add to the body of knowledge regarding the safety and effectiveness of acupuncture-related therapies in the treatment of MCI. The findings of this study would also assist in the choice of clinical guidelines that optimize acupuncture treatment for patients with MCI. However, these researchers stated that more robust comparative evidence is needed to confirm this conclusion. They noted that that more high-quality, large-sample, multi-center RCTs are needed to confirm these findings.

The authors stated that this study had several drawbacks. First, there was insufficient information on the effectiveness of acupuncture-related treatment for MCI due to the small sample sizes, the small number of patients in each trial, and the limited assessment of MMSE and MoCA data currently accessible. Second, a small number of RCTs revealed potential bias due to the small number of subjects and the use of voluntary reporting. Fortunately, no apparent inconsistency or heterogeneity was found in this network meta-analysis. However, it was possible that some of the included studies over-stated the effectiveness of therapies, which could have had an impact on the findings if they had been excluded. Third, these researchers excluded several complementary and alternative therapy interventions due to the selection criteria for outcome measures, which may have impacted the quality of the evidence in this area. Fourth, as for the use of nimodipine and donepezil in conventional therapy for MCI, some evidence from Chinese clinical articles showed that donepezil and nimodipine were used in the clinical treatment of MCI in some regions of China; but are rarely used in the U.S. and other areas of the world. Fifth, although the included studies were subjected to a full assessment using the Cochrane Collaboration's risk of bias methodology, the overall quality of the trials was not very good. This may be a result of using solitary blindness as a high-risk condition. Due to the particular nature of acupuncture research, single-blind designs are frequently used, which may have increased the possibility of bias in the study findings.

Menopause-Associated Vasomotor Symptoms

The 2015 North American Menopause Society’s position statement on "Non-hormonal management of menopause-associated vasomotor symptoms" (No authors listed) stated that clinicians need to be well informed about the level of evidence available for the wide array of non-hormonal management options currently available to midlife women to help prevent underuse of effective therapies or use of inappropriate or ineffective therapies. It recommended cognitive-behavioral therapy and, to a lesser extent, clinical hypnosis, which have been shown to be effective in reducing VMS. Paroxetine salt is the only non-hormonal medication approved by the Food and Drug Administration (FDA) for the management of VMS, although other selective serotonin reuptake/norepinephrine reuptake inhibitors, gabapentinoids, and clonidine showed evidence of efficacy. It recommended with caution some therapies that may be beneficial for alleviating

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VMS: weight loss, mindfulness-based stress reduction, the S-equol derivatives of soy isoflavones, and stellate ganglion block; but noted that additional studies of these therapies are needed. It did not recommend the following interventions for managing VMS: cooling techniques, avoidance of triggers, exercise, yoga, paced respiration, relaxation, over-the-counter supplements and herbal therapies, acupuncture, calibration of neural oscillations, and chiropractic interventions because there are negative, insufficient, or inconclusive data on these interventions.

Multiple Sclerosis

Branas et al (2000)

I. identified current treatments for fatigue in multiple sclerosis (MS) and their evidence-base, and
II. systematically reviewed the evidence for those treatments that have been investigated in more than 1 rigorous study, in order to determine their effectiveness and cost-effectiveness.

The review was carried out in 2 stages:

I. a formal scoping review (to assess the range of interventions used by people with MS, and
II. a systematic review for treatments that had been identified as promising and that had been investigated in clinical trials (as identified in the scoping review).

A systematic review of research on costs and cost-effectiveness of those interventions identified as promising was also performed. Electronic databases, including Medline and Embase, were searched for the period 1991 to June 1999 (scoping review) and 1966 to December 1999 (systematic review). Reference lists from publications were also searched, and experts were contacted for any additional information not already identified. Interventions identified for the treatment of fatigue in MS

I. Behavioral advice. This is the main element of initial clinical management and no rigorous research of its effectiveness was identified.
II. Drugs (amantadine, pemoline, potassium-channel blockers and antidepressants).
III. Training, rehabilitation and devices (cooling vests and electromagnetic fields).
IV. Alternative therapies (bee venom, cannabis, acupuncture/acupressure and yoga).

Only 2 drugs, amantadine and pemoline, met the criteria for full systematic review. One parallel and 3 cross-over trials were found, involving a total of 236 people with MS. All studies were open to bias. All studies showed a pattern in favor of amantadine compared with placebo, but there is considerable uncertainty about the validity and clinical significance of this finding. This pattern of benefit was considerably undermined when different assumptions were used in the sensitivity analysis; 1 parallel and 1 cross-over trial were found involving a total of 126 people with MS. Both studies were open to bias. There was no overall tendency in favor of pemoline over placebo and an excess of reports of adverse effects with pemoline. The drug costs of amantadine and pemoline are modest (pound 200 and pound 80 per annum, respectively). No economic evaluations were identified in the systematic review, and available data were insufficient to allow modelling of cost-effectiveness in this rapid review. The authors concluded that there is insufficient evidence to allow people with MS, clinicians or policy makers to make informed decisions on the appropriate use of the many treatments on offer. Only amantadine appeared to have some proven ability to alleviate the fatigue in MS, though only a proportion of users will obtain benefit and then only some of these patients will benefit sufficiently to take the drug in the long term. The frequency, severity and impact of fatigue, the poverty of available research, and the absence of any ongoing research, suggest that new research is an urgent priority. People with MS, clinicians and policy makers should work together to ensure that the evidence required is collected as quickly as possible by encouraging involvement in rigorous research. Research should not be restricted to the 2 drugs reviewed in depth in this report. All interventions identified in the scoping review (see above) should be considered, as should basic scientific research into the underlying mechanism of fatigue in MS.

Campbell et al (2016) evaluated the effectiveness of physiotherapy interventions, including exercise therapy, for the rehabilitation of people with progressive MS. Five databases (Cochrane Library, Physiotherapy

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Evidence Database [PEDro], Web of Science Core Collections, Medline, Embase) and reference lists of relevant articles were searched. Randomized experimental trials, including participants with progressive MS and investigating a physiotherapy intervention or an intervention containing a physiotherapy element, were included. Data were independently extracted using a standardized form, and methodological quality was assessed using the PEDro scale. A total of 13 studies (described by 15 articles) were identified and scored between 5 and 9 out of 10 on the PEDro scale; 8 interventions were assessed: exercise therapy, multi-disciplinary rehabilitation, functional electrical stimulation, botulinum toxin type A injections and manual stretches, inspiratory muscle training, therapeutic standing, acupuncture, and body weight-supported treadmill training. All studies, apart from 1, produced positive results in at least 1 outcome measure; however, only 1 article used a power calculation to determine the sample size and because of drop-outs the results were subsequently underpowered. The authors concluded that the findings of this review suggested that physiotherapy may be effective for the rehabilitation of people with progressive MS. However, they stated that further appropriately powered studies are needed.

Furthermore, an UpToDate review on "Symptom management of multiple sclerosis in adults" (Olek et al, 2016) states that "Complementary and Alternative Medicine – To manage their symptoms, patients with MS often employ a variety of complementary and/or alternative treatments such as exercise, meditation, yoga, relaxation techniques, acupuncture, cannabis, massage, dietary modification, vitamins, herbs, and mineral supplements. However, there are few high-quality data regarding the utility of these interventions".

Neuropathic Pain

In a Cochrane review, Ju and associates (2017) evaluated the analgesic efficacy and AEs of acupuncture treatments for chronic neuropathic pain in adults. These investigators searched CENTRAL, Medline, Embase, 4 Chinese databases, ClinicalTrials.gov and World Health Organization (WHO) International Clinical Trials Registry Platform (ICTRP) on February 14, 2017. They also cross-checked the reference lists of included studies; RCTs with treatment duration of 8 weeks or longer comparing acupuncture (either given alone or in combination with other therapies) with sham acupuncture, other active therapies, or treatment as usual, for neuropathic pain in adults were selected for analysis. These researchers searched for studies of acupuncture based on needle insertion and stimulation of somatic tissues for therapeutic purposes, and they excluded other methods of stimulating acupuncture points without needle insertion. They searched for studies of manual acupuncture, electro-acupuncture or other acupuncture techniques used in clinical practice (such as warm needling, fire needling, etc.). These investigators used the standard methodological procedures expected by Cochrane. The primary outcomes were pain intensity and pain relief. The secondary outcomes were any pain-related outcome indicating some improvement, withdrawals, participants experiencing any AE, serious adverse events (SAEs) and QOL. For dichotomous outcomes, these researchers calculated RR with 95 % CI, and for continuous outcomes they calculated the MD with 95 % CI. They also calculated number needed to treat for an additional beneficial outcome (NNTB) where possible. These researchers combined all data using a random-effects model and assessed the quality of evidence using GRADE to generate "Summary of findings" tables. A total of 6 studies involving 462 participants with chronic peripheral neuropathic pain (442 completers (251 male), mean ages 52 to 63 years) were included in this review. The included studies recruited 403 participants from China and 59 from the UK. Most studies included a small sample size (fewer than 50 participants per treatment arm) and all studies were at high risk of bias for blinding of participants and personnel. Most studies had unclear risk of bias for sequence generation (4 out of 6 studies), allocation concealment (5 out of 6) and selective reporting (all included studies). All studies investigated manual acupuncture, and these reviewers did not identify any study comparing acupuncture with treatment as usual, nor any study investigating other acupuncture techniques (such as electro-acupuncture, warm needling, fire needling). One study compared acupuncture with sham acupuncture. The reviewers ware uncertain if there was any difference between the 2 interventions on reducing pain intensity (n = 45; MD -0.4, 95 % CI: -1.83 to 1.03, very low-quality evidence), and neither group achieved "no worse than mild pain" (VAS, 0 to 10) average score was 5.8 and 6.2, respectively in the acupuncture and sham acupuncture groups, where 0 = no pain). There were limited data on QOL, which showed no clear difference between groups. Evidence was not available on pain relief, AEs or other pre-defined secondary outcomes for this comparison; 3 studies compared acupuncture alone versus other therapies (mecobalamin combined with nimodipine, and inositol). Acupuncture may reduce the risk of "no

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clinical response" to pain than other therapies (n = 209; RR 0.25, 95 % CI: 0.12 to 0.51), however, evidence was not available for pain intensity, pain relief, AEs or any of the other secondary outcomes; 2 studies compared acupuncture combined with other active therapies (mecobalamin, and Xiaoke bitong capsule) versus other active therapies used alone. These reviewers found that the acupuncture combination group had a lower VAS score for pain intensity (n = 104; MD -1.02, 95 % CI: -1.09 to -0.95) and improved QOL (n = 104; MD -2.19, 95 % CI: -2.39 to -1.99), than those receiving other therapy alone. However, the average VAS score of the acupuncture and control groups was 3.23 and 4.25, respectively, indicating neither group achieved "no worse than mild pain". Furthermore, this evidence was from a single study with high risk of bias and a very small sample size. There was no evidence on pain relief and the reviewers identified no clear differences between groups on other parameters, including "no clinical response" to pain and withdrawals. There was no evidence on AEs. The overall quality of evidence was very low due to study limitations (high risk of performance, detection, and attrition bias, and high risk of bias confounded by small study size) or imprecision. The reviewers have limited confidence in the effect estimate and the true effect is likely to be substantially different from the estimated effect. The authors concluded that due to the limited data available, there is insufficient evidence to support or refute the use of acupuncture for neuropathic pain in general, or for any specific neuropathic pain condition when compared with sham acupuncture or other active therapies. Moreover, they noted that 5 studies are still ongoing and 7 studies are awaiting classification due to the unclear treatment duration, and the results of these studies may influence the current findings.

Non-Alcoholic Fatty Liver Disease

In a systematic review and meta-analysis of published RCTs, Chen and colleagues (2021) examined the safety and effectiveness of acupuncture treatment (AT) or acupuncture plus conventional medicine (CM) versus CM alone for the treatment of patients with non-alcoholic fatty liver disease (NAFLD). A total of 8 databases were searched independently from inception to April 30, 2020; RCTs were included if they contained reports on the use acupuncture or the use of acupuncture combined with CM and compared with the use of CM. Summary OR and 95 % CIs were used to calculate the overall clinical efficacy. Secondary outcomes, namely aspartate aminotransferase, alanine aminotransferase, total cholesterol, triglyceride, high-density lipoprotein cholesterol, low-density lipoprotein cholesterol, and body mass index (BMI), were calculated by mean difference with 95 % CIs. After the final screening, 8 RCTs with 939 patients were included. This meta-analysis showed that AT was superior to CM in improving overall clinical efficacy (OR = 3.19, 95 % CI: 2.06 to 4.92, $p < 0.00001$). In addition, AT plus CM could significantly improve overall clinical efficacy compared to treatment with CM alone (OR = 5.11, 95 % CI: 2.43 to 10.75, $p < 0.0001$). Moreover, the benefits were also demonstrated in other outcomes, including alanine aminotransferase, aspartate aminotransferase, total cholesterol, triglyceride, high-density lipoprotein cholesterol, and low-density lipoprotein cholesterol indexes. However, AT plus CM could not decrease BMI levels in comparison with CM. The safety profile of acupuncture therapy was satisfactory. Taichong, Zusanli, Fenglong, and Sanyinjiao were major acupoints on NAFLD treatment. The authors concluded that acupuncture may be a safe and effective approach for treatment of NAFLD; however, due to insufficient methodological quality and sample size, large-scale, long-term RCTs with rigorous methodological input are needed to clarify the role of acupuncture in this population.

The authors stated that this systematic review had several drawbacks. First, the quality of the studies was poor because most studies had low methodological quality. Due to the particularity of AT methods, the blind method was difficult to implement clinically. Only 1 study was a multi-center study and used a single-blind method. Only 3 studies used random number tables to allocate participants. All the included studies did not report follow-up visits and the treatment duration ranged from 1 month to 6 months, which were insufficient to examine the safety and long-term effect of acupuncture in the treatment of NAFLD. Furthermore, none of the included studies reported recurrence rate and only 3 trials reported adverse effects, which may have led to uncertain results and could not truly reflect overall trends. All subjects were Chinese in the included studies, this limited geographical distribution and low-grade research were difficult to conduct in future large-sized experiments. There were no studies comparing acupuncture versus no intervention or placebo/sham acupuncture, so the specific effect of acupuncture for NAFLD was unclear. In addition, these investigators found that most of the studies were flawed according to the Cochrane Collaboration risk of bias tool.

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Obesity

In a systematic review and meta-analysis, Wujie et al (2022) examined the effect of acupuncture and acupoint catgut embedding in treatment of abdominal obesity. A total of 10 databases were searched as of August 2022: the English databases PubMed, Embase, Cochrane Library, Web of Science, Wiley, and Scopus and the Chinese databases China National Knowledge Infrastructure Database, China Science and Technology Journal Database, Wanfang, and SinoMed/Chinese Biomedical Literature Database. RCTs of acupuncture and acupoint catgut embedding as the main interventions to treat abdominal obesity were extracted. These researchers imported the citations into EndNote version X9.1 for deduplication, screening, extraction, and integration. The risk of bias in the included RCTs was assessed according to the Cochrane Handbook. RevMan 5.4 software was used to conduct a meta-analysis of RCTs that met the inclusion criteria. A total of 13 RCTs (1,069 patients) were included in this study, and the data of 11 RCTs (966 patients) were include in the meta-analysis. The results showed that acupoint catgut embedding could significantly change the weight and waist circumference of patients with abdominal obesity when compared to sham acupuncture or no treatment (MD = 2.32, 95 % CI: 1.88 to 2.76, $p < 0.000\ 01$); MD = 3.47, 95 % CI: 1.99 to 4.94, $p < 0.000\ 01$). The change in hip circumference after acupuncture was also significant (MD = 0.89, 95 % CI: 0.12 to 1.66, $p = 0.02$]. The authors concluded that this study found that acupuncture and acupoint catgut embedding could effectively treat abdominal obesity; thus, these interventions could be used as clinical supplements and alternative therapies. Moreover, these researchers stated that it will be necessary to improve the clinical study protocols and expand the sample size to further validate the reliability of the results obtained of this study.

The authors stated that this study had several drawbacks. First, it included several reference diagnoses, which were outdated, and new diagnostic criteria have not yet appeared. Second, regarding the selection of control measures, the inconsistent direction of each investigator and the absence of authoritative criteria may be the principal reasons for the high heterogeneity of the analysis. Third, although there were records of adverse reactions, only some studies described the adverse reactions for observation group and control group separately; thus, the safety of acupuncture and catgut embedding could not be objectively examined. Fourth, the courses of treatment included in the studies were relatively short, and most of them were not followed-up. The long-term effects of acupuncture and catgut embedding on abdominal obesity are still unclear. To draw more objective conclusions, large, high-quality, well-designed, multi-center RCTs are needed. Investigators should adopt the correct method of randomization, random sequence generation, and allocation concealment as well as the diagnostic criteria and the intervention measures of the control group need to be more unified and standardized. At the same time, it is necessary to follow-up the patients and examine the long-term effects of the interventions for a more thorough and accurate determination of the value of acupuncture and acupoint catgut embedding in the treatment of abdominal obesity.

Quan et al (2023) stated that the effectiveness of acupuncture in improving the in metabolic outcomes in children and adolescents is unclear. In a meta-analysis, these researchers examined this correlation. They carried out a comprehensive search using multiple databases, including Medline, Cochrane, Embase, Web of Science, Chinese Biomedical Literature Database, China National Knowledge Infrastructure, Chinese Scientific Journal Database, and Wan-fang Data, to identify relevant RCTs published before February 1, 2023. General information and data for the descriptive and quantitative analyses were extracted. A total of 15 RCTs of 1,288 obese/overweight children and teenagers were included. All the trials were carried out in China and South Korea. Regarding quality assessment, no other significant risk of bias was found. The acupuncture groups were more likely to have improved metabolic indicators of obesity/overweight than the control groups, in terms of BMI (SMD = -0.45, 95 % CI: -0.69 to -0.21, $I2 = 71.4$ %), body weight (SMD = -0.48, 95 % CI: -0.92 to -0.05, $I2 = 84.9$ %), and serum leptin (SMD = -0.34, 95 % CI: -0.58 to -0.10, $I2 = 91.8$ %). The subgroup analysis showed that for BMI, the results were consistent regardless of the intervention duration, body acupuncture or auricular acupuncture combined with other interventions. The authors concluded that these findings suggested that acupuncture was effective in improving metabolic outcomes of obese/overweight children and adolescents. Moreover, these researchers stated that as a consequence of the limited number of studies included in this meta-analysis, the results should be interpreted with caution. They stated that there is still a need for high-quality RCTs to successfully complete more evidence-based studies.

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Th authors stated that this study had several drawbacks. First, the number of RCTs involved in this study was small, and the included trials were mostly from China. There were low amounts of original data from other regions; thus, the representativeness of the effectiveness of acupuncture to treat obese/overweight children and adolescents was not strong. Second, most RCTs had low methodological quality because of inadequate allocation concealment and blinding. Third, differences in treatment duration may have affected the final evaluation of effectiveness. Fourth, most studies lacked post-treatment follow-up; therefore , these investigators could not determine the long-term effectiveness of acupuncture, which may add other biases. In addition, the auricular acupuncture method is a non-invasive type of acupuncture that has less stimulation and is less painful. Its effectiveness may be weaker than that of other types of acupuncture; and it was mostly used for children and adolescents. Therefore, relatively speaking, the effect of body acupuncture was better in the treatment of obesity and overweight in children and adolescents. The comparative advantages of different acupuncture intervention modes need to be performed via network meta-analysis.

Obstructive Sleep Apnea

In a systematic review and meta-analysis, Wang and colleagues (2020) examined the safety and efficacy of acupuncture for the treatment of OSA patients with various severities of the disorder. A total of 8 data-bases including PubMed, Cochrane Library, Embase, Web of Science, China National Knowledge Infrastructure (CNKI), Chongqing VIP (CQVIP), Wanfang Data, and Chinese Biomedical Literature Database (CBM) were searched till July 2019; RCTs testing acupuncture in the treatment of OSA were eligible for inclusion. Studies were selected for inclusion, and data were extracted by 2 authors independently. The Cochrane Collaboration's Risk of Bias Assessment Tool and RevMan software (version 5.3) were used to assess the quality of studies and conduct statistical analysis. A total of 9 RCTs with 584 subjects were included. The trials covered acupuncture and electropuncture. Acupuncture caused clinically significant reductions in AHI (MD: -6.18; 95 % CI: -9.58 to -2.78; $Z = 3.56$, $p = 0.0004$) as well as in ESS (MD: -2.84; 95 % CI: -4.80 to -0.16, $Z = 2.09$, $p = 0.04$). AHI was reduced more in the subgroup analysis of moderate OSA patients (MD: -9.44; 95 % CI: -12.44 to -6.45; $Z = 6.18$, $p < 0.00001$) and severe OSA patients (MD: -10.09; 95 % CI: -12.47 to -7.71; $Z = 8.31$, $p < 0.00001$). ESS was also reduced more in the subgroup analysis of moderate OSA patients (MD: -2.40; 95 % CI: -3.63 to -1.17; $Z = 3.83$, $p = 0.0001$) and severe OSA patients (MD: -4.64; 95 % CI: -5.35 to -3.92; $Z = 12.72$, $p < 0.00001$). Besides, acupuncture had a beneficial effect on LSaO2 (MD: 5.29; 95 % CI: 2.61 to 7.97; $Z = 3.86$, $p = 0.0001$). The outcome of AHI and LSaO2 yielded consistent results following sensitivity analysis; however, the direction of the outcome of ESS was reversed; and the quality of evidence was mainly low-to-very low. The authors concluded that acupuncture therapy was effective for OSA patients in reducing AHI and ESS and in improving the LSaO2 of various severities, especially in moderate and severe OSA patients. Moreover, these researchers stated that further high-quality RCTs are needed to confirm the safety and efficacy of acupuncture for OSA patients.

The authors stated that this review had several drawbacks. First, acupuncture treatment for OSA had some bias and heterogeneity. The reasons may be the various interventions of acupuncture. Second, in this review, OSA patients were categorized as mild, moderate, and severe according to AHI before treatment. AHI is an important measure thatis used to diagnose and categorize disease severity of OSA patients. However, there are inherent limitations with using the AHI calculated from 1 night of sleep to categorize disease severity, because the AHI is influenced by many factors and may vary over time and even across consecutive nights. Third, the methodological quality of the included RCTs was generally low (e.g., most of the included studies had a high risk of performance bias).

Occipital Neuralgia

Zheng et al (2022) stated that acupuncture is used to treat patients with occipital neuralgia, which is one of the main causes of occipital pain; however, its effectiveness is conflicting. In a systematic review and meta-analysis, these investigators examined the safety and effectiveness of acupuncture in the treatment of occipital neuralgia. They searched PubMed, Embase, OVID, China National Knowledge Infrastructure, Cochrane Library, Chinese Biomedical Literature Database, Wanfang databases, and Google Scholar until July 2021. A total of 15 studies that examined the effectiveness of acupuncture in the treatment of occipital neuralgia were included. Human-related trials were considered in different languages. The size of the study

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was not considered a limit for its inclusion and the study intervention should focus on comparing the impact of acupuncture in the intervention group compared with the control group. The OR and the MD with 95 % CIs were calculated with a random or fixed-effect model for different subgroup analyses. Publication bias was assessed using the Egger test, while the risk of bias was assessed using the Review manager software. Acupuncture had a significantly higher effective rate of treatment (OR, 5.40; 95 % CI: 2.48 to 11.77, $p < 0.001$) compared to control in the treatment of occipital neuralgia and lower VAS (MD, -2.45; 95 % CI: -2.69 to -2.21, $p < 0.001$). Acupuncture plus medication had a significantly higher effective rate of treatment (OR, 3.96; 95 % CI: 2.10 to 7.47, $p < 0.001$) compared to medication in the treatment of occipital neuralgia. Acupuncture analysis for safety issues showed a significant reduction of AEs compared with the medication group. The authors concluded that acupuncture alone or acupuncture plus medication had a significantly beneficial effect on the effectiveness of treatment, safety and VAS compared to medication in the treatment of occipital neuralgia. Moreover, these researchers stated that high-quality, multi-center studies are needed to validate these findings.

The authors stated that this study had several drawbacks. First, since many of the studies discovered were not included in the systematic review and meta-analysis, there may be a selection bias in this study. But those studies didn't meet the inclusion criteria; thus, these investigators couldn't include them in the meta-analysis. Second, there was also no way to tell if the outcomes had anything to do with the acupuncture sites, the acupuncture type, the age, the gender, or the ethnicity of the patients. Research into the effectiveness of acupuncture for occipital neuralgia relied on data from other studies, which could introduce bias due to missing information. Third, there was a small number of subjects in each of the 15 studies used in the meta-analysis, with the total number of subjects in the included studies being below 100. Fourth, other potential sources of bias were patients' age, gender, compliance, ethnicity, and nutritional status. The pooled effect could be inaccurate due to the presence of unreported research and gaps in data. Fifth, patients were using a wide variety of healthcare providers, therapeutic protocols, medication regimens, and administration timings. The fact that this meta-analysis was conducted on trials with inadequate blinding or few subjects, reduced the weight of the evidence it provided. There may have been a restriction in the methodology because these researchers counted acupuncture points on both the left and right sides of the body as though there was no difference between them. Although acupuncture is used by certain Korean traditional medicine doctors, there has been surprisingly little study of the topic outside of China; thus, the analysis of relevant literature from that country was limited. Hence, it was not possible to provide certain recommendations according to the recruited studies.

Oligoasthenozoospermia

You and colleagues (2020) noted that oligoasthenozoospermia is a common factor leading to male infertility. Acupuncture has been applied for treating male infertility for several thousand years in China, but clinical evidence of its safety and efficacy in treating oligoasthenozoospermia is yet to be established. These researchers systematically examined the evidence on the effect of acupuncture in men with oligoasthenozoospermia. Databases (PubMed, Embase, SINOMED, CNKI, Wanfang database and Cochrane Library) were searched to identify related studies published before June 30, 2019. The Cochrane risk of bias tool and Jadad score was adopted to assess the methodological quality of included studies. A total of 12 RCTs with 1,088 subjects were included in this review. These investigators attempted to perform a meta-analysis, but it was not possible due to considerable clinical heterogeneity among the included studies. According to the narrative analysis, acupuncture or acupuncture combined with another intervention was effective in improving the semen quality based on the included studies. However, this result should be interpreted with caution due to high risk. The methodological quality of most included studies was low. The authors concluded that the current evidence on acupuncture for oligoasthenozoospermia is inadequate to draw a solid conclusion due to the poor methodological quality. These researchers stated that rigorous full-scale RCTs are needed to validate the safety and efficacy of acupuncture in the treatment of oligoasthenozoospermia.

Oral Ulcer

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Yan et al (2022) noted that oral ulcer (OU) is a common oral mucosal disease manifested with obvious pain. In some studies, the effectiveness of acupuncture in OU has been confirmed; however, systematic reviews and meta-analyses for them are lacking. In a systematic review and meta-analysis, these investigators examined the effectiveness of acupuncture in the treatment of OU. They searched the literature from 8 databases from their inception to December 2021; RCTs of acupuncture for the treatment of OU were included. The meta-analysis was performed using Review Manager 5.3 and Stata 16.0. The main outcomes were the effective rate and the recurrence rate, the secondary outcomes were the VAS and the ulcer healing time. A total of 18 studies were =included in the meta-analysis, including 1,422 patients. In meta-analyses, these researchers found that in comparison with Western medicine, acupuncture improved effective rate (OR = 5.40, 95 % CI: 3.40 to 8.58), reduced the ulcer recurrence rate (OR = 0.21, 95 % CI: 0.13 to 0.33), and relieved the ulcer pain (MD = -2.26, 95 % CI: -4.27 to -0.24). Furthermore, compared with Western medicine, acupuncture plus Western medicine improved effective rate (OR = 2.95, 95 % CI: 1.48 to 5.85). Compared with the Chinese medicine, the acupuncture plus Chinese medicine improved the effective rate (OR = 8.26, 95 % CI: 3.61 to 18.88) and relieved the ulcer pain (MD = -1.85, 95 % CI: -2.51 to -1.19). The authors concluded that acupuncture may be more effective than Western medicine in terms of effectiveness, and acupuncture combined with Western or Chinese medicine may have the potential to reduce the recurrence of ulcer and relieve the ulcer pain. However, these researchers stated that due to limited evidence, high-quality, large sample, randomized, double-blind, placebo-controlled, multi-center trials are needed to validate these findings.

The authors stated that this study had several drawbacks. First, the included 18 studies had methodological flaws and were assessed as being of low quality. These researchers speculated for the following reasons. In the included literature, only 2 studies mentioned blinding of subjects, and other studies did not clearly address randomization protocols, blinding methods, and allocation concealment, which may have led to selection, performance, and detection biases. Second, there was significant heterogeneity in the VAS and ulcer healing time as secondary outcome measures. These investigators considered that both of these measures were subjective and easily influenced by the experience of clinicians and reviewers. Simultaneously, the number of clinical studies was less in 2 outcome measures. There were only 5 studies in the outcome measure of the VAS and 4 studies in the outcome measure of ulcer healing time, some results still need further confirmation. Third, the authors considered publication bias in the article. They used Egger's test to detect the effective rate and found that there was publication bias in the acupuncture versus Western medicine group. The specific reasons need to be further analyzed. Above all, the operation of acupuncture was subjective; and some treatment standards were difficult to be unified. This study only focused on the stimulation method of acupuncture and did not analyze the differences in acupuncture point selection, manipulation depth, and intervention time. Fourth, the literature search strategy only searched 4 English databases and 4 Chinese databases, grey literature was not taken into consideration. Then, all the studies were published in China and there was potential publication bias in the included studies. Fifth, a subgroup analysis of ulcer classification would be more relevant in clinical practice, these researchers did not carry out a subgroup analysis of oral ulcer classification due to a small number of cases and incomplete data.

Osteoporosis

Luo and co-workers (2018) stated that warm needle acupuncture (WNA) is commonly used in primary osteoporosis (OP) management in China. The evidence of its effectiveness needs to be systematically reviewed. In a meta-analysis, these investigators examined if using WNA alone or combined with conventional medicine benefits primary OP. PubMed, Embase, the Cochrane Central Register, Medline, China National Knowledge Infrastructure, Wanfang and VIP databases were searched from their inception through June 30, 2016; RCTs applying WNA independently or as an adjunct to conventional medicine, compared with conventional medicine alone, were included. Primary outcomes were bone mineral density (BMD) of the lumbar vertebrae, femoral neck, Ward's triangle and greater trochanter. The secondary outcome was chronic pain measured by VAS score. Meta-analysis was conducted using RevMan V.5.3 software. A total of 9 RCTs involving 572 participants were included. When WNA was used as an adjunct to conventional medicine, meta-analysis revealed a statistical difference in favor of increasing BMD of the lumbar vertebrae (MD = 0.06, 95 % CI: 0.03 to 0.08, $p < 0.001$); WNA increased BMD of the femoral neck (MD 0.14, 95 % CI: 0.08 to 0.21, $p < 0.001$) and greater trochanter (MD 0.09, 95 % CI: 0.04 to 0.15, p <

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0.001) when used alone, and additionally decreased VAS scores (MD = -1.10, 95 % CI: -1.14 to -1.06, p < 0.001) when used as an adjunct to conventional medicine. However, the safety of WNA was not specifically reported. The authors concluded that WNA may have beneficial effects on BMD and VAS scores of patients with primary OP. However, all included trials were at high risk of bias and of low quality. They stated that further rigorous studies are needed to determine the effectiveness of WNA for primary OP treatment.

The authors stated that this study had several drawbacks. First, all of the included trials inadequately reported randomization and none of them reported the details of random allocation or allocation concealment, which may have led to selection bias. Second, none of the included trials mentioned whether a blinding method was used, although it could be assumed there was no masking given the absence of a sham acupuncture control capable of blinding participants. Third, all of the trials used a small sample size and had significant methodological limitations. Fourth, only 1 of them mentioned AEs. Thus, these researchers could not draw a definitive conclusion as to whether WNA might be a promising (i.e., safe and effective) method of attenuating osteoporotic syndromes such as low BMD and related chronic pain.

Osteoporotic Vertebral Compression Fracture-Induced Pain

Li and associates (2021) stated that osteoporotic vertebral compression fractures (OVCFs) are common health issues in the elderly; they cause chronic pain in over 1/3 of patients. In a systematic review and meta-analysis, these researchers examined the safety and effectiveness of acupuncture for alleviating pain caused by OVCFs. They searched 8 electronic databases for publications from inception to March 30, 2021. Eligible studies were RCTs that evaluated the effect of acupuncture for the treatment of OVCFs. Two investigators evaluated literature quality and extracted data independently. RevMan V.5.4.1 was used for data analyses, with pooled risk estimates presented as MD or RR along with corresponding 95 % CIs, as appropriate. A total of 14 RCTs involving 1,130 patients were included in this meta-analysis. Compared with the control group, acupuncture showed a greater benefit on pain reduction caused by OVCFs (1 week: MD = -1.26, 95 % CI: -1.82 to -0.70); 1 month: MD = -1.63, 95 % CI: -1.82 to -1.43); 6 months: MD = -1.13, 95 % CI: -1.55 to -0.70)). Acupuncture treatment was also associated with fewer AEs, lower ODI index, and higher bone density than the control group (safety: (RR: 0.30, 95 % CI: 0.12 to 0.75); ODI: MD = -3.19, 95 % CI: -5.20 to -1.19); bone density: MD = 0.15, 95 % CI: 0.05 to 0.26). The GRADE quality of these results was assessed as low or very low. The authors concluded that compared with the control treatment, acupuncture was safer and more effective in relieving the pain caused by OVCF and made a greater improvement on patient's ODI score and bone density. Moreover, these researchers stated that given the low level of the study evidence, future high-quality studies are needed to validate these findings.

The authors stated that this study had several drawbacks. First, all the included studies were single-center RCTs conducted in China; therefore, the results may not be generalizable to patients from other regions or countries. Second, except that the study itself was difficult to implement blinding, all studies lacked blinding of allocation methods and evaluators, and no studies were registered in advance before trial conducted. These might have largely reduced the quality of evidence generated from this analysis. Third, the heterogeneity between studies was large. Subgroup analysis and sensitivity analysis showed that acupuncture combined with different treatment methods, retention time of acupuncture, and frequency of acupuncture were possible sources of heterogeneity. These investigators only analyzed the heterogeneity at 1 point in time and did not analyze the heterogeneity in other time-points because of the small number of studies. Fourth, the acupuncture points used in the studies included in this meta-analysis were different. This might have led to the differences in effectiveness of acupuncture between studies and contributed to the between-study heterogeneity. Finally, the level of evidence of these findings was from low to very low.

Parkinson's Disease-Related Fatigue

Kluger et al (2016) noted that fatigue is a common and debilitating non-motor symptom of Parkinson's disease (PD). Because preliminary evidence suggested that acupuncture improves fatigue in other conditions, these researchers examined its effectiveness in the treatment of PD-related fatigue. A total of 94 PD patients with moderate-to-high fatigue were randomized to receive 6 weeks of bi-weekly real or sham acupuncture. The primary outcome was change on the Modified Fatigue Impact Scale at 6 weeks; secondary

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outcomes included sleep, mood, QOL, and maintenance of benefits at 12 weeks. Both groups showed significant improvements in fatigue at 6 and 12 weeks, but with no significant between-group differences. Improvements from baseline in mood, sleep, and QOL were noted without between-group differences. Overall, 63 % of patients reported noticeable improvements in their fatigue; no serious AEs were observed. The authors concluded that acupuncture may improve PD-related fatigue, but real acupuncture offered no greater benefit than sham treatments. They stated that PD-related fatigue should be added to the growing list of conditions that acupuncture helps primarily through non-specific or placebo effects

Corbin et al (2016) described:

I. considerations for designing a double-blinded, randomized, placebo-controlled clinical trial of acupuncture for fatigue in PD, and
II. their experience in the implementation and early conduct of this trial.

Relevant literature was also reviewed to provide guidance for other researchers seeking to perform clinical research relevant to PD and related disorders. Trial design should be driven by a well-defined research question and sufficient detail to meet Standards for Reporting Interventions in Clinical Trials of Acupuncture criteria when a trial is complete. Important items for review include: randomization and blinding; recruitment and participant selection; sham methodology choice; staff training; and practical implementation of study procedures. Sample forms used for the current authors' trial are shared. The authors concluded that high-quality clinical trials of acupuncture can provide valuable information for clinicians, patients, and policymakers. Acupuncture trials differ in critical ways from pharmaceutical trials and might require additional considerations regarding design and implementation. Adequate preparation for the unique challenges of acupuncture studies can improve trial implementation, design, efficiency, and impact.

Plantar Fasciitis

Thiagarajah (2017) stated that plantar fasciitis is a commonly seen out-patient condition that has numerous therapeutic modalities of varying degrees of effectiveness. In a systematic review, these investigators examined the effectiveness of acupuncture in reducing pain due to plantar fasciitis. Online literature searches on the PubMed and Cochrane Library databases were done for studies on the use of acupuncture for pain due to plantar fasciitis. Studies designed as RCTs that compared acupuncture with standard treatments or had real versus sham acupuncture arms were selected. The Delphi List was used to evaluate the methodological quality of the studies retrieved. A total of 3 studies that compared acupuncture with standard treatment and 1 study on real versus sham acupuncture were found. These showed that acupuncture significantly reduced pain levels in patients with plantar fasciitis, as measured on the VAS and the Plantar Fasciitis Pain/Disability Scale. These benefits were noted between 4 and 8 weeks of treatment, with no further significant reduction in pain beyond this duration; AEs were found to be minimal. The authors concluded that although acupuncture may reduce plantar fasciitis pain in the short-term, there is insufficient evidence for a definitive conclusion regarding its long-term effectiveness. They stated that further research is needed to strengthen its acceptance among healthcare providers.

Peptic Ulcer

Tian and associates (2017) evaluated the clinical efficacy of acupuncture for peptic ulcer; RCTs of acupuncture for peptic ulcer were searched from China National Knowledge Infrastructure (CNKI), WanFang Database, Chinese Scientific and Technological Journals (VIP), China Biomedicine (CBM), PubMed and the Cochrane Library from the establishment time of databases to September, 2016. Data extraction and quality evaluation were implemented for the literature which met the inclusive criteria. The RevMan 5.3 software was used to make meta-analysis. A total of 16 papers including 1,570 patients of peptic ulcer were included. The results of meta-analysis showed that there was no statistical significance between acupuncture and western medicine in the effective rate, the healing rate of ulcer area and the H. pylori (HP) negative rate (all $p > 0.05$); the recurrence rate of acupuncture was significantly lower than that of western medicine [RR = 0.35, 95 % CI: 0.14 to 0.84), $p < 0.05$]. Acupuncture plus western medicine was significantly different from simple western medicine in the effective rate, the healing rate of ulcer area and

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the recurrence rate [RR = 1.20, 95 % CI: 1.04 to 1.38, p = 0.01; RR = 1.29, 95 % CI: 1.06 to 1.58), p = 0.01; RR = 0.27, 95 % CI: 0.16 to 0.45), p < 0.00001]. The analysis of evidence grade (GRADE) presented that the healing rate of ulcer area and the HP negative rate of acupuncture were "low grade", and others were "extremely low grade". The authors concluded that acupuncture combined with western medicine had some advantages for peptic ulcer compared with the conventional western medicine; these findings need to be confirmed because of the lower evidence grade. They stated that larger RCTs with high quality are highly recommended.

Post-Operative Ileus

In a systematic review and meta-analysis, Cheong et al (2016) examined the effectiveness of acupuncture and common acupoint selection for post-operative ileus (POI). Randomized controlled trials comparing acupuncture and non-acupuncture treatment were identified from the databases PubMed, Cochrane, EBSCO (Academic Source Premier and Medline), Ovid (including Evidence-Based Medicine Reviews), China National Knowledge Infrastructure, and Wanfang Data. The data from eligible studies were extracted and a meta-analysis performed using a fixed-effects model. Results were expressed as RR for dichotomous data, and 95 % CI were calculated. Each trial was evaluated using the CONSORT (Consolidated Standards of Reporting Trials) and STRICTA (STandards for Reporting Interventions in Controlled Trials of Acupuncture) guideline. The quality of the study was assessed using the GRADE approach. Of the 69 studies screened, 8 RCTs were included for review. Among these, 4 RCTs (with a total of 123 patients in the intervention groups and 124 patients in the control groups) met the criteria for meta-analysis. The meta-analysis results indicated that acupuncture combined with usual care showed a significantly higher total effective rate than the control condition (usual care) (RR 1.09, 95 % CI: 1.01 to 1.18; p = 0.02). Zusanli (ST 36) and Shangjuxu (ST 37) were the most common acupoints selected. However, the quality of the studies was generally low, as they did not emphasize the use of blinding. The authors concluded that the results suggested that acupuncture might be effective in improving POI; however, a definite conclusion could not be drawn because of the low quality of trials. They stated that further large-scale, high-quality RCTs are needed to validate these findings and to develop a standardized method of treatment.

Post-Prandial Distress Syndrome

Yang and colleagues (2020) noted that post-prandial distress syndrome (PDS) is the most common subtype of functional dyspepsia. Acupuncture is commonly used to treat PDS; however, its effect is uncertain because of the poor quality of prior studies. In a randomized, 2-group, multi-center study, these researchers examined the effectiveness of acupuncture versus sham acupuncture in patients with PDS. Subjects were Chinese patients aged 18 to 65 years meeting Rome IV criteria for PDS; they were given 12 sessions of acupuncture or sham acupuncture over 4 weeks. The 2 primary outcomes were the response rate based on overall treatment effect and the elimination rate of all 3 cardinal symptoms: post-prandial fullness, upper abdominal bloating, and early satiation after 4 weeks of treatment. Subjects were followed until week 16. Among the 278 randomly assigned subjects, 228 (82 %) completed outcome measurements at week 16. The estimated response rate from generalized linear mixed models at week 4 was 83.0 % in the acupuncture group versus 51.6 % in the sham acupuncture group (difference of 31.4 percentage points [95 % CI: 20.3 to 42.5 percentage points]; p < 0.001). The estimated elimination rate of all 3 cardinal symptoms was 27.8 % in the acupuncture group versus 17.3 % in the sham acupuncture group (difference of 10.5 percentage points [95 % CI: 0.08 to 20.9 percentage points]; p = 0.034). The effectiveness of acupuncture was maintained during the 12-week post-treatment follow-up. There were no serious AEs. The authors concluded that among patients with PDS, acupuncture resulted in increased response rate and elimination rate of all 3 cardinal symptoms compared with sham acupuncture, with sustained effectiveness over 12 weeks in patients who received thrice-weekly acupuncture for 4 weeks. The main drawbacks of this study were lack of objective outcomes and daily measurement, high drop-out rate, and inability to blind acupuncturists.

Post-Stroke Hiccup

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Wang et al (2023b) stated that central hiccups following a stroke are a frequent complication, exerting adverse effects on both the stroke condition and the patient's daily life. Existing treatments exhibit limited effectiveness and significant side effects. Acupuncture has been studied as a supplementary intervention in clinical practice. In a systematic review and meta-analysis, these investigators examined the effectiveness of acupuncture for post-stroke hiccups. They identified RCTs addressing post-stroke hiccups treatment; comprehensive searches were conducted across PubMed, the Cochrane Library, EMBASE, Web of Science, Chinese Biological Medical (CBM), Wanfang Database, and China Science and Technology Journal (VIP). Furthermore, these researchers scrutinized ClinicalTrials.gov and the Chinese Clinical Trial Registry. Employing Cochrane Handbook 5.1.0 and Review Manager 5.4 software, 3 authors independently reviewed literature, extracted data, and evaluated study quality. Data analysis was carried out using Stata 16.0 and Review Manager 5.4. A total of 18 studies were included in the analysis. In comparison to standard treatment, acupuncture showed a significant enhancement in treatment effectiveness (RR: 1.27, 95 % CI: 1.21 to 1.33; $p < 0.00001$). Notably, Hiccup Symptom Score displayed a considerable decrease (WMD: -1.28, 95 % CI: -1.64 to -0.93; $p < 0.00001$), concurrent with a noteworthy improvement in the QOL (WMD: 8.470, 95 % CI: 7.323 to 9.617; $p < 0.00001$). In addition, the incidence of adverse reactions decreased (RR: 0.45, 95 % CI: 0.16 to 1.25; $p = 0.13$), and there was a significant reduction in SAS (WMD: -7.23, 95 % CI: -8.47 to -5.99; $p < 0.00001$). The authors concluded that these findings suggested that acupuncture could prove effective in the treatment of post-stroke hiccup; however, due to concerns regarding the quality and size of the included studies, higher-quality RCTs are needed to validate the effectiveness of acupuncture in treating post-stroke hiccup.

Post-Stroke Shoulder Pain

Lee and Lim (2016) evaluated the evidence on the effectiveness of acupuncture in relieving post-stroke shoulder pain. A total of 7 databases were searched without language restrictions. All RCTs that evaluated the effects of acupuncture for post-stroke shoulder pain compared with controls were included. Assessments were performed primarily with the VAS, Fugl-Meyer Assessment (FMA), and effective rates. In all, 188 potentially relevant articles were identified; 12 were RCTs that met inclusion criteria. Meta-analysis showed that acupuncture combined with rehabilitation treatment appeared to be more effective than rehabilitation treatment alone for post-stroke shoulder pain, as assessed by VAS (WMD, 1.87; 95 % CI: 1.20 to 2.54; $p < 0.001$); FMA (WMD, 8.70; 95 % CI: 6.58 to 10.82; $p < 0.001$); and effective rate (RR, 1.31; 95 % CI: 1.18 to 1.47; $p < 0.001$). The authors concluded that although there is some evidence for an effect of acupuncture on post-stroke shoulder pain, the results are inconclusive. They stated that further studies with more subjects and a rigorous study design are needed to confirm the role of acupuncture in the treatment of post-stroke shoulder pain.

Premature Ejaculation

Zhao and colleagues (2018) stated that premature ejaculation is a common sexual dysfunction disease in adult males. It can be divided into primary and secondary premature ejaculation. Acupuncture is widely used in the treatment of premature ejaculation in China. There are many clinical trials confirmed that acupuncture can prolong the ejaculation latency in the vagina. These investigators carried out a meta-analysis to evaluate the safety and efficacy of acupuncture for premature ejaculation. They would systematically search all RCTs by electronic and manual search, until June 31, 2018. Electronic retrieval of the database included Medline, Embase, the Cumulative Index to Nursing and Allied Health Literature, Allied and Complementary Medicine Database, the Cochrane Library, the Chinese BioMedical Literature Database, the China National Knowledge Infrastructure (CNKI), the China Science and Technology Journal database (VIP), and the Wanfang database. Manual search would retrieve gray literature, including unpublished conference articles. The primary outcomes include the intravaginal ejaculatory latency time (IELT). At the same time, premature ejaculation diagnostic tool (PEDT), Arabic index of premature ejaculation (AIPE), index of premature ejaculation (IPE) would be the secondary outcomes. Two reviewers would independently read the articles, extract the data information, and give the assessment of risk of bias. Data analysis would use the special software like RevMan (version 5.3) and EndNote X7. The authors concluded that this systematic review would evaluate the safety and efficacy of acupuncture for premature ejaculation.

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Furthermore, an UpToDate review on "Treatment of male sexual dysfunction" (Khera and Cunningham, 2018) does not mention acupuncture as a therapeutic option.

Leisegang et al (2023) noted that ejaculatory dysfunction (EjD) is a common male sexual disorder that includes premature ejaculation (PE), delayed ejaculation, retrograde ejaculation, and anejaculation. Although psychological treatments and pharmacotherapies are available, traditional, complementary, and alternative medicine (TCAM) is reportedly used. However, the evidence for TCAM in EjD remains unclear. In a systematic review, these investigators examined the use of TCAM for the treatment of patients with EjD. They carried out a systematic review of the literature following the PRISMA guidelines by searching Scopus and PubMed databases. Controlled clinical trials examining a cohort of men diagnosed primarily with EjD and undergoing any TCAM intervention compared to any comparison group were included. Quality of the studies was assessed using the Cochrane Risk of Bias tool for RCTs. Following article screening, a total of 22 studies were included. Of these, 21 examined TCAM in PE, and only 1 examined TCAM in retrograde ejaculation. Different TCAM categories included studies that examined lifestyle, exercise and/or physical activities (n = 7); herbal medicine supplements (n = 5); topical herbal applications (n = 4); acupuncture or electroacupuncture (n = 3); vitamin, mineral and/or nutraceutical supplements (n = 1); hyaluronic acid (HA) penile injection (n = 1); and music therapy (n = 1). Only 31.8 % (n = 7) of the included studies were found to have a low risk of bias. The available studies were widely heterogenous in the TCAM intervention investigated and comparison groups used; however, the included studies generally showed improved outcomes intra-group and when compared to placebo. The authors concluded that different TCAM interventions may have an important role especially in the management of PE; however, these investigators stated that further investigations with well-designed and controlled studies using standardized interventions are needed.

As this review focused on a broad definition of TCAM, the interventions included for analysis are heterogeneous. The review identified a small number of studies included for EjD (n = 22), with the majority of these investigating PE. Therefore, the effects of TCAM in EjD remain unclear and under-investigated. The eligibility criteria did not limit to a specific outcome related to EjD. However, most of the included studies investigating PE used the IELT as a standardized measure, with or without a validated self-reported questionnaire, improving the validity of the included studies. The included studies focused on a cohort of patients with primary EjD diagnosis, with most studies on PE. However, there is some variation in the definition of the cohort for PE and inclusion criteria for the different studies. Therefore, the participants in the included PE studies were varied across different forms such as psychogenic organic, lifelong and/or acquired. Although English, Italian and Spanish full-text articles would be included, there is a limitation on other language publications.

Premenstrual Syndrome / Premenstrual Dysphoric Disorder

Ina Cochrane review, Armour and colleagues (2018) evaluated the safety and effectiveness of acupuncture or acupressure for women with premenstrual syndrome (PMS), or premenstrual dysphoric disorder (PMDD). These investigators searched the Cochrane Gynaecology and Fertility Specialised Register, Cochrane Central Register of Studies Online (CENTRAL CRSO), Medline, Embase, AMED, PsycINFO, CINAHL (from inception to September 21, 2017), 2 clinical trial databases (from their inception to September 21, 2017), and 4 electronic databases in China (from their inception to October 15, 2017): Chinese Biomedical Literature database (CBM), China National Knowledge Infrastructure (CNKI), VIP information/ Chinese Scientific Journals database and WANFANG. Reference lists from included articles were hand-searched. These researchers included studies if they randomized women with PMS and associated disorders (PMDD and late luteal phase dysphoric disorder/LPDD) to receive acupuncture or acupressure versus sham, usual care/waiting-list control or pharmaceutical interventions mentioned by the International Society for Premenstrual Disorders (ISPMD). If acupuncture or acupressure were combined with another therapy, these studies were also included where the additional therapy was the same in both groups. Cross-over studies were eligible for inclusion, but only data from the first phase could be used. Two review authors independently selected the studies, assessed eligible studies for risk of bias, and extracted data from each study. Study authors were contacted for missing information. The quality of the evidence was assessed using GRADE. The primary outcomes were overall premenstrual symptoms and AEs; secondary outcomes

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included specific PMS symptoms, response rate and QOL. A total of 5 trials (277 women) were included in this review. No trials compared acupuncture or acupressure versus other active treatments. The number of treatment sessions ranged from 7 to 28. The quality of the evidence ranged from low to very low quality, the main limitations being imprecision due to small sample sizes and risk of bias related to detection bias and selective reporting. Acupuncture may provide a greater reduction in mood-related PMS symptoms (MD -9.03, 95 % CI: -10.71 to -7.35, 1 RCT, n = 67, low-quality evidence) and in physical PMS symptoms (MD -9.11, 95 % CI: -10.82 to -7.40, 1 RCT, n = 67, low-quality evidence) than sham acupuncture, as measured by the Daily Record of Severity of Problems scale (DRSP). The evidence suggested that if women have a mood score of 51.91 points with sham acupuncture, their score with acupuncture would be between 10.71 and 7.35 points lower and if women had a physical score of 46.11 points, their score with acupuncture would be between 10.82 and 7.4 points lower. There was insufficient evidence to determine whether there was any difference between the groups in the rate of AEs (RR 1.74, 95 % CI: 0.39 to 7.76, 3 RCTs, n = 167, I2 = 0 %, very low-quality evidence). Specific PMS symptoms were not reported. There may be little or no difference between the groups in response rates. Use of a fixed-effect model suggested a higher response rate in the acupuncture group than in the sham group (RR 2.59, 95 % CI: 1.71 to 3.92; participants = 100; studies = 2; I2 = 82 %), but owing to the high heterogeneity the authors tested the effect of using a random-effects model, which provided no clear evidence of benefit for acupuncture (RR 4.22, 95 % CI: 0.45 to 39.88, 2 RCTs, n = 100, I2 = 82 %, very low-quality evidence). Acupuncture may improve QOL (measured by the WHOQOL-BREF) compared to sham (MD 2.85, 95 % CI: 1.47 to 4.23, 1 RCT, n = 67, low-quality evidence). Due to the very low quality of the evidence, these researchers were uncertain whether acupuncture reduces PMS symptoms compared to a no treatment control (MD -13.60, 95 % CI: -15.70 to -11.50, 1 RCT, n = 14). No AEs were reported in either group. No data were available on specific PMS symptoms, response rate or QOL outcomes. These investigators found low-quality evidence that acupressure may reduce the number of women with moderate-to-severe PMS symptoms at the end of the trial compared to sham acupressure (RR 0.64 95 % CI: 0.52 to 0.79, 1 RCT, n = 90, low-quality evidence). The evidence suggested that if 97 women out of 100 in the sham acupressure group had moderate-to-severe PMS symptoms, the number of women in the acupressure group with moderate-to-severe symptoms would be 50 to 76 women. Acupressure may improve both physical (MD 24.3, 95 % CI: 17.18 to 31.42, 1 RCT, n = 90, low-quality evidence) and mental (MD 17.17, 95 % CI: 13.08 to 21.26, 1 RCT, n = 90, low-quality evidence) QOL. No data were available on AEs, specific symptoms or response rates. The authors concluded that the limited evidence available suggested that acupuncture and acupressure may improve both physical and psychological symptoms of PMS when compared to a sham control. There was insufficient evidence to determine whether there was a difference between the groups in rates of AEs. There was no evidence comparing acupuncture or acupressure versus current ISPMD recommended treatments for PMS such as selective serotonin reuptake inhibitors (SSRIs). They stated that further research is needed, using validated outcome measures for PMS, adequate blinding and suitable comparator groups reflecting current best practice.

Furthermore, an UpToDate review on "Treatment of premenstrual syndrome and premenstrual dysphoric disorder" (Casper and Yonkers, 2018) does not mention acupuncture as a therapeutic option.

Renal Colic Caused by Urinary Calculi

Qu et al (2022) stated that acute renal colic caused by urinary calculi has a considerable impact on the QOL. Pain relief is the primary objective in the management of patients with acute renal colic caused by urinary calculi. Currently, there is no systematic evaluation of the safety and effectiveness of manual acupuncture in the treatment of acute renal colic caused by urinary calculi in adults. In a systematic review and meta-analysis, these investigators examined the safety and effectiveness of manual acupuncture in the treatment of acute renal colic caused by urinary calculi in adults. Databases of PubMed, Embase, Cochrane Library, China National Knowledge Infrastructure (CNKI), Wanfang Medical, VIP Database for Chinese Technical Periodicals (VIP), and China Biomedical Literature (SinoMed) were searched for literature and other randomized controlled registration platforms. They searched to identify the relevant RCTs from the establishment of the database to February 9, 2022. Only RCTs of manual acupuncture as the therapy for acute renal colic caused by urinary calculi in adults were included, whether or not the blind method was used. The patients were adults diagnosed with urinary calculi and renal colic. The control group was treated with commonly used analgesics and antispasmodics. The experimental group was treated with acupuncture

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as a monotherapy or as an adjuvant therapy (manual acupuncture combined with analgesics and antispasmodics). Two review authors independently assessed titles and abstracts for relevance and extracted data on study design, participants, interventions, and outcomes from potentially relevant articles. Cochrane risk bias assessment tool was used to evaluate the quality of the included study, and RevMan5.4 software was used for meta-analysis. The primary outcomes were response rate and time duration before pain remission. Secondary outcomes were the time of complete pain relief, pain variation, need for rescue analgesia, and AEs. Out of 1,123 records identified, 15 were found to be of relevance to this study, and 1,210 participants were included in the meta-analysis. The meta-analysis of the results showed that, in terms of response rate, compared with the control group, acupuncture as a monotherapy appeared to have a slight advantage (RR = 1.10 (95 % CI: 1.03 to 1.18), I2 = 28 %, p = 0.004), while acupuncture as an adjuvant therapy had no advantage (RR = 1.06 (95 % CI: 0.95 to 1.20), I2 = 77 %, p = 0.30). In terms of duration before pain relief, acupuncture as a monotherapy had an advantage over the control group (MD = -10.28 (95 % CI: -14.40 to -6.17), I2 = 93 %, p < 0.00001). Acupuncture as a monotherapy was similar to positive medication in terms of complete pain relief (MD = -7.13 (95 % CI: -20.19 to 5.94), I2 = 95 %, p = 0.28). Pain variation: VAS scores at 10 mins, acupuncture as a monotherapy (MD = -2.47 (95 % CI: -3.40 to -1.53), I2 = 84 %, p < 0.00001) or as an adjuvant therapy (MD = -3.38 (95 % CI: -4.33 to -2.43), I2 = 60 %, p < 0.00001) was better than the control group. VAS scores at 30 mins, compared with the control group, there was no difference between acupuncture as a monotherapy (MD = -0.27 (95 % CI: -1.43 to 0.88), I2 = 88 %, p = 0.64) and acupuncture as an adjuvant therapy (MD = -1.17 (95 % CI: -3.15 to 0.81), I2 = 96 %, p = 0.25). VAS scores at 60 mins, compared with the control group, there was no difference in the acupuncture as a monotherapy (MD = 0.58 (95 % CI: -0.28 to 1.45), I2 = 77 %, p = 0.19), while acupuncture as an adjuvant therapy was better (MD = -1.22 (95 % CI: -1.93 to -0.51), I2 = 72 %, p = 0.0007). VAS scores at 120 mins, there was no difference in acupuncture as a monotherapy compared to the control group (MD = -0.24 (95 % CI:- 1.22 to 0.75), I2 = 0, p = 0.64). One study reported on rescue analgesia. Fewer AEs occurred in the experimental group compared to the control group. The authors concluded that in the course of manual acupuncture treatment of acute renal colic caused by urinary calculi in adults, available evidence suggested that manual acupuncture was as effective as positive treatment drugs, either as a monotherapy or as an adjunctive therapy, with the advantage of acupuncture being its rapid onset of action. However, these researchers stated that the number of existing clinical studies was small, and the quality of evidence was generally low; thus, it is recommended to use it with caution. These investigators stated that in order to further verify the above conclusions, more high-quality clinical RCTs clinical trials with large samples, multi-center design, and select internationally recognized outcome indicators need to be carried out.

The authors stated that this study had several drawbacks. First, the meta-analysis reported here combined data across studies in order to estimate treatment effects with more precision than is possible in a single study. Thus, the patient population, acupuncture as a monotherapy or acupuncture as an adjuvant therapy regimen, and the outcome definitions were not the same across studies. Second, most of the included studies were Chinese with a relatively small sample size. The quality of research was uneven. Only 3 studies reported comparatively complete details of needling. In some studies, the key information for judging the risk of bias was often not clearly explained. For example, most studies did not explicitly state that data analysis followed the intention-to-treat principle, which may have led to the over-estimation of treatment effects in these trials. Third, the evidence for these findings was low. Some literatures were too old, and the selected primary outcome indicators were too subjective and lacked objectivity. Fourth, the included literature comparison methods were not uniform enough, which may be one of the reasons for the heterogeneity.

Restless Leg Syndrome

In a systematic review and meta-analysis, Huang and colleagues (2021) examined the effectiveness of acupuncture in the management of restless leg syndrome (RLS). These researchers carried out a comprehensive literature survey in electronic databases to identify studies examining the effectiveness of acupuncture in the treatment of RLS. Cure, marked effect, effective, ineffective, and total effective rates of individual studies were pooled to achieve their respective overall estimates and a meta-analysis of mean change from baseline in International Restless Legs Syndrome Rating Scale (IRLSRS) was performed. A total of 18 studies were included in this meta-analysis, among which 640 RLS patients were treated with

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acupuncture alone or combined with other therapies (acupuncture group) and 447 RLS patients were treated with non-acupuncture therapies (control group). Cure, marked effect, and effective rates were 47.8 % [95 % CI: 38.3 % to 57.3 %], 27.4 % (95 % CI: 20.3 % to 34.8 %), and 24.2 % (95 % CI: 16.9 % to 31.5 %) in patients treated with acupuncture either alone or combination with other treatments, and 21.7 % (95 % CI: 20.7 % to 22.7 %), 28.0 % (95 % CI: 20.0 % to 32.9 %), and 22.3 % (95 % CI: 17.0 % to 27.6%) in patients treated with non-acupuncture therapies, respectively. The ineffective rate was 4.7 % (95 % CI: 4.3 % to 5.0 %) in the acupuncture group; and 32.9 % (95 % CI: 22.2 % to 43.7 %) in non-acupuncture groups. IRLSRS scores improved significantly after acupuncture treatment [mean change from baseline -9.45 (95 % CI: -18.42 to -0.49); p = 0.04]. The authors concluded that although the overall quality of the included studies was low, the results of this meta-analysis suggested that acupuncture was an effective therapeutic option for RLS. Moreover, these researchers stated that RCTs especially those with better designs and suitable controls with a sham-acupuncture arm are needed to confirm these findings.

The authors stated that among the major drawbacks of this trial, less availability of non-Chinese literature was an important consideration. These investigators observed high statistical heterogeneity in the pooled analysis that could be attributed to the use of many types of controls and many therapeutic combinations with acupuncture. To test sources of heterogeneity, these researchers conducted meta-regression analyses but neither study sample size nor follow-up duration, or year of study publication was significantly associated with total effective rate. Definitions of effectiveness rates were not given by most of the included studies and there may have been some between-studies deviation in measuring the actual effect of therapy.

Rheumatoid Arthritis

Lu and colleagues (2022) noted that one new type of acupuncture and related techniques (ACNRT) is increasingly used by rheumatoid arthritis (RA) patients to control their disease and improve their QOL; however, the effectiveness of using ACNRT in combination with Western medicine (WM) for this purpose remains unknown. These investigators searched for RCTs of ACNRT and WM treatments for RA from January 1, 2000 to January 31, 2021 using databases PubMed, Embase, Medline, and the Cochrane Central Register of Controlled Trials, as well as 3 Chinese databases: China National Knowledge Infrastructure, Wanfang Data, and Airiti Library. The primary outcomes were inflammatory markers including C reactive protein (CRP), erythrocyte sedimentation rate (ESR) and rheumatoid factor (RF). The secondary outcomes were clinical characteristics including pain VAS score; Disease Activity Score (DAS-28); swollen-joints count (SJC); tender-joints count (TJC); morning stiffness (MS); and the results of a health assessment questionnaire (HAQ). The 3 types of ACNRT used in the focal trials were acupuncture, moxibustion, and electro-acupuncture. Two qualified researchers extracted data from these trials' results and independently examined their risk of bias. Statistical analyses were carried out using Comprehensive Meta-Analysis V3 software. A total of 12 RCTs with 874 patients met the inclusion criteria. As compared with the patients who received WM treatment alone, those who were given integrated ACNRT/WM treatment showed greater reductions in CRP (weighted mean difference [WMD]: -6.299; 95 % CI: -9.082 to -3.517), ESR (WMD: -6.563; 95 % CI: -8.604 to -4.522), VAS (WMD: -1.089; 95 % CI: -1.575 to -0.602), DAS-28 (WMD: -0.633; 95 % CI: -1.006 to -0.259), SJC (WMD: -1.921; 95 % CI:- 3.635 to -0.207), TJC (WMD: -1.491; 95 % CI: -2.941 to -0.042). The authors concluded that this meta-analysis of RA provided reliable evidence in favor of ACNRT plus WM; however, longer-term, high-quality, repeatable, multi-center RCTs with larger sample sizes are needed.

Senile Insomnia

In a systematic review and meta-analysis, Hu et al (2023) examined the effectiveness of acupuncture or moxibustion therapy in the treatment of patients with senile insomnia. These investigators carried out a comprehensive literature search using 7 electronic databases to identify RCTs reported on the use of acupuncture or moxibustion therapy in insomnia. The time frame was set from database establishment to March 11, 2023. The RevMan (version 5.3) and STATA (version 17.0) software were used to evaluate the quality of the included RCTs and performed a meta-analysis. The methodological quality of the included studies was assessed using the Cochrane risk-of-bias tool. Subgroup analysis was carried out based on different intervention methods. The I2 statistic was employed to evaluate heterogeneity among studies. A

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total of 20 studies conducted between 2007 and 2022 were included, involving 1,677 patients with senile insomnia. In terms of effectiveness, acupuncture or moxibustion alone was significantly better than Western drugs (RR = 1.12; 95 % CI: 1.06 to 1.20), acupuncture combined with drugs was better than drugs alone (RR = 1.20; 95 % CI: 1.12 to 1.29), and acupuncture combined with cognitive behavior therapy intervention (CBT-I) was significantly better than CBT-I alone (RR = 1.52; 95 % CI: 1.07 to 2.17). In terms of Pittsburgh Sleep Quality Index (PSQI) scores, acupuncture or moxibustion alone was more effective than Western drugs (MD = -1.82; 95 % CI: -2.37 to -1.26), acupuncture combined with drugs was more effective than drugs alone (MD = -3.10; 95 % CI: -4.25 to -1.95), and acupuncture was significantly more effective than sham acupuncture (MD = -4.18; 95 % CI: -5.85 to -2.51) and psychological intervention (MD = -3.54; 95 % CI: -4.33 to -2.75) in improving sleep quality. The authors concluded that this meta-analysis showed that acupuncture or moxibustion alone or combination with other therapies (drugs, CBT-I or psychological intervention) had high clinical effectiveness and could improve the sleep quality of patients with senile insomnia. Moreover, these researchers stated that further well-designed studies are needed to verify these findings.

The authors stated that this meta-analysis had several drawbacks. First, the heterogeneity of PSQI scores was high; thus, a definite conclusion could not be drawn. Second, the methodological quality of the included studies was low. An unclear or high risk of bias was observed in terms of blinding, random sequence generation, and allocation concealment in the included studies. Therefore, it is necessary to record all relevant information to minimize performance and assessment bias and improve the quality of study design in the future. Third, the criteria for evaluating the rate of improvement were different among the included studies, which increased bias in the pooled rate of improvement in sleep quality to some extent. However, owing to an insufficient number of studies focusing on senile insomnia, studies involving the rate of improvement were included. A standardized criterion should be provided for more robust conclusions regarding the effectiveness of each treatment method in relieving the symptoms of insomnia. Fourth, subjects in the included studies were selected from a single center in China, which may have led to a lack of representativeness. Moreover, owing to the lack of follow-up data, the long-term effects of acupuncture could not be examined. These researchers stated that these drawbacks should be addressed in future studies.

Sexual Dysfunction

Abdi and colleagues (2021) noted that sexual dysfunction can adversely affect the QOL and inter-personal relationships. To-date, a lot of attention is paid to traditional Chinese medicine with better curative effects and less AEs. Recent studies have implied the promising effect of acupuncture on sexual function. In a systematic review, these investigators examined the safety and effectiveness of acupuncture in the treatment of female and male sexual dysfunction. They searched PubMed, Cochrane Central of Controlled Trials (CENTRAL), Embase, Web of Science, China National Knowledge Infrastructure (CNKI), Chinese Biomedical Literature Database (CBM), Scopus, and Google Scholar up to 2021; no limitation to language and date was applied. The methodological quality of the studies was evaluated using the Mixed Methods Appraisal Tool (MMAT). Of 160 initially assessed papers, 13 articles were included. The findings showed that acupuncture improved sexual dysfunction in domains of desire, libido, erectile dysfunction and impotency. Most studies did not report any serious side effects from acupuncture, and only 3 studies reported minor AEs. The authors concluded that available evidence indicated that acupuncture has positive on improvement of sexual dysfunction with no serious side effects. Moreover, these researchers stated that although acupuncture has gained increasing popularity for the management of sexual dysfunction, high methodological quality evidence regarding its effectiveness is lacking.

Sleep Disturbance in Cancer Patients

Cheung et al (2023) noted that both acupuncture and acupressure have been suggested beneficial for reducing sleep disturbance in cancer patients. While acupuncture is invasive involving needle insertion, acupressure is non-invasive. Their comparative effectiveness is unclear, hindering clinical recommendations. In a systematic review and network meta-analysis, these investigators compared the effectiveness of acupuncture and acupressure on sleep in cancer patients. A total of 8 key English and Chinese databases were searched; 24 RCTs entailing 2,002 cancer patients comparing the effects of 6 treatments (manual acupuncture, electro-

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acupuncture, acupressure, sham, enhanced usual care, and no treatment) on sleep were found. Compared with enhanced supportive care, acupressure demonstrated the largest effect size for reducing self-reported sleep disturbance (SMD = -2.67, 95 % CrI: -3.46 to -1.90; GRADE = moderate), followed by acupuncture (SMD = -1.87, 95 % CrI: -2.94 to -0.81, GRADE = moderate) and electro-acupuncture (SMD = -1.60, 95 % CrI: -3 to -0.21; GRADE = low). The surface under the cumulative ranking curve indicated that acupressure was most likely to rank highest. The authors concluded that based on available evidence, acupressure could be recommended as the optimal treatment for reducing sleep disturbance in cancer patients; however, these researchers stated that more rigorous studies are needed to confirm whether different forms of acupuncture or acupressure have different effects on sleep in cancer patients. Particularly, studies examining acupuncture interventions alone instead of in combination with other therapies are needed.

Teding Dianci Pu (TDP) Lamp as an Adjunct to Acupunctur

The Teding Diancibo Pu (TDP) lamp is a specific electromagnetic lamp that produce far infrared (below visible light) emissions (2 to 50 um range). "Teding Diancibo Pu" loosely translated means specific electromagnetic spectrum. The lamp is used in several hospitals in China and Japan to increase microcirculation, to loosen fascia and muscles, and to accelerate the natural healing processes of the body. The TDP lamp is different from other far-infrared heating devices. It contains a heated mineral plate made of a proprietary formula of 33 trace elements, which stimulates one's body to heal naturally.

In a randomized, controlled, single-blind, phase-III clinical trial, Zhang and Yuan (2020) compared the efficacy of honey mouthwash 12.5 % and chlorhexidine solution 0.2 % to reduce the rate of oropharyngeal bacterial colonization in mechanically ventilated patients. This study included 60 patients newly admitted to internal and trauma intensive care units (ICUs) of the 2 educational hospitals of Sanandaj City affiliated with Kurdistan University of Medical Sciences. Subjects were selected by convenience sampling and allocated to 2 groups of 30 patients using random blocks design. In each group, the mouthwash was applied twice-daily for 4 consecutive days. Swab samples were taken from the mouth and throat of all patients thrice-daily (pre intervention, 2 days, and 4 days after the intervention) and then the samples were transferred onto the blood agar and eosin methylene blue (EMB) culture plates and examined for bacterial growth and colonization after 24 to 48 hours. The findings showed that oropharyngeal colonization was not significantly different between the 2 groups, pre-intervention, 2 days, and 4 days after the intervention ($p > 0.05$). Rinsing with honey mouthwash 12.5 % led to the inhibition of Staphylococcus aureus and Pseudomonas aeruginosa on the 4th day of the intervention in all samples. The authors concluded that none of the studied solutions contributed to the reduction of oropharyngeal bacterial colonization. It appeared that the growth inhibition of Staphylococcus aureus and Pseudomonas aeruginosa by the honey 12.5 % mouthwash in mechanically ventilated patients need further investigation. Teding Dianci Pu therapy was one of the key words of this study.

Thoracic Back Pain

In a systematic review, Southerst et al (2015) examined available evidence on the effectiveness of non-invasive interventions, excluding pharmacotherapies, for musculoskeletal thoracic pain; RCTs, cohort studies, and case-control studies evaluating the effectiveness of non-invasive interventions were eligible. These investigators searched Medline, Embase, PsycINFO, and the Cochrane Central Register of Controlled Trials accessed via Ovid Technologies, Inc, and CINAHL Plus with full text accessed via EBSCOhost from 1990 to 2015. The search strategies combined controlled vocabulary relevant to each database (e.g., MeSH for Medline) and text words relevant to the research question and the inclusion criteria. Random pairs of independent reviewers screened studies for relevance and critically appraised relevant studies using the Scottish Intercollegiate Guidelines Network criteria. Studies with a low risk of bias were synthesized following best evidence synthesis principles. These researchers screened 6,988 articles and critically appraised 2 studies. Both studies had a low-risk of bias and were included in the synthesis. One RCT compared thoracic spinal manipulation, needle acupuncture, and placebo electrotherapy for recent thoracic spine pain. There were statistically significant but clinically non-important short-term reductions in pain favoring manipulation. There were no differences between acupuncture and placebo electrotherapy. Another RCT compared a multi-modal program of care and a session of education for recent musculoskeletal chest

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wall pain. The multi-modal care resulted in statistically significant but clinically non-important short-term reductions in pain over education; however, subjects receiving multi-modal care were more likely to report important improvements in chest pain. The authors concluded that quality evidence on the management of musculoskeletal thoracic pain was sparse. Available evidence suggested that compared to placebo, spinal manipulation was associated with a small and clinically non-important reduction in pain intensity and that acupuncture resulted in similar outcomes as placebo. Additionally, a multi-modal program of care (i.e., manual therapy, soft tissue therapy, exercises, heat/ice, and advice) and a single education session resulted in similar pain reduction for recent-onset musculoskeletal chest wall pain. However, patients who receive multi-modal care were more likely to report pain improvements.

Furthermore, an UpToDate review on "Overview of the clinical uses of acupuncture" (Yang and Yeh, 2023) does not mention thoracic back pain as an indication of acupuncture.

Tinnitus

In a systematic review, Xu et al (2022) examined the evidence on the effectiveness of acupuncture in the treatment of tinnitus. These researchers evaluated the methodological quality, reporting quality, and evidence quality of systematic reviews/meta-analyses (SRs/MAs) of acupuncture in the treatment of tinnitus. From inception to March 2022, these investigators carried out a detailed and comprehensive search of 8 electronic databases in Chinese and English. The AMSTAR-2, the PRISMA checklist, and the GRADE were used to evaluate methodological quality, reporting quality and evidence quality for inclusion in SRs/MAs, respectively. A total of 14 published SRs/MAs met the inclusion criteria and were included in the study – 11 studies reported that acupuncture was effective in treating tinnitus, and 3 studies reported that no firm conclusions could be drawn regarding the effectiveness of acupuncture in treating tinnitus. The results of the AMSTAR-2 assessment showed that the methodological quality of the included studies was relatively low in general, with 1 being moderate quality and the rest being very low quality. The PRISMA checklist evaluation results showed that no studies fully reported checklists, with protocol registration and search strategies being the main reporting weaknesses. The GRADE assessment showed that no results were of high-quality evidence, 17 results were moderate-quality evidence, 25 results were low-quality evidence, and 12 results were very low-quality evidence. The authors concluded that acupuncture appeared to be a positive and effective treatment for tinnitus; however, the methodological quality and quality of evidence for SRs/MAs in the included studies were generally low, and this result must be viewed with caution. These researchers stated that more high-quality, large-scale, multi-center RCTs are needed in the future to verify the effectiveness of acupuncture in the treatment of tinnitus.

The authors stated that this was the 1st overview of a systematic review of acupuncture for tinnitus and the 1st to provide a comprehensive evidence base for the clinical practice of acupuncture for tinnitus using the AMSTER-2 and GRADE assessment tools. However, these investigators acknowledged several drawbacks. First, examining the methodological quality and quality of evidence is subjective. Even if they evaluated each item of the evaluation system in detail and objectively, guidelines or authoritative 3d parties judged the disputes. The results may still be partially different. Second, these researchers only included studies published in full text in both Chinese and English, which was likely to have a certain degree of risk of bias. Third, the authors did not carry out advance review registration on the PROSPERO website, which may have led to the risk of bias. Fourth, due to the disease characteristics of tinnitus, the primary outcome measure of the RCTs included in this study was subjective clinical evaluation, which may have an impact on treatment outcomes.

Ji et al (2023) provided evidence for medical management of tinnitus based on an assessment of the evidence concerning the effectiveness of acupuncture as a treatment for tinnitus using network meta-analysis (NMA). These investigators carried out a systematic literature review by searching 8 national and international databases (inception to February 2023) for RCTs for tinnitus. Only RCTs that recruited participants aged over 18 years and diagnosed with tinnitus, and that evaluated acupuncture or acupuncture in combination with conventional Western medical therapy were included. These researchers used response rate and tinnitus handicap inventory (THI) to examine effectiveness. They carried out NMA with random effects, and the rate ratio or MD with its 95 % CI was calculated. Furthermore, these investigators ranked all treatments via their SUCRA and examined the quality of evidence according to the GRADE criteria. A total of 2,575 patients

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were included in the study. The main findings of the current NMA were that acupoint injection combined with warm acupuncture was the most effective for response rate, followed by warm acupuncture and acupoint injection combined with Western medical treatment. Acupuncture combined with Western medical treatment was the most effective for THI, followed by electro-acupuncture combined with warm acupuncture and acupuncture combined with moxibustion. The authors concluded that acupuncture appeared to be a better trend treatment for tinnitus. Moreover, these researchers stated that further rigorous RCTs that include direct comparisons for different acupuncture-related treatments are encouraged to provide the most promising evidence for patients with tinnitus.

The authors stated that this review had several drawbacks. First, this NMA may have been under-powered due to the heterogeneity of the participants (e.g., co-morbidities, baseline severity of tinnitus, duration of tinnitus onset, treatment duration in each study, and follow-up duration). Second, tinnitus is a subjective symptom and there is no objective measurement to measure its severity or determine the effectiveness of its treatment; effectiveness of tinnitus therapy was assessed using a subjective questionnaire, which may have resulted in heterogeneity of the reported results, which was an unavoidable limitation of any NMA examining subjective outcomes. Objective measures of tinnitus are also being researched, and the results of these studies may lead to a major break-through in the diagnosis and management of tinnitus. Furthermore, the specificity and complexity of the acupuncture-related treatment limited the application of the blinding of participants and intervenors, which may have some impact on the effectiveness, as tinnitus is a condition closely related to psychological factors, and psychological comfort could not be excluded. Third, given the relatively small number of patients and RCTs, the key findings of this study should probably be conservatively applied to clinical practice. Fourth, these researchers did not examine AEs, because there were too few studies providing data on AEs to form a network. Additionally, to further clarify the better treatments to the tinnitus patients, more rigorous RCTs with high-grade evidence and greater sample sizes that include direct comparisons for different acupuncture-related treatments are needed. The authors stated that they still need to update the literature to incorporate higher quality studies in the future to find more effective interventions for tinnitus.

Vascular Dementia

In a systematic review protocol, Ye and colleagues (2017) provided the methods used to evaluate the safety and effectiveness of acupuncture in the treatment of vascular dementia (VaD). The following 8 databases will be searched from inception to July 2017: Cochrane Central Register of Controlled Trials, PubMed, Medline, Embase, China National Knowledge Infrastructure, Chinese Biomedical Literature Database, VIP Database and Wanfang Database. All RCTs in English or Chinese related to acupuncture for vascular dementia will be included. Outcomes will include change in cognitive function and ADL. The incidence of AEs will be assessed for safety evaluation. Study inclusion, data extraction and quality assessment will be performed independently by 2 reviewers. Assessment of risk of bias and data synthesis will be performed using Review Manager software. The authors stated that this systematic review will provide an assessment of the current state of acupuncture treatment for VaD. Conclusions drawn from this review may benefit patients with VaD, clinicians and policy makers. The process of conducting this review will be divided into 4r parts:

I. identification,
II. study inclusion,
III. data extraction, and
IV. data synthesis.

This review has 2 main potential limitations:

I. various forms of acupuncture may cause considerable heterogeneity, and
II. the quality of included reports might be poor, which will limit the ability to generate conclusions based on high confidence.

The findings of this systematic review will be disseminated through peer-reviewed publication or conference presentations.

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Dry Needling

Dry needling is considered by many professional societies and organizations as a type of acupuncture, both involving insertion of solid filiform needles into parts of the body to treat various health conditions. However, there are differences between the two methods based on the theory, technique for how and where the needles are placed, and professional training.

"It's important to note dry needling is not the same as acupuncture. It uses similar tools, but that's where the similarities end. Dry needling is performed by different practitioners with different training" (Mayo Clinic, 2017).

Acupuncture has been around for the past 2000 years and is based on the philosophy of traditional Chinese medicine (TCM). Acupuncture follows the principles of energy flow as a guide to where the needles will be inserted into points along meridian lines (energy-carrying channels). During acupuncture, very thin needles are placed in the skin at various acupoints which is usually left in place for 15 to 30 minutes. This technique is permformed by an acupuncturist who is appropriately trained and credentialed in order to practice acupuncture in the United States. Credentialing requirements vary based on state and institution regulations.

Dry needling (also known as intramuscular manual stimulation, or intramuscular needling) is considered by many as a relatively newer teatment modality which is no longer based on TCM concepts and belongs to a subcategory of western medical acupuncture (Hu et al, 2018). "Dry Needling (DN) referred originally to a hypodermic needle on an empty syringe, inserted into the body without injection," has now expanded to include filiform needles as used in acupuncture (McIntyre, 2016). Dry needling involves needles that are inserted directly into a myofascial trigger point, in an attempt to inactivate it, thereby decreasing the associated pain. Dry needling or trigger point needling is typically performed by physical therapists to help patients with acute or chronic pain.

Dommerholt (2011) states that "Although various needling approaches are commonly referred to as 'dry needling', it is important to realize that there are significant differences between schools of dry needling, their specific needling techniques, underlying philosophy or rationale, and duration of training programs. Each approach appears to address particular aspects of the total picture. Different dry needling techniques have been promoted to treat various forms of soft tissue dysfunction".

An increasing number of clinical trials have been performed to explore the efficacy of dry needling for treating pain; however, the conclusions were conflicting. The real effectiveness of dry needling remains controversial (Hu et al, 2018).

Several dry needling models have been developed including a radiculopathy model, myofascial trigger point (MTrP) model, a neurological model, and superficial dry needling model. Superficial dry needling involves inserting a needle at a depth up to 10 mm or into the subcutaneous tissue and may be combined with manipulation of the needle while in situ. Deep needle penetration involves inserting a needle through the skin, beyond subcutaneous tissue, and into muscular or other connective tissue structures and can be combined with needle manipulation. The majority of dry needling studies involve deep needling into MTrP for pain relief; however, deep needle penetration has not consistently demonstrated superior results compared to superficial needle insertion (Griswold et al, 2019).

Position statements of professional organizations (not an all-inclusive list) regarding the practice of "dry needling":

- National Certification Commission for Acupuncture and Oriental Medicine (NCCAOM) states that dry needling is considered acupuncture. "Dry needling is a recently coined name for an acupuncture technique that involves the insertion of acupuncture needles directly into muscles and "trigger points" for the relief of musculoskeletal pain". "Some physical therapists and other healthcare providers have claimed that "dry needling" is "new" and "not acupuncture", because the point locations and needling style are based on anatomical structures and physiological function rather than on traditional Chinese

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medicine acupuncture theory. In fact, licensed acupuncturists receive training in the application of both traditional foundations of acupuncture and modern biomedical theories and have done so since long before the term “dry needling” was invented".

- American Academy of Medical Acupuncture (AAMA) state, "Dry needling, like acupuncture, involves the use of solid needles (contrasted with the use of hollow hypodermic needles that are used for injections) to treat muscle pain by stimulating and breaking muscular knots and bands. Unlike trigger point injections used by physicians and licensed acupuncturists for the same purpose, no anesthetics are used in dry needling. There is controversy regarding the definition of dry needling. Licensed medical physicians and licensed acupuncturists consider dry needling as Western Style Acupuncture or Trigger Point Acupuncture whereby the insertion sites are determined by tender painful areas and tight muscles. These sites may be treated alone or in combination with known acupuncture points. Other practitioners take the position that dry needling is different from acupuncture in that it is not an holistic procedure and does not use meridians or other Eastern medicine paradigms to determine the insertion sites. Regardless of the theory, it is incontrovertible that dry needling is an invasive procedure. Needle length can range up to 4 inches in order to reach the affected muscles. It is critical to understand that dry needling, in the hands of minimally educated practitioners can cause extreme harm. Any invasive procedure has associated and potentially serious medical risks and is safe only if performed by a properly educated, trained and experienced health professional. The technique of dry needling frequently involves needling of muscular structures that may be deep and/or hidden under layers of other muscles and tissues and close to sensitive structures and organs including blood vessels, nerves and organs as, for example, the lungs. The patient can develop painful bruises after the procedure and adverse sequelae may include hematoma, pneumothorax, nerve injury, vascular injury and infection. Angle the needle incorrectly and, for example, the lung may be punctured. Post procedure analgesic medications may be necessary (usually over the counter medications are sufficient). In the worse case scenario, vital organs can be pierced, resulting in complex medical situations or even death". "The AAMA strongly believes that, for the health and safety of the public, this procedure should be performed only by practitioners with extensive training and familiarity with routine use of needles in their practice and who are duly licensed to perform these procedures, such as licensed medical physicians or licensed acupuncturists. In our experience and medical opinion, it is inadvisable legally to expand the scope of physical therapists to include dry needling as part of their practice" (NCCAOM, 2016).
- American Academy of Physical Medicine and Rehabiliation (AAPM&R, 2012) "recognizes dry needling as an invasive procedure using acupuncture needles that has associated medical risks. Therefore, the AAPMR maintains that this procedure should only be performed by practitioners with standard training and familiarity with routine use of needles in their practice, such as licensed acupuncturists or licensed medical physicians".
- American Association of Acupuncture and Oriental Medicine (AAAOM) "has determined that dry needling and any of its alternate designations, including intramuscular manual therapy, trigger point needling, functional dry needling, intramuscular stimulation or any other method by which a needle is inserted to effect therapeutic change, is, by definition, the practice of acupuncture".
- American Medical Association (AMA, 2016) recognizes dry needling as an invasive procedure and maintains that dry needling should only be performed by practitioners with standard training and familiarity with routine use of needles in their practice, such as licensed medical physicians and licensed acupuncturists.

Dry needling, even though it targets a trigger point, differs from a trigger point injection, as there is no injection of medication or fluid.

Dry Needling and Myofascial Trigger Points

Trigger points have been treated with dry needling. A myofascial trigger point is a discrete focal tenderness, 2-5 mm in diameter that is located in distinct tight bands or knots of skeletal muscle (AHFMR, 2002). When palpated, these hyper-irritable areas cause pain in distant areas, or referred pain zones, which are specific for each trigger point. Trigger point injection, or direct wet needling, involves injection of fluid directly into the trigger point located in the taut muscle band. The main objective of trigger point injection is fast pain relief

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and elimination of muscle spasm in order to break the pain cycle. This facilitates physical therapy aimed at reducing muscle contracture and increasing range of motion. Trigger point injection is rarely used in isolation but is generally part of a multi-disciplinary approach aimed at treating both the trigger points and reducing all contributing factors (Scott and Guo, 2005; AHFMR, 2002; Sanders et al, 1999). Thus, treatment may also include patient education, psychosocial support, oral medications, and physical therapy to improve the strength and flexibility of the affected musculoskeletal systems. An assessment conducted by the Alberta Heritage Foundation for Medical Research (Scott and Guo, 2005) found that the evidence for the effectiveness of trigger point injections when used as the sole treatment for patients with chronic head, neck, and shoulder pain and whiplash syndrome was inconclusive, regardless of whether sterile water, saline, or botulinum toxin is injected. The assessment found that the combined use of dry needling and trigger point injection with procaine offers no obvious clinical benefit in the treatment of chronic craniofacial pain, while the effectiveness of trigger point injection for the treatment of cervicogenic headache is unknown. In contrast, the assessment found that trigger point injection with lidocaine may be useful in the treatment of joint pain caused by osteoarthritis (Scott and Guo, 2005). The assessment found no proof that triggers point injection is more effective than other less invasive treatments, such as physical therapy and ultrasound, in achieving pain relief, and there is some suggestion that the only advantage of injecting anesthetic into trigger points is that it reduces the pain of the needling process (Scott and Guo, 2005). Usually, approximately 3 treatments are necessary to abolish a trigger point completely (AHFMR, 2002). A number of trigger points may be injected in 1 session, but rarely more than 5. Repeated injections in a particular muscle are not recommended if 2 or 3 previous attempts have been unsuccessful (Alvarez and Rockwell, 2002; Sanders et al, 1999). The pain relief may last for the duration of the anesthetic to many months, depending on the chronicity and severity of the trigger points and the concomitant treatment of perpetuating factors. According to available guidelines, use of trigger point injections should be short-term and part of a comprehensive rehabilitation program. Available guidelines indicate that, while there are a number of uncontrolled case studies using trigger point injections in more acute pain presentations, there is virtually no consistent evidence for its application with chronic non-malignant pain syndrome patients to date (Sanders et al, 1999; AHFMR, 2002).

Kietrys et al (2013) state that dry needling involves inserting an acupuncture-like needle into a myofascial trigger point, with the goal of reducing pain and restoring range of motion in persons with myofascial pain syndrome (MPS). The authors explored the evidence regarding the effectiveness of dry needling to reduce pain in patients with MPS of the upper quarter. An electronic literature search was performed using the key word dry needling. The initial search yielded 246 articles. Twelve RCTs were ultimately selected. The methodological quality scores ranged from 23 to 40 points, with a mean of 34 points (scale range, 0-48; best possible score, 48). The findings of 3 studies that compared dry needling to sham or placebo treatment provided evidence that dry needling can immediately decrease pain in patients with upper-quarter MPS, with an overall effect favoring dry needling. The findings of 2 studies that compared dry needling to sham or placebo treatment provided evidence that dry needling can decrease pain after 4 weeks in patients with upper-quarter MPS, although a wide confidence interval for the overall effect limits the impact of the effect. Findings of studies that compared dry needling to other treatments were highly heterogeneous, most likely due to variance in the comparison treatments. There was evidence from 2 studies that lidocaine injection may be more effective in reducing pain than dry needling at 4 weeks. The authors concluded that based on the best current available evidence (grade A), they recommend dry needling, compared to sham or placebo, for decreasing pain immediately after treatment and at 4 weeks in patients with upper-quarter MPS. However, due to the small number of high-quality RCTs published to date, additional well-designed studies are needed to support this recommendation.

In a 2017 review on trigger point dry needling found in the Journal of Orthopaedic & Sports Physical Therapy, the authors analyzed 13 randomized controlled studies that examined the effectiveness of dry needling on musculoskeletal pain. The authors found that, to date, most of the evidence is of very low to moderate quality. There are also risks of bias in the available research. Further, very little evidence exists regarding the longer-term benefits of dry needling, or that guides optimal treatment techniques and dosing.

Dry Needling for the Treatment of Low Back Pain

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UpToDate reviews on "Subacute and chronic low back pain: Nonpharmacologic and pharmacologic treatment" (Chou, 2020a) and "Treatment of acute low back pain" (Knight et al, 2020) do not mention dry needling as a management / therapeutic option.

Furthermore, an "Subacute and chronic low back pain: Nonsurgical interventional treatment" (Chou, 2020b) states that "Local or trigger point injection -- A systematic review found no clear differences between local or trigger point injections with a local anesthetic, with or without a corticosteroid, and control interventions (saline or dry needle injections, or ethyl chloride plus acupressure) for short-term (7 days to 2 months) pain relief in 3 trials of patients with subacute or chronic low back pain. All trials had methodological shortcomings and evaluated heterogeneous injection methods. One trial evaluated an injection over the iliac crest, one evaluated injections over the iliolumbar ligament, and one evaluated trigger point injections. The limited benefit observed in heterogeneous, low-quality studies does not support their widespread use".

Acupotomy for the Treatment of Herpes Zoster, Low Back/Shoulder Pain, Paresthesia of Skin, and Trigger Finger

You et al (2019) noted that acupotomy therapy, which entails the addition of a scalpel function to the conventional acupuncture treatment, is often used for pain management; however, the effectiveness of acupotomy on shoulder adhesive capsulitis (SAC) is still uncertain. In a systematic review and meta-analysis, these investigators examined the safety and effectiveness of acupotomy therapy for SAC. They searched 7 electronic databases to collect RCTs of acupotomy for SAC published before April 2019. A meta-analysis was carried out according to the Cochrane systematic review method by using RevMan 5.3 software. A total of 8 RCTs involving 501 patients were enrolled. Meta-analysis showed that acupotomy was significantly better than the control group in lowering the VAS score (MD = −0.97, 95 % CI: −1.49 to −0.45],) and improving the Constant-Murley Score (CMS) (MD = 8.46, 95 % CI: 1.04 to 15.87), and there was no significant difference in AEs (OR = 1.24, 95 % CI: 0.34 to 4.52) between the 2 groups. The authors concluded that acupotomy therapy was a safe and effective treatment for SAC, and this treatment could be recommended for the management of SAC. Moreover, these researchers stated that as a consequence of the low-quality and small sample size of the included studies, further investigations with more rigorously designed, high-quality, large-scale RCTs are needed.

Jeong and colleagues (2020) stated that acupotomy has recently been applied as a conservative therapy for the treatment of patients with lumbar disc herniation (LDH). In a randomized-controlled, assessor-blinded, multi-center study, these investigators examined the safety and effectiveness of acupotomy, compared to manual acupuncture, for the treatment of patients with LDH. A total of 146 patients diagnosed with LDH were randomly assigned to either the acupotomy group or the manual acupuncture group at a 1:1 ratio. Subjects in both groups received 4 sessions of each intervention over 2 weeks. Outcome assessments based on the VAS, Roland Morris Disability Questionnaire (RMDQ), Modified-Modified Schober Test (MMST), EQ-5D, clinically important difference (CID), and patient global impression of change (PGIC) were performed at baseline and at 2, 4, and 6 weeks post-randomization. The acupotomy group showed significant improvement in VAS and MMST at 2, 4, and 6 weeks than did the manual acupuncture group. RMDQ was significantly different between the 2 groups at 2 and 6 weeks. In EQ-5D, there was no significant difference between the 2 groups. The proportion of patients with 15 mm or greater decrease on the VAS (minimal CID) was significantly higher in the acupotomy group at weeks 2 and 4. Better improvement in the PGIC at week 4 was also observed in the acupotomy group. Post-intervention muscle pain was reported, but there was no serious AE related to interventions. The authors concluded that 4 sessions of acupotomy treatment were found to be effective in improving the pain intensity and range of motion (ROM) of the lumbar region in patients with LDH. Despite post-treatment muscle pain, acupotomy treatment can be considered a preferred treatment method over manual acupuncture.

The authors stated that this study had several drawbacks. First, the follow-up period of 4 weeks post-intervention was not enough to confirm the long-term effects of acupotomy. Second, neither the subjects nor practitioners were blinded, which could act as a potential bias, such as the expectation by the subject and the reliability of the assigned intervention. These researchers stated that in the future, it is necessary to conduct large-scale, long-term follow-up clinical trials comparing other conservative interventions with acupotomy.

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Furthermore, studies examining the synergistic effects of other treatments with acupotomy treatment are also needed.

In a systematic review and meta-analysis, Zhang et al (2023b) examined the safety and effectiveness of acupuncture-related therapy in the treatment of herpes zoster (HZ). PubMed, Cochrane Library, Embase, Web of Science, CNKI, VIP database, Wanfang Database, and SinoMED Chinese and English databases were systematically searched for RCTs on acupuncture-related therapy for HZ until February 2023. Literature screening, data extraction, and risk of bias assessment were independently conducted by 2 researchers. R4.2.1 and Stata15.1 software were employed for data analysis. A total of 26 RCTs entailing 1,929 HZ patients were included, and the NMA results showed that acupotomy appeared to be most effective in terms of improving VAS scores and the time of incrustation; SUCRA cumulative probability ranking showed the best effectiveness of blood-letting in improving efficiency and the time of ceasing new blisters, acupuncture in improving the time of decrustation, and moxibustion in reducing the incidence of post-herpetic neuralgia (PHN). The authors concluded that via a comprehensive comparison of 9 different acupuncture therapies, acupotomy was the optimal treatment to improve VAS pain scores and the time of incrustation; blood-letting, acupuncture, fire needle, and moxibustion could be effective intervention measures to improve the time of ceasing new blisters and decrustation, and reduce the incidence of PHN; Western medicine was closely related to adverse reactions. Moreover, these researchers stated that these findings need to be further validated using prospective, large-scale, multi-center RCTs due to the limitations of the study.

The authors stated that this study had 2 main drawbacks. First, only 1 study was performed outside of China. The vast majority of patients in the studies were yellow races, and there was a lack of effectiveness evaluation for different races. Second, the included studies differed in experimental design, random allocation concealment, blind implementation, safety reporting, etc., which limited the reliability of evidence.

Liang et al (2023) noted that trigger finger is a common condition in the hand, these investigators examined the safety and effectiveness of US-guided acupotomy for the treatment of trigger finger. They searched for relevant studies in the Cochrane Library, China National Knowledge Infrastructure (CNKI), Embase, PubMed, Chinese Biomedical Literature Database (CBM), Wanfang Data, and other resources from their inception to January 2023; RCTs of US-guided acupotomy for trigger finger were included. A meta-analysis was performed using Review Manager 5.4 and Stata 15.1. A total of 15 studies with 988 patients were included. The experimental group was treated with US-guided acupotomy, and the control group received traditional acupotomy, traditional operation, or injection of medication. Meta-analysis showed that the overall clinical effectiveness (OR = 4.83; 95 % CI: 2.49 to 9.37; I2 = 73.1 %; $p < 0.001$) in the experimental group was significantly better than that of the control group. And the VAS score (WMD = - 1; 95 % CI: - 1.24 to - 0.76; I2 = 99 %; $p < 0.001$), the QuinneII classification (WMD = - 0.84; 95 % CI: - 1.28 to - 0.39; I2 = 99.1 %, $p < 0.001$), the incidence of complications (RR = 0.26; 95 % CI: 0.11 to 0.63; I2 = 0 %, p = 0.003), and the recurrence rate (RR = 0.14; 95 % CI: 0.03 to 0.74; I2 = 0 %; p = 0.021) were significantly lower in the experimental group. The authors concluded that this systematic review and meta-analysis could prove the safety and effectiveness of US-guided acupotomy in the treatment of trigger finger. Moreover, these researchers stated that these findings still need to be validated by a clinical standard large sample test.

The authors stated that this study had several drawbacks. First, the small number and the low quality of the studies included in this review may have affected the results. Second, the control group was treated with acupuncture, surgery, and corticosteroid injections, which may have been biased. And only a few articles have studied adverse reactions, so the safety of US-guided acupotomy needs to be further examined.

Furthermore, a StatPearls review on “Acupuncture” (Van Hal et al, 2023) does NOT mention “acupotomy”.

Combined Acupuncture Therapy and Pharmacotherapy for the Treatment of Systemic Lupus Erythematosus

Wang et al (2023a) noted that currently, the mainstream treatments for systemic lupus erythematosus (SLE) are based on glucocorticoids and immunosuppressants, which are known to have considerable adverse effects. In a meta-analysis, these researchers examined the safety and effectiveness of acupuncture therapy in

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combination with traditional medications in the treatment of SLE. Multiple databases were searched for RCTs using acupuncture therapy in combination with conventional pharmacotherapy for the treatment of SLE, from the establishment of the database to March 2023. Study selection, data collection, as well as quality assessment were conducted by 2 reviewers independently. RevMan 5.4 and Stata 17 software were used for meta-analysis. A total of 7 eligible studies involving 514 patients with SLE were included. Meta-analysis showed that in SLE patients, extra treatment with acupuncture was superior to drug therapy alone in improving the overall response rate (ORR) (RR = 1.20, 95 % CI: 1.11 to 1.29, $p < 0.00001$, heterogeneity $p = 0.69$, $I2 = 0$ %), and regulating immunological indicators (C3, C4, IgG, T lymphocyte subpopulation, IL-6, ds-DNA, ESR) while reducing traditional Chinese medicine (TCM) symptom scores, the SLE Disease Activity Index (SLEDAI) and the incidence of AEs on treatment ($p \leq 0.05$). Furthermore, it was able to reduce blood urea nitrogen and serum creatinine (BUN, Scr) and 24 hours urine protein, suggesting that acupuncture treatment had a protective effect on the kidneys. The authors concluded that acupuncture therapy combined with conventional pharmacotherapy was a safe and effective way in the treatment of SLE; however, the conclusions drawn from this meta-analysis had several drawbacks due to the small number and uneven quality of the included studies, resulting in heterogeneity and bias. These researchers stated that more well-designed, well-executed, high-quality RCTs with large sample sizes are needed to further verify the long-term safety and effectiveness of acupuncture in the treatment of SLE.

The authors stated that this meta-analysis had several drawbacks. First, quality of included studies: The quality of included RCTs is variable, which could affect the reliability and validity of the overall results. Studies with methodological flaws, insufficient blinding, or small sample sizes may introduce biases that affect the precision of estimated treatment effects. Second, publication bias: There was some publication bias in the included studies, probably due to the existence of non-published negative results. Studies on acupuncture for SLE were mainly initiated by Chinese scholars, and most of the study subjects were also mainly Chinese, making the possibility of bias in the results. Third, limited number of studies: The number of relevant RCT was limited, and some ongoing experiments were not included in this meta-analysis, which may have limited the statistical power and generalizability of the findings. However, these investigators hoped that this meta-analysis will inspire more researchers to carry out relevant experiments to provide more credible evidence for this new therapy in treating SLE. Fourth, lack of long-term data: The follow-up was inadequate, as the long-term outcome is essential for examining the treatment means due to the complexity and recurrence of SLE. Fifth, heterogeneity: Heterogeneity in this meta-analysis stemmed mainly from the modality of the intervention (with or without herbal intervention, type of drug administered, frequency), the type of acupuncture (1 study used acupoint application and 1 study used acupoint injections; the acupoints used in each study were not exactly the same), and the participant (1 research involved participant of patient with lupus nephritis). Sixth, limited scope of outcomes: In addition to the outcomes involved in this meta-analysis, IgA, IL-4, Th1/Th2, IFN-gamma, and other outcomes are also important indicators reflecting the development of SLE. Due to insufficient number of included studies, meta-analysis of the above indicators was not applied. These researchers stated that the afore-mentioned drawbacks adversely affected the credibility of this meta-analysis.

Transcutaneous Electronic Acupoint Stimulation for Improvement of Post-Operative Recovery After Gynecologic Surgery

In a systematic review and meta-analysis, Ge and Zheng (2023) examined the effectiveness of transcutaneous electronic acupoint stimulation (TEAS) for the improvement of post-operative recovery after gynecologic surgery. These investigators carried out a thorough search of 6 databases until March 2023, identifying 12 RCTs that met the pre-defined inclusion criteria and encompassed a total of 1,510 patients. For continuous outcomes, these researchers calculated the WMD, and for binomial outcomes, they used the risk ratio (RR). They evaluated heterogeneity among the included studies using Cochran I2 and Q statistics, employing a random-effects model when the I2 value exceeded 50 %. To assess publication bias, these researchers employed Egger test. The analysis found that TEAS significantly reduced the risk of post-operative nausea (RR: 0.60, 95 % CI: 0.43 to 0.83, $p = 0.002$) and post-operative vomiting (RR: 0.54, 95 % CI: 0.43 to 0.67, $p < 0.001$), VAS (WMD: -0.47, 95 % CI: -0.76 to -0.17, $p = 0.002$), as well as shortened the time to 1st bowel movement (WMD: -18.43, 95 % CI: -20.87 to -15.99, $p < 0.001$) and time to 1st flatus (WMD: -8.98, 95 % CI: -12.46 to -5.51, $p < 0.001$) compared to the conventional group. The authors concluded that these

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findings suggested that TEAS may improve post-operative recovery following gynecologic surgery. Moreover, these researchers stated that to confirm these findings, larger RCTs encompassing a more diverse range of patient populations are needed.

Acupuncture for the Treatment of Coronary Heart Disease

Cui et al (2024) stated that dyslipidemia is considered a substantial risk factor for coronary heart disease (CHD). Blood lipid levels in patients with CHD influence cardiovascular events; thus, it is important to monitor and control these levels. As a traditional, complementary alternative therapy, acupuncture regulates blood lipid levels in patients with CHD. More acupuncture-related treatments are currently available; however, systematic reviews or evidence-based summaries have yet to be carried out. In a comprehensive review and network meta-analysis, these investigators examined the effects of acupuncture-related therapies on blood lipid levels in CHD patients. The outcome indicators were changes in the total cholesterol (TC), triglycerides (TGs), low-density lipoprotein cholesterol (LDL-C), and high-density lipoprotein cholesterol (HDL-C) levels. Two independent reviewers extracted data from PubMed, Web of Science, Embase, Cochrane Library, Medline (Ovid), China Knowledge Network, Wanfang, Vipshop, and SinoMed databases until January 7, 2024. The literature quality was assessed using RevMan 5.4.1. The data were analyzed using the Frequentist network meta-analysis with STATA 15.1 software. Network meta-analysis was used to compare direct and indirect evidence, and a cumulative ranking curve (SUCRA) was used to examine the curative effect. For consistency, global inconsistency and ring-inconsistency test evaluations were used. The Cochrane bias risk assessment tool was used to evaluate quality. A total of 14 studies with 1,416 patients were included in the study. The evidence network demonstrated that there were many studies on acupuncture and warm acupuncture. According to the direct comparison and SUCRA ranking, acupuncture reduced TC [-1.82 (-3.36 to -0.28)], heat-sensitized moxibustion lowered TG [-2.12 (-3.55 to -0.69)], LDL-C [-1.37 (-2.09 to -0.66)], and increased HDL-C [0.87 (0.52 to 1.22)]. These 2 interventions were the first in the SUCRA ranking. The inconsistency analysis showed that direct evidence corresponded with indirect evidence. There were some methodological defects in the included studies, and some studies did not implement blinding methods, had small sample sizes or other problems. The authors concluded that heat-sensitized moxibustion significantly lowered TG and LDL-C levels and elevated HDL-C levels. Acupuncture significantly reduced TC levels. These researchers stated that heat-sensitized moxibustion was a more effective intervention than other acupuncture-related treatments for regulating blood lipid levels in patients with CHD.

The authors stated that this study had several drawbacks. First, many studies did not specify randomization methods or allocation concealment, which may have influenced the testing power of the results. Second, there was some heterogeneity in the included studies regarding standard treatment drugs in terms of type and dosage, acupuncture-related intervention selection points, and intervention cycles. Third, some publication bias may have affected the results. Fourth, individual intervention studies had a small sample size, and some had only one literature report, which may have limited the accuracy of these findings. Fifth, AEs were reported less frequently; 4 studies reported 0 AEs, and safety warrants further investigation.

Acupuncture for the Treatment of Parkinson's Disease-Related Depression

Xiao et al (2024) stated that depression is a common non-motor symptom of PD that markedly affects the QOL of patients with PD. The principal treatment for depression in PD (DPD) is pharmacotherapy; however, pharmacotherapy has a long cycle and many adverse reactions. Acupuncture as a non-pharmacological intervention method, has been widely used in the treatment of patients DP and patients with depression in China; thus, the study of acupuncture in the treatment of early PD with mild-to-moderate depression has important practical significance. In a randomized, single-blinded, and placebo-controlled study, these researchers examined the effects of acupuncture for patients with early PD with mild-to-moderate depression. A total of 88 patients with DPD will be randomly assigned to either an acupuncture group or a control group in parallel in a 1:1 allocation ratio. Each group will receive 30 mins of acupuncture treatments or sham acupuncture treatments, 3 times/week for 12 weeks, followed by a 36-week follow-up period. The primary outcome is the response rate of the Hamilton Depression Rating Scale (HDRS)-17 at 12 weeks. Data will be collected at baseline, at the end of the 12-week treatment period, and during the 12-week and

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36-week follow-up. The authors stated that this trial hypothesized that acupuncture may treat DPD by restoring pathological alterations in brain neural activity. The findings will provide scientific evidence for acupuncture in the treatment of early PD with mild-to-moderate depression.

Acupuncture for the Treatment of Post-Surgical Neuropathic Pain After Breast Cancer Surgery

Relf and Brahmanand (2024) stated that chronic post-surgical neuropathic pain (CPSNP) and related static mechanical allodynia (SMA) is recognized as a significant long-term complication after breast cancer surgery (BCS). In a retrospective, observational study, these investigators examined the effectiveness of a novel laser acupuncture technique in the treatment of CPSNP and SMA post-BCS while patients underwent standard cancer treatments. A total of 30 patients with significant breast area CPSNP and SMA (DN4 of 4 or higher) were treated with 60-milliwatt (mW), 658-nm laser acupuncture at Austin Hospital between January 2021 and 2024. Patient data were analyzed retrospectively: 29 with CPSNP, 22 with SMA. The cohort had exposures to pain aggravating cancer therapies (e.g., chemotherapy, radiotherapy, and anti-hormonal medications). An online patient questionnaire assessed treatment outcomes qualitatively. Patients received an average of 6.4 treatments and their average CPSNP showed significant reduction from 6.86 to 1.29 ($p < 0.001$), with a mean 5.57 point decrease (95 % CI: 4.90 to 6.24). Average SMA showed significant reduction from an average of 6.77 to 1.75 ($p < 0.001$), with a mean 5.02-point decrease (95 % CI: 4.09 to 5.96). Pain reductions occurred despite NP aggravating treatment and ongoing usage by 40 % of the cohort. A total of 92 % of patients reported acute reductions in SMA in the online questionnaire. The authors concluded that the significant pain reductions observed may indicate effectiveness of laser acupuncture in reducing both CPSNP and SMA in breast cancer patients post-BCS. This was a small ($n = 30$) retrospective study; these preliminary findings need to be validated by well-designed studies.

Dry Needling for the Treatment of Cervical Spondylosis

Mengi and Bulut (2024) noted that there are no data on the additional contribution of DN for trigger points (TPs) accompanying patients with cervical spondylosis (CS). In a prospective RCT, these researchers examined the contribution of DN applied to concomitant active TPs in the upper trapezius muscle on the treatment outcomes of physiotherapy in patients with CS. A total of 70 patients with CS and active TPs in the upper trapezius muscle were included. The 1st group received physiotherapy for 5 days/week for 3 weeks. The 2nd group received DN with the same program. All subjects were examined before treatment (day 0) and at the end of treatment (day 21) in terms of pain, functional status, QOL, anxiety/depression scores, and number of TPs. A total of 33 patients in the 1st group and 32 patients in the 2nd group completed the study. While the change over time was found significant in all variables, the change was not different between groups. The group-time interaction effect was not found to be statistically significant in any variable. Percentage changes of all variables were similar between the 2 groups. The authors concluded that DN treatment added to the physiotherapy did not contribute to recovery in patients with CS.

Dry Needling for the Treatment of Low Back Pain

In a RCT, Alrawaili et al (2024) examined the effectiveness of using acupuncture dry needles in the short-term for the treatment of chronic mechanical LBP. This trial included 30 subjects with non-specific LBP; age of 20 to 45 years. Subjects were divided into 2 parallel groups using a random process: 15 individuals in group A had acupuncture treatment with a dry needle placed to specified locations on their backs, whereas 15 individuals in group B participated in muscle strengthening and stretching exercises. Both before and after therapy, researchers examined the lumbar ROM in 4 directions: flexion, extension, bilateral side bending, as well as pain intensity and functional impairment. These investigators examined the data for normality and co-variance homogeneity using the Shapiro-Wilk and Box's tests, respectively. The variables of interest that were examined were compared across various test groups and measurement periods using a 2x2 mixed design-multivariate analysis of variance (MANOVA). They employed a significance threshold of 0.05. When comparing the 2 groups following treatment, it was found that the flexion and extension ROM increased significantly ($p < 0.05$), while the pain intensity, bending to the right and left, and functional

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impairment decreased significantly ($p < 0.05$). The findings of multiple pair-wise comparisons showed a note-worthy difference in pain severity, functional debility, and back ROM between the 2 groups before and after treatment ($p < 0.05$), with group A showing an advantage over group B following treatment. The authors concluded that based on these findings, dry needle acupuncture may be a beneficial modality in the treatment of chronic mechanical LBP by relieving pain intensity, improving functional debility, and improving ROM. Moreover, these researchers stated that future research should examine comparative effectiveness by evaluating acupuncture dry needles against other conventional treatments for chronic mechanical LBP. This approach would aid in establishing the relative effectiveness and potential advantages of acupuncture in comparison to existing therapeutic modalities.

Dry Needling for the Treatment of Headaches

Kandeel et al (2024) stated that headaches pose a significant burden on individuals and society, necessitating both pharmacotherapies and non-pharmacological treatments; DN has surfaced as a notable non-pharmacological alternative for addressing headaches. In a systematic review and meta-analysis, these researchers compared DN intervention with the control group in the management of different types of headaches. This meta-analysis encompassed a systematic exploration of 5 electronic databases up to May 2023, with studies assessed for validity using the Cochrane risk of bias tool. Eligible studies included those with a control group, which could consist of sham interventions, physiotherapy, or pharmacotherapies. Outcomes related to disability, headache intensity, and frequency were extracted and analyzed. Out of 4,304 studies screened, 13 RCTs were included in this systematic review, of them 10 RCTs were included in the meta-analysis. A significant decline in the disability score and headache intensity was evident in the DN group compared to the control group, 1- and 3-month follow-ups. Specifically, substantial reductions in headache frequency were prominent after 1 and 3 months post-intervention. The authors concluded that despite its relative safety and effectiveness in improving disability, pain intensity and headache frequency, the considerable heterogeneity and potential biases in the incorporated studies necessitate a cautious interpretation of the outcomes. These disparities underscored the need for more uniform, high-quality research to fortify the understanding of DN's optimal application conditions and protocols. These investigators stated that while DN significantly mitigated headache frequency and intensity, its overarching effectiveness in modifying disability scores is yet inconclusive at early interventions, necessitating further investigations to refine its use in multi-faceted pain management strategies for chronic and recurrent conditions.

Dry Needling for the Treatment of Neck Pain

In a biblio-metric analysis, Kalia and Rajput (2024) analyzed the available evidence on the use of DN in the management of neck pain (NP). These investigators carried out a biblio-metric analysis of publications from January 1, 2002 to December 31, 2022. Using the Scopus database, studies on the use of DN in NP were retrieved. Scopus analyzer was employed to obtain results such as "documents by year", "source", "country" and so on. VOS viewer Version 1.6.18 was used to analyze various units such as "co-authorship", "co-occurrences" as well as "citation analysis". The search yielded 125 studies; most of them were published in 2022, followed by 2021. A network analysis of various parameters showed that the topic exhibits a high research potential. The authors concluded that this biblio-metric analysis provided an understanding of the literature on DN for the treatment of NP. The number of studies in this sector has increased over the past 10 years; however, there is a gap in research from low- and middle-income countries on use of DN in the management of NP. Moreover, these researchers stated that this study emphasized the rising interest and involvement of nations, organizations, and writers in the field of DN for the treatment of NP revealing that there is growing awareness of the potential advantages of DN in the management of NP.

Sami Alattar and Alzahrani (2024) noted that MPS is a chronic condition caused by sensitive pressure regions within the muscles known as myofascial trigger points (MTrPs). In a RCT, these researchers examined the effectiveness of adding DN to activate MTrPs in the upper trapezius muscle compared with usual physiotherapy among individuals with chronic NP. A total of 30 subjects (mean age of 29.7 ± 4.4 years) were recruited from a private clinic in Saudi Arabia. Participants were randomized into 2 groups: the experimental group (application of DN to the MTrPs coupled with usual physiotherapy ($n = 15$)) and the

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control group (usual physiotherapy alone (n = 15)). The primary outcomes were pain (assessed using the VAS) and disability (Neck Disability Index [NDI]), and the secondary outcomes were neck active ROM (AROM; assessed using cervical ROM) and depression (BDI). Significant between-group difference in pain intensity was observed immediately post-intervention. Subjects in the experimental group had significantly higher pain (mean difference = 1.27, 95 % CI: 0.20 to 2.33, p = 0.022, Cohen's d = 0.889) than those in the control group. There was no significant difference between both groups in pain intensity during the follow-up. There were no between-group differences in disability immediately post-intervention. However, there was a between-group difference in disability at follow-up; subjects in the experimental group had significantly lower disability (MD = -3.13, 95 % CI: -5.07 to -1.20, p = 0.003, Cohen's d = 1.211) than those in the control group. Immediately post-intervention, the experimental group showed greater flexion AROM compared to the control group, with no differences in other AROM measures. At follow-up, the experimental group exhibited significantly higher neck AROM in extension, flexion, right and left side bending, and lower depression, while no differences were observed in right- and left-rotation AROMs between groups. The authors concluded that the addition of DN to standard physiotherapy effectively improved disability, AROM (extension, flexion, and side bending), and depression among patients with chronic NP. Moreover, these investigators stated that future research should examine long-term effects, include other cervical muscles, and consider individual patient characteristics to clarify DN's benefits.

Dry Needling for the Treatment of Patellar Tendonitis (Jumpers' Knee)

Wang and Lyu (2024) stated that patellar tendonitis is a common degenerative disease of the patellar tendon that markedly affects the sports careers of elite athletes and ADL of sports enthusiasts. Injection therapy is a prevalent invasive treatment for patellar tendonitis. In a systematic review and meta-analysis, these investigators examined various injection treatments for patellar tendonitis that consider agent type and characteristics, frequency, and assessment time-points. Data sources included PubMed, Embase, and the Cochrane Library; RCTs examining the effect of various injection strategies on the treatment of patients with patellar tendonitis were considered. First author, year of publication, research location, registration information, patient type, sample size, age, sex, intervention, control treatment, and follow-up period in each study were extracted. A total of 19 RCTs were included in the analysis. In the network meta-analysis of Victorian Institute of Sports Assessment-Patellar (VISA-P) outcomes, polidocanol (SMD, 6.52; 95 % CI: 4.75 to 8.30; p < 0.01), tenocyte-like cells (SMD, 4.08; 95 % CI: 2.92 to 5.25; p < 0.01), and leukocyte-poor platelet-rich plasma (LP-PRP) plus high-volume image-guided injection (HVIGI) (SMD, 1.56; 95 % CI: 0.62 to 2.50; p < 0.01) were significantly superior to non-injection conservative treatment, mainly at the 6-month follow-up timepoint. For VAS results, multiple DN (SMD, -1.78; 95 % CI: -2.56 to -1.00; p < 0.01), LP-PRP (SMD, -0.71; 95 % CI: -1.31 to -0.12; p = 0.02), and LP-PRP plus HVIGI (SMD, -1.31; 95 % CI: -2.22 to -0.39; p < 0.01) were significantly superior to blank, which was also mainly at the 6-month time-point. The authors concluded that injection-related treatments: polidocanol, tenocyte-like cells, LP-PRP, and multiple DN showed potential short-term (1 to 3 months) or medium-term (6 months) treatment benefits. Moreover, there is still no evidence for injection interventions with long-term therapeutic benefit. Level of Evidence = II.

Dry Needling for the Treatment of Scars

In a systematic review, Trybulski et al (2024) synthesized existing data on the evidence gap in scar treatment and examined the effectiveness of acupuncture and DN in the treatment of scars and related symptoms. The review adhered to the PRISMA 2020 statement for recommended reporting elements in systematic reviews. The inclusion criteria followed the population, intervention, comparison and outcome (PICO) methodology. The literature search was carried out using databases including PubMed, Cochrane Library, Semantic Scholar, Europe PubMed Central, and Google Scholar. Studies on acupuncture and DN for scar treatment were included. Because of the diversity of the studies' results and methodologies, a systematic review was performed to organize and describe the findings without attempting a numerical synthesis. A total of 19 studies relevant to the article's theme were identified, with 11 selected for detailed review. The studies included 2 case reports on DN, 1 case series DN, 5 case reports on acupuncture, 2 RCTs on acupuncture, and 1 case report on Fu's subcutaneous needling. A quality assessment was carried out using the JBI CAT and PEDro scales – 4 case reports scored 7 points, 1 case scored 8 points, 3 cases were rated 6 points or lower,

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the case series was rated 6 points, and the RCTs scored 8 and 5 points. Most studies showed a desired therapeutic effect in scar treatment with acupuncture and DN; however, the level of evidence varied across studies. The analysis did not conclusively support the use of acupuncture and DN to improve scar conditions. The authors concluded that although DN and acupuncture techniques are popular in physiotherapy, adequate scientific evidence is currently not available to support their effectiveness in scar treatment. These researchers stated that there are gaps in the research methodology, a lack of randomized trials, as well as significant heterogeneity in the assessment of effects.

Dry Needling for the Treatment of Stroke

Esmaeeli et al (2024) noted that stroke is a leading cause of disability globally and is often accompanied by complications such as spasticity. Static stretching (SS) is a common physiotherapy intervention for reducing spasticity, whereas DN is a novel approach. However, the combined effects of DN and SS on spasticity have not been thoroughly examined. Given the pivotal effect of spasticity on daily activities, mitigating spasticity can significantly contribute to restoring patient independence. These researchers described the protocol of a randomized, double-blind, sham-controlled study that will examine the impact of DN plus SS on spasticity, alpha motor neuron excitability, overall function, and QOL in patients with chronic stroke. This trial will be carried out in patients with post-stroke spasticity in the plantar flexor muscles. A total of 28 subjects will be randomly assigned to either an intervention or control group. The intervention group will receive DN (60s × 3 days/week; 1 week) plus SS (20 mins × 5 days/ week; 1 week). The control group will undergo sham DN (60s × 3 days/week; 1 week) and SS (20 mins × 5 days/week; 1 week). Both groups will be evaluated at baseline, immediately post-treatment, and after 1 week of follow-up. Outcome measures will include the Modified Ashworth Scale, H-reflex latency, Hmax/Mmax ratio, active and passive ankle dorsiflexion ROM, timed up and go (TUG) test, and the EuroQol questionnaire. Results from this randomized, sham-controlled trial will provide evidence for the effectiveness of DN in combination with SS for the treatment of spasticity. The authors stated that the additional impact of DN in conjunction with SS remains unclear and warrants investigation. This study, with a high level of evidence, aims to address this knowledge gap.

Appendix

Documentation Requirements

Acupuncture should be provided in accordance with an ongoing, written plan of care. The purpose of the written plan of care is to assist in determining medical necessity and should include the following:

The written plan of care should be sufficient to determine the medical necessity of treatment, including:

I. The diagnosis along with the date of onset or exacerbation of the disorder/diagnosis:

   A. A reasonable estimate of when the goals will be reached;
   B. Long-term and short-term goals that are specific, quantitative and objective;
   C. Acupuncture evaluation;
   D. The frequency and duration of treatment; *and*
   E. The acupuncture protocol to be used in treatment.

II. Signatures of the patient's attending physician and/or acupuncturist

   The plan of care should be ongoing, (i.e., updated as the member's condition changes), and treatment should demonstrate reasonable expectation of improvement (as defined below):

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A. Acupuncture services are considered medically necessary only if there is a reasonable expectation that acupuncture will achieve measurable improvement in the member's condition in a reasonable and predictable period of time.
B. The member should be reevaluated regularly, and there should be documentation of progress made toward the goals of acupuncture.

The treatment goals and subsequent documentation of treatment results should specifically demonstrate that acupuncture services are contributing to such improvement.

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The above policy is based on the following references:

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